Quantum Error Correction Codes using Rydberg Facilitation

MASTER THESIS
in Advanced Quantum Physics

by

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Prof. Dr. Michael Fleischhauer

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Abstract

Quantum information processing exploits superposition and entanglement to perform tasks that are intractable for classical devices. Neutral atoms trapped in optical tweezers, with interactions activated via Rydberg excitation, offer a compelling route to scalable quantum computing. This thesis develops hardware-aware error-correction workflows tailored to such platforms. We study two primitive three-qubit repetition codes — the bit-flip and phase-flip codes emphasizing two-level and three-level atomic control, respectively. We begin by designing the arrangement of Rydberg atoms by proposing compact geometries that leverage facilitation and blockade effects. We propose using different Rydberg atoms to enable conditional operations while suppressing unwanted interactions. We then develop control protocols to realize the necessary quantum gates. For two-level control, we employ rapid adiabatic passage (RAP) and for three-level control, we use Raman transitions to achieve robust single-qubit rotations and entangling operations. More importantly, we incorporate realistic noise models capturing the dominant error channels in Rydberg systems, including decay and dephasing. Performance is evaluated with Monte Carlo wave-function simulations including time-dependent drives, yielding trajectory-resolved fidelity distributions through encode–store–correct–decode cycles.

Overall, the work bridges control-level design and code-level performance for Rydberg arrays, providing circuit blueprints, parameter windows, and simulation evidence to guide near-term demonstrations of logical protection on neutral-atom hardware.

Contents

List of Abbreviations

BFEC	Bit-Flip Error Correction
FRR	Fully-Resonant Raman
MCWF	Monte Carlo Wave-Function
MS	Morris-Shore
PFEC	Phase-Flip Error Correction
QEC	Quantum Error Correction
RAP	Rapid Adiabatic Passage

Part I

Introduction

To be an Error and to be Cast out is a part of God's Design

William Blake
The Marriage of Heaven and Hell

Quantum computation and quantum information is the study of information processing tasks that can be accomplished using quantum mechanical systems. The field came to prominence in the 1980s with two key proposals: Paul Benioff (1980) described the idea of a quantum mechanical model of a Turing machine [1], and Richard Feynman (1982) independently proposed the idea of a quantum computer capable of efficiently simulating quantum systems, a task that is exponentially hard for classical computers [2].

Designing such quantum mechanical systems led to the introduction of quantum bits, or *qubits*, the quantum analogue of classical bits. A qubit is a controllable two-level system with computational basis $\{|0\rangle, |1\rangle\}$. Unlike a classical bit, a qubit can exist in a superposition $\alpha|0\rangle + \beta|1\rangle$ with complex amplitudes α, β satisfying $|\alpha|^2 + |\beta|^2 = 1$. Geometrically, pure qubit states exist in a two-dimensional complex vector space that form the Bloch sphere; by contrast, a classical bit is either 0 or 1. Qubit state manipulations are implemented by unitary *quantum gates*, broadly classified into single-qubit and multi-qubit gates. Single-qubit gates effect rotations $R_{\vec{n}}(\theta)$ on the Bloch sphere, while multi-qubit gates (e.g., CNOT, CZ) create entanglement. Single-qubit rotations together with a two-qubit entangling gate form a *universal set* of gates [3].

Today, quantum information processing promises computational and communication tasks infeasible for classical devices, from factoring large integers [4] and simulating strongly correlated matter to secure key distribution [3, 5]. Turning that promise into practice requires hardware platforms that implement universal gates with high fidelity, scale to large registers, and more importantly mitigate noise through quantum error correction (QEC) [6, 7].

Programmable arrays of neutral (Rydberg) atoms in optical tweezers have emerged as strong contenders for scalable quantum computing hardware [8]. Their key advantage is a native, fast, switchable two-qubit interaction via Rydberg excitation: strong van der Waals interactions generate conditional multi-qubit gates on microsecond timescales while leaving idling qubits essentially non-interacting [9]. This thesis work leverages those features to design, analyze, and simulate error-correcting primitive algorithms tailored to neutral-atom constraints.

Despite their advantages, Rydberg-atom processors face prominent error channels, including spontaneous decay of Rydberg population, optical dephasing from intensity and phase noise, motional Doppler shifts, and cross-talk in dense arrays. QEC provides a route to suppress such errors below a certain threshold by encoding a *logical qubit* into several physical qubits and repeatedly extracting error syndromes. The central objective of this thesis is to develop and evaluate error-correction workflows tailored to neutral-atom hardware: (i) repetition-style bit-flip and phase-flip codes (BFEC and PFEC), and (ii) the nine-qubit Shor code as combined protection against both X- and Z-type errors. We emphasize architectures and pulse protocols that are realistic for current experimental capabilities, and we quantify performance through dynamical simulations.

In this thesis, we develop hardware-aware designs of bit-flip and phase-flip protocols on Rydberg arrays, including geometry, gate sequences, and timings consistent with facilitation and blockade effects. We propose a "two-type" choice (different Rydberg states) for atoms used in the bit-flip and phase-flip codes and a "three-type" choice for the Shor code. We present two complementary control toolboxes depending on the chosen atom states manifold: (i) rapid adiabatic passage (RAP) across a Rydberg-induced avoided crossing to implement CNOT gates in a two-level system, and (ii) Raman transition based single-qubit rotations and CNOT gates in a three-level Λ-system. We provide compact, realistic noise models for Rydberg systems and quantify code performance via dynamical simulations using the Monte Carlo wave-function (MCWF) method [10, 11].

The thesis is organized as follows. Following this introduction (Part I), Part II lays out the necessary theoretical background, from Rydberg-atom physics (Ch. 1) and open quantum systems (Ch. 2) to quantum error correction (Ch. 3). Part III is dedicated to the bit-flip error-correction code, where we develop system architecture (Ch. 4), the corresponding control toolbox (Ch. 5), and implementation with simulation results (Ch. 6). Part IV focuses on the phase-flip error correction code, where we adapt the same systematic as in Part III, from architecture design (Ch. 7), control protocols (Ch. 8), to implementation and results (Ch. 9). Finally, Part V concludes with a summary and outlook.

Part II

Theoretical Foundations

1 Rydberg Physics

1.1 Introduction

In 1862, Ander Jonas Ångström discovered a series of spectral lines in the emission spectrum of hydrogen [12]. These were later explained by Johannes Rydberg in 1888 using an empirical formula, for spectral lines emitted by atoms that have single outer shell electron [13]. This is given by

$$E_n = \frac{-R_\mathrm{H}}{n^2}, \quad n \in \mathbb{N} \tag{1.1}$$

where R_H is the Rydberg constant for hydrogen, and n is the principal quantum number. The formula describes wavelengths of spectral lines of hydrogen and other elements, leading to the concept of *Rydberg states*. These are highly excited states of atoms with one or more electrons, more precisely in high principal quantum number orbitals. They exhibit exaggerated atomic properties, such as large electric dipole moments and long lifetimes, making them ideal for studying quantum phenomena and interactions. Recently, Rydberg atoms have gained traction in many fields spanning quantum information processing, quantum simulation, and quantum optics. This is primarily due to interactions of Rydberg atoms, which enable strong and controllable interactions between atoms over relatively large distances, such as *Rydberg blockade* and *Rydberg facilitation*. This made them a suitable candidate for neutral atom quantum computing platforms [8, 9]. Also with the advent of optical tweezer technology, availablity of commercial high-power diode lasers, and high resolution imaging systems, it is now possible to trap and manipulate individual neutral atoms with high precision [14–17]. Combining these advancements in control and switchability of Rydberg interactions, has established applications for quantum simulation, quantum communication, and quantum sensing, paving the way for future developments in quantum technologies.

1.2 Lifetime of Rydberg Atoms

The spatial overlap between the small ground state orbital and the large Rydberg orbital is very small, leading to a very small electric dipole matrix element between these two states. This results in Rydberg states being long-lived. The Rydberg state lifetime τ is primarily determined by spontaneous emission and blackbody radiation-induced transitions to lower-lying states, which scale with the principal quantum number as

$$\tau \propto n^3 \tag{1.2}$$

thus, reaching lifetimes in the range of hundreds of microseconds (μs) for $n \sim 60$[18]. One would also expect the increased overlap between the Rydberg orbital and its neighbouring orbitals, which would lead to rapid spontaneous decay to these states. The dipole matrix element between such states scales as

$$\langle nS_{1/2}| \, \hat{d} \, |(n+1)P_{3/2}\rangle \propto n^2 \tag{1.3}$$

where $\hat{d}$ is the electric dipole operator. However, thankfully for the stability of Rydberg states, the number of vacuum modes available for spontaneous decay goes down even faster [18].

1.3 Interactions between Rydberg Atoms

As discussed before, Rydberg atoms exhibit strong interactions between each other. These interactions can be explained by the exchange of (*virtual*) *photons* between the atoms and occur over relatively large distances, typically on the order of several μm. The two primary types of interactions are the resonant *dipole-dipole interaction* and the non-resonant *van der Waals interaction*.

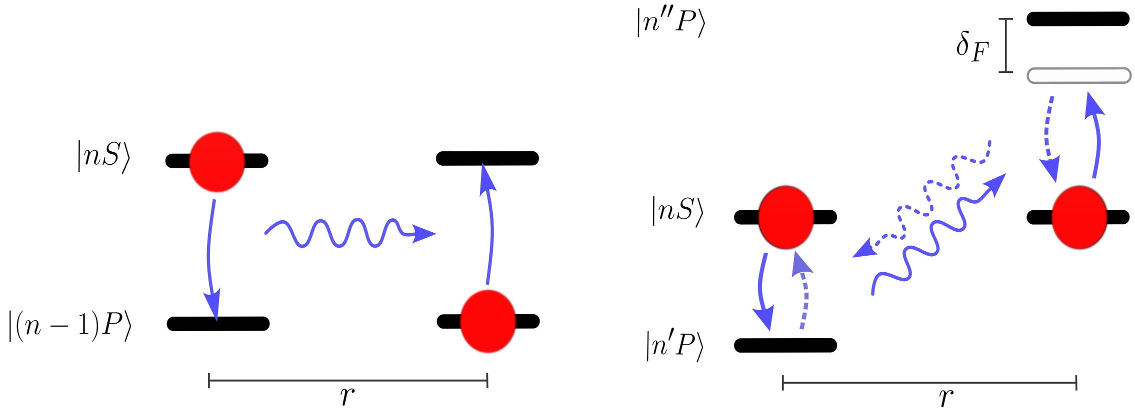


Figure 1.1. Schematic representation of the dipole-dipole and van der Waals interactions between two Rydberg atoms. (Left) Dipolar c_3 interaction between two Rydberg atoms in different states $|nS\rangle$ and $|(n-1)P\rangle$ allowing resonant energy exchange via a virtual photon. (Right) Van der Waals c_6 interaction between two Rydberg atoms in the same state $|nS\rangle$ via off-resonant energy exchange through virtual photons, due to the energy shift δ_F between $|n''P\rangle$ and $|nS\rangle$ compared to the $|nS\rangle$ and $|n'P\rangle$ states. Figure is adapted from [18].

1.3.1 Dipole-Dipole c_3 Interaction

Consider two Rydberg atoms, A and B, separated by a distance R, such that they are in states $|r_A\rangle = |nS\rangle$ and $|r_B\rangle = |(n-1)P\rangle$ respectively. In the two-atom basis, the states $|r_A r_B\rangle = |nS, (n-1)P\rangle$ and $|r'_A r'_B\rangle = |(n-1)P, nS\rangle$ are degenerate and can interact via a coherent exchange of a virtual photon. This exchange leads to a resonant dipole-dipole interaction, where atom A can transfer its excitation to atom B and vice versa.

Under the multipole expansion, the leading term in the interaction potential is the dipole-dipole term, which dominates at large distances and is given by

$$\hat{V}_{dd}(r) = \frac{1}{4\pi\epsilon_0}\left(\frac{\hat{\mathbf{d}}_A \cdot \hat{\mathbf{d}}_B - 3(\hat{\mathbf{d}}_A \cdot \hat{\mathbf{r}})(\hat{\mathbf{d}}_B \cdot \hat{\mathbf{r}})}{r^3}\right) \tag{1.4}$$

where $\hat{\mathbf{d}}_A$ and $\hat{\mathbf{d}}_B$ are the electric dipole operators of atoms A and B, respectively, and $\hat{\mathbf{r}}$ is the unit vector along the line connecting the two atoms [19]. The strength of this interaction scales as $1/r^3$.

This potential is quadratic in the dipole operators, and thus scales as $(n^2)^2 = n^4$ and is characterized by the coefficient c_3, which depends on the specific Rydberg states involved and their dipole moments. We can rewrite the interaction potential as the semi-classical expression

$$V_{dd}(r) = \frac{c_3}{r^3}(1 - 3\cos^2\theta) \tag{1.5}$$

where θ is the angle between the dipole moment and the interatomic axis. The coefficient c_3 can be calculated to as

$$c_3 = \frac{1}{4\pi\epsilon_0} \langle nS, (n-1)P| \, \hat{\mathbf{d}}_A \cdot \hat{\mathbf{d}}_B \, |(n-1)P, nS\rangle \propto n^4. \tag{1.6}$$

Thus, this interaction is also known as the c_3 interaction.

1.3.2 Van der Waals c_6 Interaction

When two Rydberg atoms are in the same state, for example, both in the state $|r_A\rangle = |r_B\rangle = |nS\rangle$, they cannot always interact via the resonant dipole-dipole interaction because there are no degenerate states to couple to. However, they can still interact through a non-resonant exchange of virtual photons. That is, the atom A decays to a lower-lying state $|n'P\rangle$, emitting a virtual photon that is then absorbed by atom B to excite, but there likely exists no state $|n''P\rangle$ such that the two-atom states $|nS, nS\rangle$ and $|n'P, n''P\rangle$ are degenerate. Thus, the atom B reemits the virtual photon and both the atoms return to their original states. The energy difference between the two-atom states $|nS, nS\rangle$ and $|n'P, n''P\rangle$ is called the *Förster defect* δ_F.

In the basis of the two-atom states $|nS, nS\rangle$ and $|n'P, n''P\rangle$, the interaction Hamiltonian can be written as

$$H = \begin{pmatrix} \delta_F & V_{dd}(r) \\ V_{dd}(r) & 0 \end{pmatrix}. \tag{1.7}$$

The eigenvalues of this Hamiltonian are given by

$$V_{int}(r) = \frac{\delta_F}{2} \pm \sqrt{\left(\frac{\delta_F}{2}\right)^2 + V_{dd}(r)^2} \tag{1.8}$$

where $V_{dd}(r)$ is the dipole-dipole interaction potential. In the limit where the Förster defect δ_F is much larger than the dipole-dipole interaction $V_{dd}(r)$, i.e., $|\delta_F| \gg |V_{dd}(r)|$, we can approximate the interaction potential by taylor expanding as

$$V_{int}(r) \approx \pm \frac{V_{dd}(r)^2}{\delta_F}. \tag{1.9}$$

This interaction scales as $1/r^6$ and is known as the van der Waals interaction. The strength of this interaction is characterized by the coefficient c_6, which depends on the specific Rydberg states involved and their dipole moments. The van der Waals interaction can be expressed as

$$V_{vdW}(r) = \frac{c_6}{r^6}. \tag{1.10}$$

The coefficient c_6 can be calculated as

$$c_6 = \frac{c_3^2}{\delta_F} \propto n^{11} \tag{1.11}$$

where the Förster defect δ_F typically scales as $1/n^3$. Thus, this interaction is also known as the c_6 interaction.

Following this, all the Rydberg-Rydberg interactions discussed in this thesis are of the van der Waals type.

1.4 Rydberg Blockade

One of the most significant phenomena arising from the strong interactions between Rydberg atoms is the *Rydberg Blockade*. Theoretically proposed by Lukin et al. in 2001 [20], and experimentally realised by Urban et al. [21] and Gaëtan et al. [22] in 2009, the Rydberg blockade effect occurs when two or more atoms are excited to Rydberg states within a certain distance, known as the *blockade radius* r_b. When one atom is excited to a Rydberg state within this radius, the strong interactions shift the energy levels of the neighbouring atoms, preventing them from being excited. This process allows for the creation of entangled states and the implementation of quantum gates, making it a crucial mechanism for quantum information processing with neutral atoms [23].

Consider two atoms, A and B separated by a distance r_b, both initially in the ground state $|g\rangle$ and a resonant laser with Rabi frequency Ω coupled to the ground state $|g\rangle$ and Rydberg state $|r\rangle$ for both atoms. In the two-atom basis states of the system: $|gg\rangle$, $|gr\rangle$, $|rg\rangle$, and $|rr\rangle$, the Hamiltonian can be written as

$$H = \frac{\hbar}{2}\begin{pmatrix} 0 & \Omega & \Omega & 0 \\ \Omega & 0 & 0 & \Omega \\ \Omega & 0 & 0 & \Omega \\ 0 & \Omega & \Omega & 2V_{vdW}(r_b) \end{pmatrix} \tag{1.12}$$

where $V_{vdW}(r_b)$ is the van der Waals interaction between the two Rydberg atoms. To simplify the problem, we introduce the symmetric and antisymmetric states: $|+\rangle = \frac{1}{\sqrt{2}}(|gr\rangle + |rg\rangle)$ and $|-\rangle = \frac{1}{\sqrt{2}}(|gr\rangle - |rg\rangle)$. One can notice that the antisymmetric state $|-\rangle$ is decoupled from the dynamics of the system, and thus we can ignore it. The Hamiltonian in the reduced basis $\{|gg\rangle, |+\rangle, |rr\rangle\}$ becomes

$$H = \frac{\hbar}{2}\begin{pmatrix} 0 & \sqrt{2}\Omega & 0 \\ \sqrt{2}\Omega & 0 & \sqrt{2}\Omega \\ 0 & \sqrt{2}\Omega & 2V_{vdW}(r_b) \end{pmatrix}. \tag{1.13}$$

Starting from the initial state $|gg\rangle$, a resonant excitation to the symmetric state $|+\rangle$ occurs with an enhanced Rabi frequency of $\sqrt{2}\Omega$. However, the doubly excited state $|rr\rangle$ is shifted out of resonance by the interaction energy $V_{vdW}(r_b)$, depending on the distance r_b between the atoms. For large distances or weakly interacting regime given by $|V_{vdW}(r_b)| \ll \Omega$, the coupling between the $|+\rangle$ and $|rr\rangle$ stays nearly resonant and the system can be excited to the doubly excited state $|rr\rangle$. However, in the strongly interacting regime where $|V_{vdW}(r_b)| \gg \Omega$, the excitation to the doubly excited state $|rr\rangle$ is suppressed, and the system effectively behaves as a two-level system between the states $|gg\rangle$ and $|+\rangle$. This decoupling of the doubly excited state is the essence of the Rydberg blockade effect. The distance r_b where this blockade occurs is given by

$$r_b = \sqrt[6]{\frac{c_6}{\Omega}}. \tag{1.14}$$

The effect also extends to ensembles of atoms, where the presence of a single Rydberg excitation can block the excitation of all other atoms within the blockade radius, leading to collective behaviour and entanglement among the atoms. This collective behaviour is called a *Rydberg superatom* [24] and is a key resource for quantum information processing and quantum simulation applications.

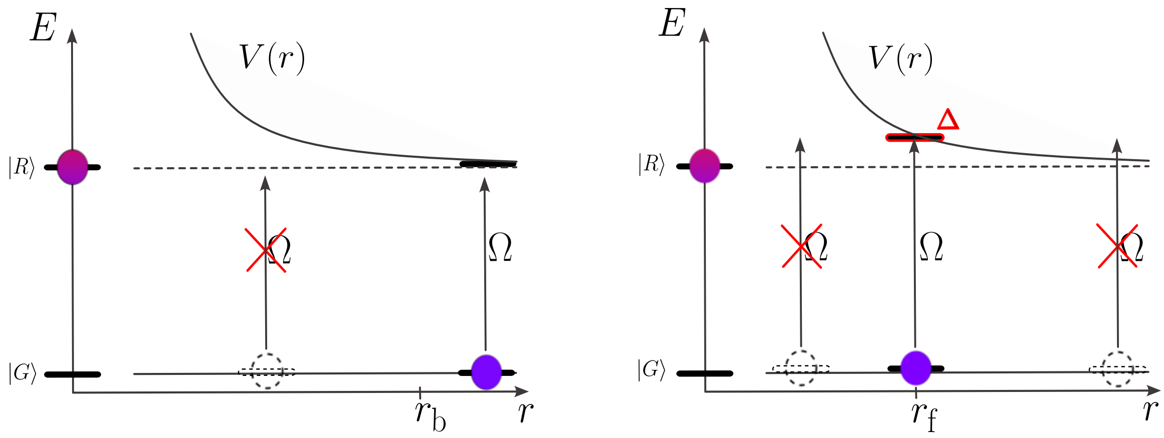


Figure 1.2. Schematic representation of the Rydberg Blockade and Rydberg Facilitation effects between two Rydberg atoms, when atom A is excited to the Rydberg state. (Left) Rydberg Blockade: For distances $r < r_b$, the strong van der Waals interaction $V_{vdW}(r)$ shifts the doubly excited state $|RR\rangle$ out of resonance, preventing atom B from being excited to the Rydberg state. (Right) Rydberg Facilitation: For distances $r \approx r_f$, the interaction-induced energy shift $V_{vdW}(r)$ compensates for the laser detuning Δ, bringing the doubly excited state $|RR\rangle$ back into resonance, allowing atom B to be excited to the Rydberg state. Figure is adapted from [18].

1.5 Rydberg Facilitation

Another intriguing phenomenon arising from the interactions between Rydberg atoms is *Rydberg Facilitation*. This is quite the opposite of the Rydberg blockade effect studied previously. If the driving laser is strongly detuned from the Rydberg state by an amount Δ i.e., $|\Delta| \gg \Omega$, the excitations from $|gg\rangle$ to $|+\rangle$, as well as from $|+\rangle$ to $|rr\rangle$, are suppressed. However, at a specific distance r_{f}, where

$$V_{vdW}(r_{\mathrm{f}}) + \Delta = 0, \tag{1.15}$$

the interaction-induced energy shift $V_{vdW}(r_{\mathrm{f}})$ compensates for the laser detuning Δ, bringing the doubly excited state $|rr\rangle$ back into resonance with the laser. This condition allows for resonant excitation of the second atom to the Rydberg state, facilitated by the presence of the first atom's Rydberg excitation. The Eq. 1.15 is called the *facilitation condition*, and the distance r_{f} is known as the *facilitation radius* and it is given by

$$r_{\mathrm{f}} = \sqrt[6]{\frac{c_6}{\Delta}}. \tag{1.16}$$

Due to the finite linewidth of the driving laser, the facilitation condition is satisfied within a certain range of distances around r_{f}, known as the *facilitation shell*. The width of this shell δr_{f} can be estimated by considering the range of distances where the interaction-induced energy shift $V_{vdW}(r)$ is comparable to the laser linewidth γ. This width is given by

$$\delta r_{\mathrm{f}} = r_{\mathrm{f}} \frac{\gamma}{2\Delta}. \tag{1.17}$$

The facilitation shell plays a crucial role in determining the spatial correlations and dynamics of Rydberg excitations in an ensemble of atoms [25].

This facilitation effect can lead to conditional excitation of neighbouring atoms, allowing for enhanced control over quantum states and facilitating the implementation of quantum gates in larger atomic ensembles. This phenomenon along with the Rydberg blockade effect, provides a versatile toolkit for manipulating and controlling quantum states in neutral atom systems, paving the way for advancements in quantum information processing and quantum simulation. In this thesis, we see how to exploit the Rydberg facilitation effect for the implementation of multi-qubit gate operations and also the Rydberg blockade effect for the realisation of a quantum error correction protocol.

2 Open Quantum Systems

2.1 Introduction

An open quantum system is a quantum system that interacts with its environment, leading to non-unitary evolution of the system's state. This interaction can significantly affect the dynamics of the system, leading to phenomena such as decoherence, dissipation, and quantum jumps, where the information contained in the system is exchanged with the environment. The study of open quantum systems is crucial for understanding real-world quantum systems, as no quantum system is perfectly isolated from its surroundings.

The theoretical framework for open quantum systems is built upon the concepts of quantum mechanics, statistical mechanics, and quantum information theory. Most commonly, the analysis of properties of such systems require numerical simulations, since analytical solutions are often not feasible because the environment we are dealing with is usually too complex, making it very difficult, if not impossible, to derive the exact solution.

In general, the time evolution of a closed quantum system is described by the *Schrödinger equation*, which is a unitary evolution. However, when a quantum system interacts with the environment, the evolution can no longer be described by unitary dynamics alone. Instead, we need to consider the effects of the environment on the system's state, which leads to a more complex description of the system's dynamics. This is typically done using the framework of *Master Equations*, which describe the time evolution of the system using the density matrix formalism. Broadly, these systems can be classified into two categories: *Markovian* and *non-Markovian* systems. In Markovian systems, the system evolves without a memory of its past states, while in non-Markovian systems, the next state of the system is determined by its past states, leading to a more complex evolution. In the following thesis, we will work with Markovian systems, which are also more commonly used in quantum optics and quantum information processing.

Under the assumption that the entire system-environment is a closed quantum system, we can therefore describe the entire dynamics by a unitary evolution. The global hamiltonian of the system-environment can be written as

$$\hat{H}_{\mathrm{SE}} = \hat{H}_{\mathrm{S}} + \hat{H}_{\mathrm{E}} + \hat{H}_{\mathrm{int}} \tag{2.1}$$

where $\hat{H}_{\mathrm{S}}$ is the Hamiltonian of the system belonging to the Hilbert space $\mathcal{H}_{\mathrm{sys}}$, $\hat{H}_{\mathrm{E}}$ is the Hamiltonian of the environment in Hilbert space $\mathcal{H}_{\mathrm{env}}$, and $\hat{H}_{\mathrm{int}}$ is the interaction Hamiltonian between the system and the environment. The global hamiltonian belongs to the Hilbert space $\mathcal{H}_{\mathrm{sys}} \otimes \mathcal{H}_{\mathrm{env}}$. This combined system-environment can thus be described by a pure state $|\Psi\rangle_{\mathrm{SE}}$, which evolves according to the Schrödinger equation as

$$i\hbar \frac{d}{dt} |\Psi(t)\rangle_{\mathrm{SE}} = \hat{H}_{\mathrm{SE}} |\Psi(t)\rangle_{\mathrm{SE}} . \tag{2.2}$$

The state of the system at time t can be described by the density matrix obtained from the partial trace over the combined system-environment state, ie.,

$$\hat{\rho}_{\mathrm{S}}(t) = \mathrm{tr}_{\mathrm{E}} \left(\hat{\rho}_{\mathrm{SE}}(t) \right) . \tag{2.3}$$

Hence, this leads to density matrix based master equations, which describe the time evolution of the system's state in the presence of the environment.

In this chapter, we will discuss the theoretical foundations of open quantum systems, focusing on the mathematical framework used to describe their dynamics. We will introduce the von Neumann equation and then extend this to the Lindblad Master equation, which is a key tool for analyzing Markovian systems. Following this, there is the quantum trajectories method, which provides a numerical approach to simulate the dynamics of open quantum systems. We will also discuss the concept of quantum jumps, which are discrete events that occur during the evolution of an open quantum system, and how they can be incorporated into the simulation framework. Finally, we will explore the relationship between the quantum trajectories method and the Lindblad Master equation, showing how the two approaches are equivalent in the limit of large numbers of trajectories.

This chapter will provide a solid foundation for understanding the dynamics of open quantum systems and the methods used to analyze them, which will be essential for the subsequent parts of this thesis.

2.2 Master Equation

Let us now proceed to the mathematical framework of quantum systems based on the density matrix formalism. We start with the von Neumann equation, which describes the time evolution of a closed quantum system. Following this, we will extend the discussion to the Lindblad Master equation [26, 27] also called Gorini-Kossakowski-Sudarshan-Lindblad (GKSL) equation, which is used to describe the dynamics of open quantum systems.

2.2.1 The Von Neumann Equation

The von Neumann equation [28] describes the time evolution of a density matrix state $\hat{\rho}$ under a Hamiltonian $\hat{H}(t)$ with no interaction from the environment. One can derive it from the Schrödinger equation,

$$\mathrm{i}\hbar\frac{d}{dt}\left|\Psi(t)\right\rangle = \hat{H}(t)\left|\Psi(t)\right\rangle. \tag{2.4}$$

Considering the density matrix $\hat{\rho} = |\Psi(t)\rangle\langle\Psi(t)|$, we can write the time derivative as

$$\begin{aligned}
\mathrm{i}\hbar\frac{\partial}{\partial t}\hat{\rho}(t) &= \mathrm{i}\hbar\frac{\partial}{\partial t}\left(\left|\Psi(t)\right\rangle\left\langle\Psi(t)\right|\right) \\
&= \mathrm{i}\hbar\left(\frac{\partial}{\partial t}\left|\Psi(t)\right\rangle\left\langle\Psi(t)\right| + \left|\Psi(t)\right\rangle\frac{\partial}{\partial t}\left\langle\Psi(t)\right|\right) \\
&= \mathrm{i}\hbar\left(-\frac{\mathrm{i}}{\hbar}\hat{H}(t)\left|\Psi(t)\right\rangle\left\langle\Psi(t)\right| + \left|\Psi(t)\right\rangle(+\frac{\mathrm{i}}{\hbar}\hat{H}^{\dagger}(t))\left\langle\Psi(t)\right|\right) \\
&= [\hat{H}(t), \hat{\rho}(t)].
\end{aligned} \tag{2.5}$$

Thus the von Neumann equation is given by

$$\mathrm{i}\hbar\frac{\partial}{\partial t}\hat{\rho}(t) = [\hat{H}(t), \hat{\rho}(t)]. \tag{2.6}$$

The time evolution of the density matrix is unitary, and the solution to the von Neumann equation can be expressed as

$$\hat{\rho}(t) = U(t)\hat{\rho}(0)U^{\dagger}(t) \tag{2.7}$$

where for time-dependent Hamiltonian, we have $U(t) = \exp\left(-\frac{\mathrm{i}}{\hbar}\int_0^t \hat{H}(t')dt'\right)$ is the time evolution operator, and for time-independent Hamiltonian, we have $U(t) = \exp\left(-\frac{\mathrm{i}}{\hbar}\hat{H}t\right)$ is the time evolution operator.

2.2.2 Lindblad Master Equation

Now, generalizing the von Neumann equation in Eq. 2.6 to include the effects of the environment and some strong assumptions, one can arrive at the Lindblad Master Equation, which is given by [1]

$$\hat{\mathcal{L}}\hat{\rho} = -\mathrm{i}[\hat{H}, \hat{\rho}] + \sum_{i=1}^{N^2-1} \gamma_i \left(\hat{A}_i\hat{\rho}\hat{A}_i^\dagger - \frac{1}{2}\left\{\hat{A}_i^\dagger\hat{A}_i, \hat{\rho}\right\}\right) \equiv \hat{\mathcal{L}}\hat{\rho}_{\mathrm{S}}(t). \tag{2.8}$$

where $\hat{A}_i$ are the jump operators, which describe the effect of the bath on the system, γ_i are the corresponding rates and $\hat{\mathcal{L}}$ is the Lindbladian superoperator. This superoperator is an element of N^2 dimensional Hilbert space, where N is the dimension of the system Hilbert space. The resulting dynamics of the system follow the following properties:

1. **Hermiticity:** $\hat{\rho}(t) = \hat{\rho}^\dagger(t)$ for all times t.

2. **Trace preserving:** $tr(\hat{\rho}(t)) = 1$ for all times t.

3. **Completely positive:** $0 \leq \hat{\rho}(t) \leq 1$ for all times t i.e., all the eigenvalues of the density matrix are semi-positive definite and bounded by 1.

Following this, one can express the time dependence of an arbitrary operator $\hat{O}$ in the Heisenberg picture as

$$\frac{d}{dt}\langle\hat{O}\rangle = \mathrm{tr}\left(\hat{O}\frac{d}{dt}\hat{\rho}\right) = \mathrm{i}\langle[\hat{H}, \hat{O}]\rangle + \sum_{i=1}^{N^2-1} \gamma_i \left(\langle\hat{A}_i^\dagger\hat{O}\hat{A}_i\rangle - \frac{1}{2}\langle\{\hat{A}_i^\dagger\hat{A}_i, \hat{O}\}\rangle\right). \tag{2.9}$$

2.3 Monte Carlo Wave Function (MCWF) Method

The Monte Carlo Wave Function (MCWF) method, also known as the *quantum jump method*, is a numerical technique used to simulate the dynamics of open quantum systems. It provides an alternative to solving the Lindblad Master Equation directly, which can be computationally expensive for large systems. The MCWF method is based on the idea of simulating the evolution of individual quantum trajectories, which are then averaged to obtain the overall dynamics of the system. This method was first introduced by Mølmer et al. in 1992 [11] and has since been widely used in various fields of quantum physics, including quantum optics, quantum information, and condensed matter physics.

In this section, we will provide complete description of the MCWF method, including its theoretical foundations, and its equivalence to the Lindblad Master Equation. At the end of this section, we will provide the algorithmic steps to implement a single trajectory of the MCWF method and finally analyze the convergence of the method by studying the statistical errors involved.

[1] The complete derivation of the Lindblad Master Equation is shown in the Sec. A.

2.3.1 Method

In this subsection, we follow the original work of Mølmer et al. [11]. The main idea of the method is to simulate the evolution of a quantum system by considering both the coherent evolution due to the system Hamiltonian and the stochastic jumps due to the interaction with the environment. In this procedure, we proceed with the evolution of the wavefunction of the system and thus making use of the Schrödinger equation.

The step by step description of the procedure is given as follows:

1. We define the effective non-Hermitian Hamiltonian, to describe the system behaviour and the dissipation into the environment, as

$$\begin{aligned}\hat{H}_{\text{eff}} &= \hat{H} - \frac{\mathrm{i}\hbar}{2}\sum_i \hat{C_i}^\dagger \hat{C_i} \\ &= \hat{H} - \frac{\mathrm{i}\hbar}{2}\sum_i \gamma_i \hat{A_i}^\dagger \hat{A_i}.\end{aligned} \tag{2.10}$$

We could now calculate $|\phi^{(1)}(t+\delta t)\rangle$, by evolving the state $|\phi(t)\rangle$ according to the Schrödinger equation, with the above Eq. 2.10 i.e.,

$$\mathrm{i}\hbar\frac{d}{dt}|\phi'(t)\rangle = \hat{H}_{\text{eff}}|\phi'(t)\rangle .$$

In the limit $\delta t \ll 1$, we have

$$|\phi^{(1)}(t+\delta t)\rangle = \mathrm{e}^{\frac{-\mathrm{i}\hat{H}_{\text{eff}}\delta t}{\hbar}}|\phi(t)\rangle . \tag{2.11}$$

Taylor expanding the exponential, we have

$$|\phi^{(1)}(t+\delta t)\rangle = \left(1 - \frac{\mathrm{i}\hat{H}_{\text{eff}}\delta t}{\hbar}\right)|\phi(t)\rangle . \tag{2.12}$$

The above time evolution of the system is non-unitary and thus we have to normalize the wavefunction after time step δt. Calculating the norm, we have

$$\begin{aligned}\sqrt{\langle\phi^{(1)}(t+\delta t)|\phi^{(1)}(t+\delta t)\rangle} &= \sqrt{\langle\phi(t)|\left(1+\frac{\mathrm{i}\hat{H}^\dagger_{\text{eff}}\delta t}{\hbar}\right)\left(1-\frac{\mathrm{i}\hat{H}_{\text{eff}}\delta t}{\hbar}\right)|\phi(t)\rangle} \\ &\approx \sqrt{\langle\phi(t)|\,\delta t\frac{\mathrm{i}}{\hbar}\left(\hat{H}^\dagger_{\text{eff}} - \hat{H}\right)|\phi(t)\rangle} \qquad (\text{since} \quad \delta t \ll 1) \\ &= \sqrt{1 - \langle\phi(t)|\sum_i \gamma_i \hat{A}^\dagger_i \hat{A}_i \delta t\,|\phi(t)\rangle}.\end{aligned} \tag{2.13}$$

We can define the probability δp_i as the probability for a jump at site i after a time step as,

$$\delta p_i = \langle\phi(t)|\,\gamma_i \hat{A}^\dagger_i \hat{A}_i \delta t\,|\phi(t)\rangle . \tag{2.14}$$

Thus, rewriting Eq. 2.13, we have

$$\begin{aligned}\sqrt{\langle\phi^{(1)}(t+\delta t)|\phi^{(1)}(t+\delta t)\rangle} &= \sqrt{1-\sum_i \delta p_i} \\ &= \sqrt{1-\delta p}.\end{aligned} \tag{2.15}$$

The magnitude of δt is chosen such that this time evolution is valid at first order i.e., in particular, we have $\delta p \ll 1$.

2. The second step is the notice of a possible quantum jump, due to the drop in the norm during the time evolution. This is interpreted as a probability for a quantum jump. In order to decide whether a jump occurs, we generate a quasi-random number r, uniformly distributed in the interval $[0, 1)$ such that,

   1. If $r < \delta p$, no quantum jump occurs:

$$|\phi(t+\delta t)\rangle = \frac{|\phi^{(1)}(t+\delta t)\rangle}{|\langle\phi^{(1)}(t+\delta t)|\phi^{(1)}(t+\delta t)\rangle|^2} = \frac{|\phi^{(1)}(t+\delta t)\rangle}{(1-\delta p)^2} \tag{2.16}$$

   2. If $r > \delta p$, a quantum jump occurs:

$$|\phi(t+\delta t)\rangle = \frac{\hat{A}_i\,|\phi(t)\rangle}{||\hat{A}_i\,|\phi(t)\rangle\,||} \tag{2.17}$$

$$\text{with conditional probability} \quad \Pi_i = \frac{\delta p_i}{\delta p}. \tag{2.18}$$

2.3.2 Algorithm for Single Trajectory Simulation

Here is the complete algorithm for the MCWF method for single trajectory simulation:

1. Initialize the system wavefunction $|\phi(t)\rangle$ at time t.

2. For each time step δt:

   a) Calculate the effective non-Hermitian Hamiltonian $\hat{H}_{\text{eff}}$ if it is time-dependent, else reuse the once calculated $\hat{H}_{\text{eff}}$.

   b) Evolve the wavefunction to $|\phi^{(1)}(t+\delta t)\rangle$ using the Schrödinger equation with $\hat{H}_{\text{eff}}$.

$$|\phi^{(1)}(t+\delta t)\rangle = \mathrm{e}^{\frac{-\mathrm{i}\hat{H}_{\text{eff}}\delta t}{\hbar}}\,|\phi(t)\rangle$$

   where $\|\phi^{(1)}(t+\delta t)\|^2 = 1 - \delta p$.

   c) Generate a random number r uniformly distributed in $[0, 1)$.

   d) Check if $r < \delta p$:

      - If true, normalize the wavefunction and set $|\phi(t+\delta t)\rangle = \frac{|\phi^{(1)}(t+\delta t)\rangle}{(1-\delta p)^2}$.

      - If false, select a jump operator $\hat{A}_i$ at position i, perform a jump and update the wavefunction to $|\phi(t+\delta t)\rangle = \frac{\hat{A}_i|\phi(t)\rangle}{||\hat{A}_i|\phi(t)\rangle||}$.

3. Repeat Step 2 until the desired time is reached.

We repeat the above algorithm over large number of trajectories to obtain the statistical average of the system behaviour.

2.3.3 Equivalence to the Lindblad Equation

We proceed to verify the equivalence of the system dynamics via MCWF method with the Lindblad Master Equation. To do so, we prove that the average density matrix representation $\bar{\rho}(t)$, over a large number of trajectories satisfies the master equation.

Consider a wavefunction $\phi(t)$ at time t evolved according to the MCWF method. At time $t+\delta t$, we can express the average density matrix over many trajectories as,

$$\bar{\rho}(t+\delta t) = (1-\delta p)\frac{|\phi^{(1)}(t+\delta t)\rangle}{(1-\delta p)^{1/2}}\frac{\langle\phi^{(1)}(t+\delta t)|}{(1-\delta p)^{1/2}} + \delta p\sum_i \gamma_i \frac{\delta p_i}{\delta p}\frac{\hat{A}_i|\phi(t)\rangle}{(\delta p_i/\delta t)^{1/2}}\frac{\langle\phi(t)|\hat{A}_i^\dagger}{(\delta p_i/\delta t)^{1/2}} \tag{2.19}$$

For visual clarity, we omit the time dependence of the wavefunctions in the following steps:

$$\begin{aligned}\bar{\rho}(t+\delta t) &= |\phi^1\rangle\langle\phi^1| + \delta t\sum_i \gamma_i\hat{A}_i|\phi\rangle\langle\phi|\hat{A}_i^\dagger \\ &= (1-\mathrm{e}^{-\mathrm{i}\hat{H}_{\text{eff}}\delta t})|\phi\rangle\langle\phi|(1-\mathrm{e}^{+\mathrm{i}\hat{H}_{\text{eff}}^\dagger\delta t}) + \delta t\sum_i \gamma_i\hat{A}_i|\phi\rangle\langle\phi|\hat{A}_i^\dagger \\ &= |\phi\rangle\langle\phi| - \mathrm{i}\hat{H}_{\text{eff}}\delta t|\phi\rangle\langle\phi| + |\phi\rangle\langle\phi|\mathrm{i}\hat{H}_{\text{eff}}\delta t + \delta t\sum_i \gamma_i\hat{A}_i|\phi\rangle\langle\phi|\hat{A}_i^\dagger \\ &= \bar{\rho}(t) - \mathrm{i}\delta t\left[\hat{H}_{\text{eff}},\bar{\rho}(t)\right] + \delta t\sum_i \gamma_i\hat{A}_i|\phi\rangle\langle\phi|\hat{A}_i^\dagger - \mathrm{i}^2\delta t^2\hat{H}_{\text{eff}}\bar{\rho}(t)\hat{H}_{\text{eff}}^\dagger\bar{\rho}(t)\end{aligned} \tag{2.20}$$

Neglecting the second order terms, we get

$$\begin{aligned}\dot{\bar{\rho}}(t) &= \mathrm{i}\left(\bar{\rho}(t)\hat{H}_{\text{eff}}^\dagger - \hat{H}_{\text{eff}}\bar{\rho}(t)\right) + \sum_i \gamma_i\hat{A}_i\bar{\rho}(t)\hat{A}_i^\dagger \\ &= \mathrm{i}\left[\bar{\rho}(t),\hat{H}\right] + \sum_i \gamma_i\left(\hat{A}_i\bar{\rho}(t)\hat{A}_i^\dagger - \frac{1}{2}\left(\bar{\rho}(t)\hat{A}_i\hat{A}_i^\dagger + \hat{A}_i\hat{A}_i^\dagger\bar{\rho}(t)\right)\right)\end{aligned} \tag{2.21}$$

which is equivalent to the Lindblad Master Equation in Eq. 2.8. This shows that our method coincides with the master equation at any time t, given that we start from the same initial state at $t=0$ for both methods.

2.3.4 Observables and Statistical Error

One of the main goals of simulating a quantum system is to calculate the expectation values of any observable at given time t. Due to the stochastic nature of the MCWF method, the convergence of the operator expectation values is subject to statistical errors, bound by the number of trajectories(N) used in the simulation. This method allows us to extract exact results only in the limit of infinite number of trajectories. However, in practice, we can obtain good approximations with a finite number of trajectories, subject to certain deviations.

For an arbitrary Hermitian operator $\hat{O}$, we can obtain the sample mean for N trajectories at time t as

$$\bar{O}_N(t) = \frac{1}{N}\sum_{i=1}^{N}\langle\phi^i(t)|\hat{O}|\phi^i(t)\rangle\,. \tag{2.22}$$

For large N, the *Central limit theorem* states that the distribution of normalized sample means approaches to normal distribution with mean $\langle\hat{O}\rangle(t)$ and variance σ^2 i.e.,

$$\text{In the limit } N\to\infty, \qquad \langle\hat{O}\rangle(t) = \bar{O}_N(t). \tag{2.23}$$

Following this, one can calculate the variance of the sample mean of the operator $\bar{O}_N(t)$ as

$$\sigma^2(t) = \frac{1}{N}\left(\langle\hat{O}^2\rangle(t) - \langle\hat{O}\rangle^2(t)\right) = \mathrm{Var}(\hat{O})(t). \tag{2.24}$$

The statistical error δO_N of the sample mean is therefore given by the standard deviation of the sample mean, which is defined as

$$\delta O_N(t) = \sigma(t) = \sqrt{\frac{\mathrm{Var}(\hat{O})(t)}{N}}. \tag{2.25}$$

3 Quantum Error Correction

3.1 Introduction

Error correction is a fundamental aspect of both classical and quantum computing to ensure reliable information processing. In modern classical computers, the components are extremely reliable as the error rates are typically below 10^{-15}. For example, the error rate for a standard CPU is typically measured in errors per billion (EPB) or errors per trillion (EPT) [29]. In contrast, the error rates for quantum computers are typically much higher due to the nature of quantum mechanics and the challenges associated with building and operating quantum systems. Noise, decoherence, and imperfections in quantum gates can cause errors in quantum computations. Today, the state-of-the-art quantum computers have error rates of the order 10^{-3} for two-qubit operations and for single qubit operations, very recently, error rates of the order 10^{-7} have been reported [30]. While these are significant individual improvements, they still do not correspond to a single quantum computing platform. For practical fault-tolerant quantum computing, error rates need to be far lower, typically below 10^{-4} for two-qubit gates and far lower for single qubit gates. Due to this reason, we are currently in the Noisy Intermediate-Scale Quantum (NISQ) era of quantum computing, where we have access to quantum devices with a limited number of qubits and relatively high error rates.

In classical computing, many different methods have been developed to detect and correct errors that occur during data transmission or storage such as parity checks, checksums, and more advanced techniques like Hamming codes [31] and Reed-Solomon codes [32]. However, quantum error correction (QEC) is more challenging due to the unique properties of quantum mechanics, such as superposition and entanglement.

Platform	**Best 1Q fidelity**	**Best 2Q fidelity**	**Metric(s)**
Trapped ions[30, 33]	99.999985%	99.914(3)%	RB (1Q), IRB (2Q)
Superconducting[34, 35]	99.998%	99.94%	RB (both)
Neutral atoms[36, 37]	$> 99.97\%$	99.54(2)%	RRB(1Q), repeat-gate(2Q)
Silicon spin qubits[38, 39]	99.992(1)%	99.8(2)%	RB (1Q), IRB (2sQ)
NV Centre[40]	99.999(1)%	99.93(5)%	GST (both)

Table 3.1. Snapshot of best reported physical (native) single- and two-qubit gate fidelities by platform. Metrics differ across works (RB = randomized benchmarking; IRB = Interleaved RB; RRB = Raman RB; GST = gate-set tomography; "repeat-gate" indicates SPAM(State Preparation And Measurement)-corrected repeated-gate methods).

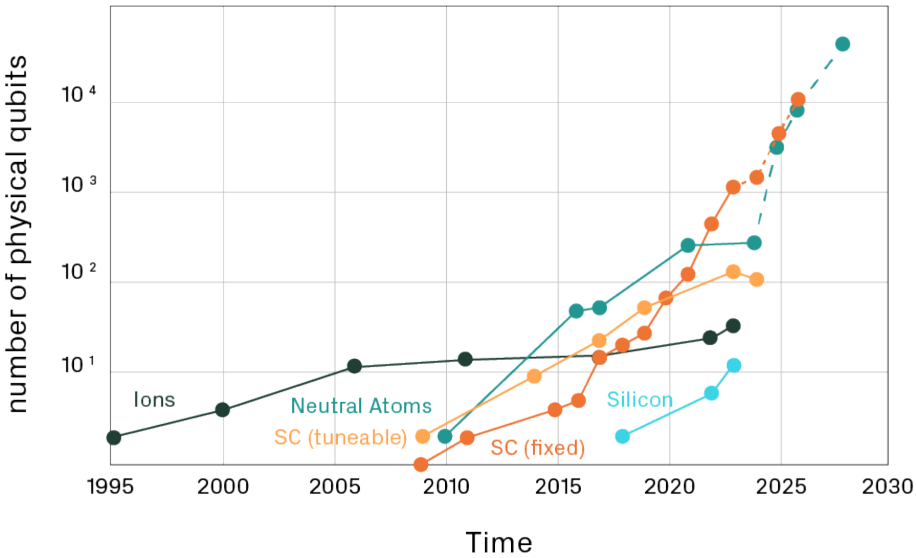


Figure 3.1. Number of physical qubits for trapped ions (Ions), neutral atoms, superconducting (SC) and silicon qubit technologies. The green and orange dashed lines for neutral atoms and SC (fixed) quantum computers show the public roadmaps of companies developing the respective technology. The plot is taken from [41].

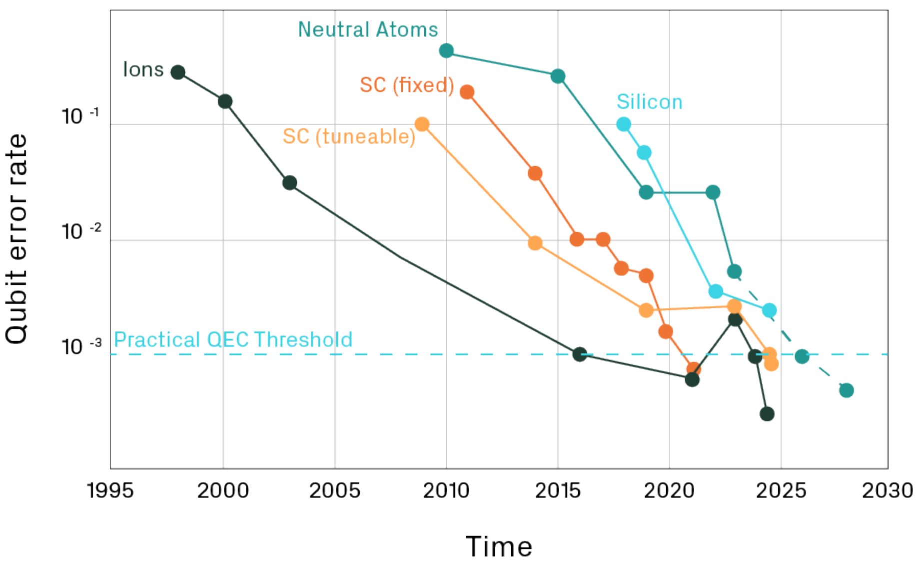


Figure 3.2. Two-qubit gate error rates (best or average) for trapped ions (Ions), neutral atoms, superconducting (SC) and silicon qubit technologies. The green dashed line for neutral atoms shows the public roadmaps of companies developing that technology. The practical QEC threshold is widely regarded as about 99.9% physical two-qubit gate fidelity (i.e. an error rate of 10^{-3}), which is when quantum hardware companies can start to implement additional QEC techniques. The plot is taken from [41].

According to the *no-cloning theorem*[2] [42], we cannot simply duplicate quantum infor-

[2] It states that, "It is impossible to create an independent and identical copy of an arbitrary unknown quantum state".

mation to protect it from errors, as it is done in classical error correction. Additionally, measuring a quantum state to check for errors can collapse the superposition, destroying the quantum information we are trying to protect. Thus, QEC works by encoding quantum information into a large Hilbert space using multiple *physical qubits* to represent a single *logical qubit* via entanglement. This redundancy allows the detection and correction of errors without directly measuring the quantum state, and is referred to as "*quantum error correcting code*". Some of the classical error correction techniques and ideas have been translated for the quantum domain such as the *Repetition codes*, along with new techniques that have been specifically designed for quantum systems such as the *Shor code* and the *Steane code* [43]. More advanced techniques include the *Stabilizer codes* [6], which provide a systematic way to construct and analyze quantum error-correcting codes, and *Topological codes* [44], which leverage the topological properties of quantum states to protect against errors.

In our work, we focus on the simplest quantum error correcting codes, the *Repetition codes*, and the *Shor code*, and utilize the knowledge of Rydberg physics to design and implement them. In further sections, we will discuss the concepts and mathematical framework behind QEC, including the use of Kraus operators to model quantum noise and discuss specific quantum error-correcting codes and their properties.

3.2 Kraus Operators and Quantum Channels

As seen in Ch. 2, in evolution of quantum systems, we often encounter situations where the system interacts with its environment, leading to noise and decoherence. Kraus operators provide a powerful way to represent the evolution of open quantum systems [3]. This framework is essential for understanding how quantum information can be corrupted and how we can protect it using error correction techniques.

More formally, a quantum channel is a completely positive, trace-preserving (CPTP) map that describes the evolution of a quantum state, including the effects of noise and decoherence. Mathematically, a quantum channel $\mathcal{E}$ acting on a density matrix ρ can be represented using a set of Kraus operators $\{E_k\}$ as

$$\mathcal{E}(\rho) = \sum_k E_k \rho E_k^\dagger, \tag{3.1}$$

where the Kraus operators satisfy the completeness relation

$$\sum_k E_k^\dagger E_k = I, \tag{3.2}$$

where I is the identity operator. Kraus operators can represent various types of quantum noise, such as bit-flip errors, phase-flip errors, and depolarizing noise. For example, a bit-flip error on a single qubit can be represented by the following Kraus operators

$$E_0 = \sqrt{1-p}I, \quad E_1 = \sqrt{p}X, \tag{3.3}$$

where p is the probability of a bit-flip error occurring, and X is the Pauli-X operator (bit-flip operator). Similarly, a phase-flip error can be represented by

$$E_0 = \sqrt{1-p}I, \quad E_1 = \sqrt{p}Z, \tag{3.4}$$

where Z is the Pauli-Z operator (phase-flip operator).

This is an equivalent representation of the *Lindbladian formalism* as we have seen in Ch. 2. Markovian open quantum system dynamics generated by a Lindbladian satisfying $\dot{\rho} = \mathcal{L}(\rho)$ yields a quantum channel over a finite time interval $[0, t]$ as,

$$\mathcal{E}_t : \rho(0) \mapsto \rho(t), \quad \rho(t) = \mathcal{T} \exp\left(\int_0^t \mathcal{L}(\tau) d\tau\right) \rho(0) \tag{3.5}$$

where $\mathcal{T}$ is the time-ordering operator. Thus, continuous-time jump rates from a master equation translate directly into discrete-time channel parameters used in circuit-level noise models. In simulations with piecewise-constant controls, the total map is the composition $\mathcal{E}_{t_n} \circ \cdots \circ \mathcal{E}_{t_1}$ and remains CPTP.

By using Kraus operators to model quantum noise, we can analyze how different error-correcting codes perform under various noise models. This understanding is crucial for designing effective QEC schemes that can protect quantum information from the detrimental effects of the environment.

3.3 Repetition Codes

In this section, we will discuss the simplest quantum error-correcting codes, originally inspired by classical repetition codes. Repetition codes were among the first quantum error-correcting codes to be proposed and provide a foundational understanding of how quantum error correction works. The idea behind repetition codes is to encode a single information into multiple copies by repeating the information and later if these copies differ, the original information can most likely be determined by *majority voting* among the copies. For example, in a classical repetition code, a single bit "0" can be encoded as "000" and a single bit "1" can be encoded as "111". If one of the bits gets flipped due to noise, we can still recover the original bit by taking the majority value.

In the quantum domain, we can use a similar approach to protect against specific types of errors, such as bit-flip errors and phase-flip errors. For quantum version of repetition codes, the encoding is done using entanglement between multiple qubits respecting the no-cloning theorem. We define any quantum error-correcting code by the notation $[[n, k, d]]$, where:

- n is the number of physical qubits used to encode the information.
- k is the number of logical qubits that are encoded.
- d is the code distance, which is the minimum number of physical qubits that need to be altered to change one logical qubit state to another. The code can correct up to $\lfloor (d-1)/2 \rfloor$ errors.

Any quantum error-correcting code encompasses the following main steps:

- **Encoding:** The process of mapping the logical qubits into a larger Hilbert space using multiple physical qubits.
- **Syndrome Measurement:** The process of measuring certain properties of the encoded state to detect the presence and type of errors without collapsing the quantum information.
- **Error Correction:** The process of applying corrective operations based on the syndrome measurements to restore the original logical qubit state.
- **Decoding:** The process of mapping the corrected physical qubits back to the logical qubit state.

Majority voting used in these codes fails if more than half of the physical qubits are erroneous. Thus, well known three-qubit repetition codes only work the best if the probability of two or more errors is less than the probability of a single error. This can be calculated as

$$\begin{aligned} P(1 \text{ error}) &= \binom{3}{1} p(1-p)^2 = 3p - 6p^2 + 3p^3 \\ P(\text{at least 2 errors}) &= \binom{3}{2} p^2(1-p) + \binom{3}{3} p^3 = 3p^2 - 2p^3. \end{aligned} \tag{3.6}$$

This gives us the *break-even point*[3] for the three-qubit repetition code as

$$3p - 6p^2 + 3p^3 > 3p^2 - 2p^3 \implies p < 0.5. \tag{3.7}$$

This means that if the probability of an error occurring on a single physical qubit is less than 0.5, the three-qubit repetition code can improve the fidelity of the logical qubit state compared to not using any error correction.
Let us explore some of the three-qubit repetition codes in detail.

3.3.1 Bit-Flip Code

In 1985, Asher Peres first translated the idea of repetition codes to the quantum domain by introducing the *three-qubit bit-flip code* [45]. This code can correct a single bit-flip error that may occur on any one of the three physical qubits and has a code distance of $d = 3$. Hence, the three-qubit bit-flip code is denoted by $[[3, 1, 3]]$.

The encoding process for the three-qubit bit-flip code is

$$|0_L\rangle = |000\rangle, \quad |1_L\rangle = |111\rangle, \tag{3.8}$$

where $|0_L\rangle$ and $|1_L\rangle$ are the logical qubit states. More generally encoding of an arbitrary qubit state is given as

$$|\psi\rangle = \alpha|1\rangle + \beta|0\rangle \mapsto |\psi\rangle_{\text{enc}} = \alpha|111\rangle + \beta|000\rangle. \tag{3.9}$$

To detect and correct errors, we perform syndrome measurements using two ancillary qubits. The syndrome measurements are designed to detect bit-flip errors without collapsing the quantum information. The syndrome measurement operators for the three-qubit bit-flip code are

$$S_1 = Z_1 Z_2, \quad S_2 = Z_2 Z_3, \tag{3.10}$$

where Z_i is the Pauli-Z operator acting on the i-th qubit. The possible outcomes of the syndrome measurements are

- $S_1 = +1, S_2 = +1$: No error.
- $S_1 = -1, S_2 = +1$: Bit-flip error on qubit 1.
- $S_1 = +1, S_2 = -1$: Bit-flip error on qubit 3.
- $S_1 = -1, S_2 = -1$: Bit-flip error on qubit 2.

[3] The "break-even" point is the physical error rate below which QEC helps, and above which QEC hurts.

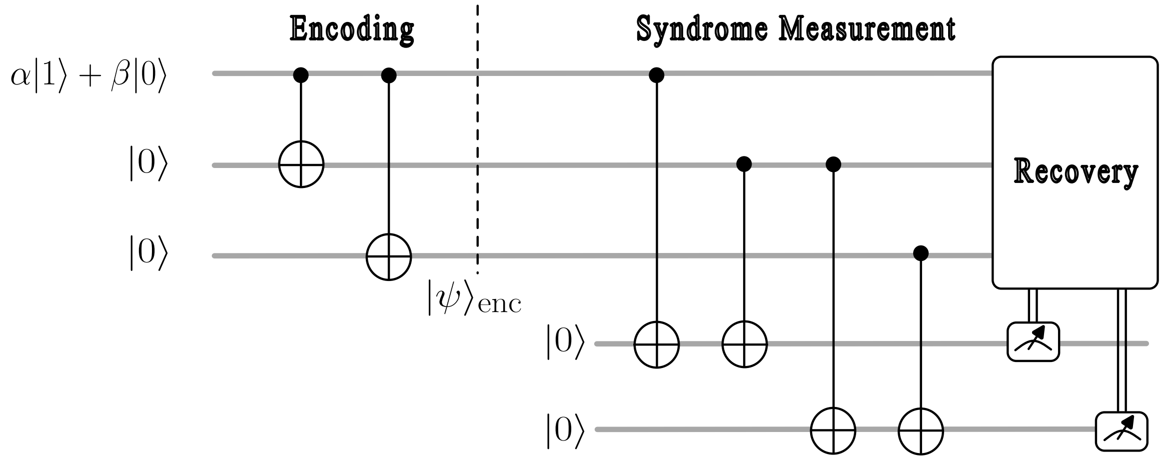


Figure 3.3. Quantum circuit representation for the three-qubit bit-flip code. The circuit includes encoding, syndrome measurement using two ancillary qubits, error correction (recovery) based on the syndrome outcomes.

These syndrome measurements operators used here are also called *parity checks* and these can be implemented using CNOT gates, where the data qubits are the control qubits and the ancillary qubits are the target qubits as shown in Fig. 3.3.

Based on the syndrome measurement outcomes, we can apply the appropriate corrective operation (recovery) to restore the original logical qubit state. The corrective operations are:

- If $S_1 = -1, S_2 = +1$: Apply X on qubit 1.
- If $S_1 = +1, S_2 = -1$: Apply X on qubit 3.
- If $S_1 = -1, S_2 = -1$: Apply X on qubit 2.

After applying the corrective operation, we can decode the logical qubit by measuring the three physical qubits and taking the majority value to determine the original logical state. After decoding, the logical qubit state can be used for further computation or error correction processes.

The minimum fidelity for the three-qubit bit-flip code with respect to when no error-correction is performed, can be calculated under the consideration that this code works perfectly if there is at most one bit-flip error. Suppose the quantum state of interest is $|\psi\rangle$ and the output state after passing through the noisy channel and applying error correction is ρ_{out}, then the minimum fidelity is given by

$$F_{min} = \langle\psi|\, \rho_{out}\, |\psi\rangle = (1-p)^3 + 3p(1-p)^2 = 1 - 3p^2 + 2p^3, \tag{3.11}$$

where p is the probability of a bit-flip error occurring on a single qubit.

3.3.2 Phase-Flip Code

The quantum systems are also susceptible to phase-flip errors, which can be corrected using a similar approach to the bit-flip code. The three-qubit phase-flip code is also denoted by $[[3, 1, 3]]$. The three-qubit phase-flip code is designed from the bit-flip code by applying a Hadamard transformation to each qubit before and after the encoding and corrective processes. The idea behind this is, the Hadamard transformation maps phase-flip errors to bit-flip errors, allowing us to use the same error correction techniques as in the bit-flip code. The encoding process for the three-qubit phase-flip code is

$$|0_L\rangle = \frac{1}{\sqrt{2}}(|000\rangle + |111\rangle), \quad |1_L\rangle = \frac{1}{\sqrt{2}}(|000\rangle - |111\rangle), \tag{3.12}$$

where $|0_L\rangle$ and $|1_L\rangle$ are the logical qubit states. More generally encoding of an arbitrary qubit state is given as

$$|\psi\rangle = \alpha|1\rangle + \beta|0\rangle \mapsto |\psi\rangle_{\text{enc}} = \alpha|---\rangle + \beta|+++\rangle. \tag{3.13}$$

To detect and correct phase-flip errors, we perform syndrome measurements using two ancillary qubits. The syndrome measurement operators for the three-qubit phase-flip code are

$$S_1 = H^{\otimes 3} Z_1 Z_2 H^{\otimes 3} = X_1 X_2, \quad S_2 = H^{\otimes 3} Z_2 Z_3 H^{\otimes 3} = X_2 X_3, \tag{3.14}$$

where X_i is the Pauli-X operator acting on the i-th qubit. The possible outcomes of the syndrome measurements are

- $S_1 = +1, S_2 = +1$: No error.
- $S_1 = -1, S_2 = +1$: Phase-flip error on qubit 1.
- $S_1 = +1, S_2 = -1$: Phase-flip error on qubit 3.
- $S_1 = -1, S_2 = -1$: Phase-flip error on qubit 2.

Based on the syndrome measurement outcomes, we can apply the appropriate corrective operation to restore the original logical qubit state. The corrective operations are

- If $S_1 = -1, S_2 = +1$: Apply Z on qubit 1.
- If $S_1 = +1, S_2 = -1$: Apply Z on qubit 3.
- If $S_1 = -1, S_2 = -1$: Apply Z on qubit 2.

This code for the phase-flip channel essentially has the same characteristics as the bit-flip channel i.e., the minimum fidelity for the phase-flip channel is the same as the bit-flip channel and we have the same threshold for break-even point to produce improvement over the case without error correction. Thus, these two channels are *unitarily equivalent*, since they can be transformed into each other by applying a unitary operation (the Hadamard transformation in this case) [3].

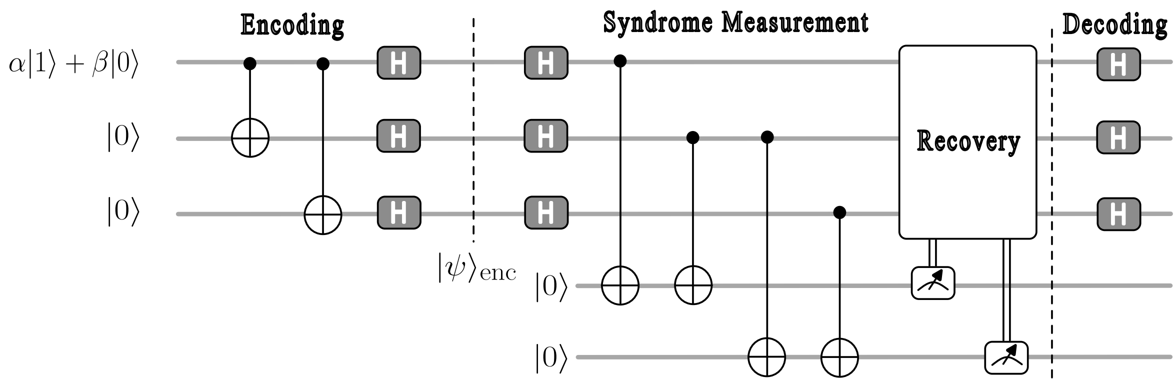


Figure 3.4. Quantum circuit representation for the three-qubit phase-flip code. The circuit includes encoding, syndrome measurement using two ancillary qubits, error correction (recovery) based on the syndrome outcomes.

As shown in Fig. 3.4, in our implementation we swap the recovery and the final Hadamards i.e., after the initial $H^{\otimes 3}$ and the CNOT parity checks, we apply recovery in the *rotated* frame (using X corrections) and only then apply the final $H^{\otimes 3}$. This does *not* change the measured syndromes. The ancilla network right after the front Hadamards measures Z_1Z_2 and Z_2Z_3, which in the original basis are the X-type stabilizers X_1X_2 and X_2X_3 via $HXH = Z$ and $HZH = X$. Applying X_i before the final Hadamards is therefore equivalent to applying Z_i after them.

3.3.3 Shor Code

The three-qubit bit-flip and phase-flip codes can only correct a single type of error. To correct any arbitrary single-qubit error, we need a more robust code. The first such code was proposed by Peter Shor in 1995 [4], known as the *Shor code*. The Shor code is a nine-qubit code that can correct any single-qubit error, including bit-flip, phase-flip, and combinations of both. The Shor code is denoted by $[[9, 1, 3]]$. The idea behind this is to first encode the logical qubit using a three-qubit phase-flip code and then encode each of them using a three-qubit bit-flip code. This *concatenation* of codes allows the Shor code to correct both types of errors. More formally, the encoding process for the Shor code is

$$
\begin{aligned}
|0_L\rangle &= \frac{1}{\sqrt{8}}(|000\rangle + |111\rangle) \otimes (|000\rangle + |111\rangle) \otimes (|000\rangle + |111\rangle) \\
|1_L\rangle &= \frac{1}{\sqrt{8}}(|000\rangle - |111\rangle) \otimes (|000\rangle - |111\rangle) \otimes (|000\rangle - |111\rangle).
\end{aligned}
\tag{3.15}
$$

For an arbitrary qubit state, the encoding is given by

$$
|\psi\rangle = \alpha|1\rangle + \beta|0\rangle \mapsto |\psi_L\rangle = \alpha|1_L\rangle + \beta|0_L\rangle. \tag{3.16}
$$

To detect and correct errors, we perform syndrome measurements using eight ancillary qubits. As shown in Fig. 3.5, we need two ancillary qubits for phase-flip error detection and two ancillary qubits for each of the three bit-flip error detections. The syndrome measurement operators for the Shor code are

$$
\begin{aligned}
S_1 &= Z_1Z_2, & S_2 &= Z_2Z_3 \\
S_3 &= Z_4Z_5, & S_4 &= Z_5Z_6 \\
S_5 &= Z_7Z_8, & S_6 &= Z_8Z_9 \\
S_7 &= X_1X_2X_3X_4X_5X_6, & S_8 &= X_4X_5X_6X_7X_8X_9,
\end{aligned}
$$

where Z_i and X_i are the Pauli-Z and Pauli-X operators acting on the i-th qubit, respectively. The possible outcomes of the syndrome measurements and corresponding corrections are summarized below.

Syndrome	Error Detected	Correction
$S_1 = +1, S_2 = +1$	No bit-flip error in the first block.	
$S_1 = -1, S_2 = +1$	Bit-flip error on qubit 1.	Apply X_1.
$S_1 = +1, S_2 = -1$	Bit-flip error on qubit 3.	Apply X_3.
$S_1 = -1, S_2 = -1$	Bit-flip error on qubit 2.	Apply X_2.
$S_3 = +1, S_4 = +1$	No bit-flip error in the second block.	
$S_3 = -1, S_4 = +1$	Bit-flip error on qubit 4.	Apply X_4.
$S_3 = +1, S_4 = -1$	Bit-flip error on qubit 6.	Apply X_6.
$S_3 = -1, S_4 = -1$	Bit-flip error on qubit 5.	Apply X_5.
$S_5 = +1, S_6 = +1$	No bit-flip error in the third block.	
$S_5 = -1, S_6 = +1$	Bit-flip error on qubit 7.	Apply X_7.
$S_5 = +1, S_6 = -1$	Bit-flip error on qubit 9.	Apply X_9.
$S_5 = -1, S_6 = -1$	Bit-flip error on qubit 8.	Apply X_8.
$S_7 = +1, S_8 = +1$	No phase-flip error.	
$S_7 = -1, S_8 = +1$	Phase-flip error on qubit block 1, 2, and 3.	Apply $Z_1Z_2Z_3$
$S_7 = +1, S_8 = -1$	Phase-flip error on qubit block 7, 8, and 9.	Apply $Z_7Z_8Z_9$
$S_7 = -1, S_8 = -1$	Phase-flip error on qubit block 4, 5, and 6.	Apply $Z_4Z_5Z_6$

Table 3.2. Syndrome measurement outcomes and corresponding correction operations for the Shor code.

The corrective operations for when both bit-flip and phase-flip errors occur can be determined by combining the individual corrections for each type of error. Infact, one can show that any *continuum* of errors on a single qubit can be expressed as a *discrete* set of errors, specifically the Pauli errors $\{I, X, Y, Z\}$. Mathematically, suppose the quantum state of interest is $|\psi\rangle$ and passing through a noisy quantum channel $\mathcal{E}$ such that the output state is $\mathcal{E}(|\psi\rangle\langle\psi|) = \sum_i E_i |\psi\rangle\langle\psi| E_i^\dagger$, then Kraus operator E_i can be expressed as a linear combination of the Pauli operators,

$$E_i = \alpha_{i0}I + \alpha_{i1}X + \alpha_{i2}Y + \alpha_{i3}Z \tag{3.17}$$

where α_{ij} are complex coefficients. Thus, the Shor code can correct any arbitrary single-qubit error by detecting and correcting both bit-flip and phase-flip errors.

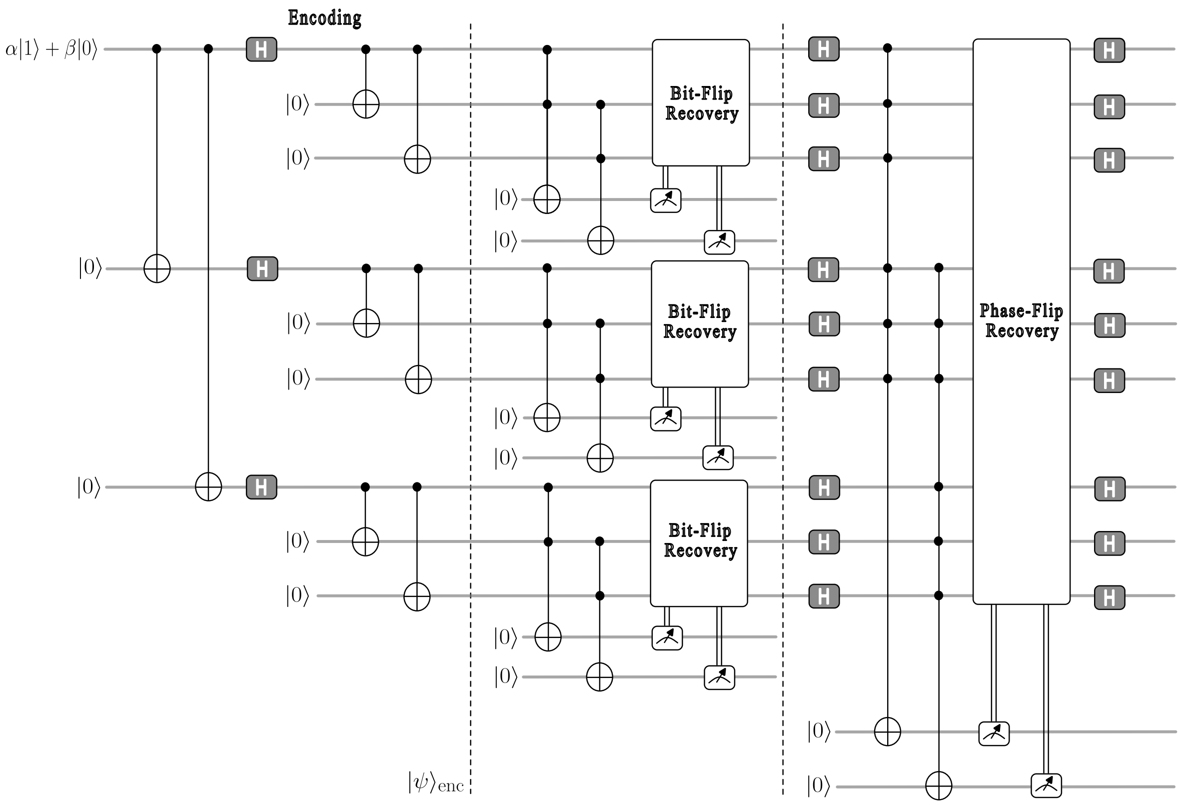


Figure 3.5. Quantum circuit representation for the Shor code. The circuit includes encoding using 9 physical qubits, followed by syndrome measurement using 8 ancillary qubits (2 for phase-flip and 6 for bit-flip error detection), and corresponding error correction (recovery) based on the syndrome outcomes. The bit-flip and phase-flip syndrome measurement and correction blocks can be applied in any order since they commute.

Part III

Bit-Flip Error Correction Protocol

4 System Architecture

In this part of the work, we will be designing the Bit-Flip Error Correction (BFEC) algorithm for the Rydberg systems. We start by defining the choice of qubit states, building the system architecture and then implementing the correction protocol through a carefully designed workflow that can be used for MCWF simulations. Further to this, we will analyze the implications of the protocol in terms of its error correction capabilities and the constraints imposed by a successful correction.

4.1 Qubit States

According to the Di-Vincenzo criteria [46], the first step in designing any quantum computer and thus any quantum algorithm is to well define the choice of qubit states. This choice is crucial as it determines the feasibility of algorithm implementation and the physical processes for gate operations involved in doing so.

The straightforward choice of qubit states is to use the ground state $|g\rangle$ and one of the Rydberg states $|r\rangle$. This choice is motivated by the fact that the Rydberg state is long-lived and can be selectively excited to, either through a direct one-photon transition[4] or through a two-photon transition via an intermediate state[5] [47]. Thus allowing for precise control over the qubit states, which is essential for implementing quantum protocols in Rydberg systems. Thus, our qubit states are defined as

$$|0\rangle \equiv |g\rangle\,, \quad |1\rangle \equiv |r\rangle\,. \tag{4.1}$$

This choice of qubit states allows us to utilize the *blockade* and *facilitation* mechanisms for implementing multi-qubit gates, which is the goal of this work. We will see in the further sections how these mechanisms can be exploited for implementing the BFEC algorithm.

4.2 Rydberg System Architecture for BFEC Algorithm

Now that we have a chosen set of qubit states, we can proceed to design the multi-atom system architecture that corresponds to the quantum circuit defined for BFEC algorithm in Fig. 3.3. At a high level, the design of the system architecture must:

1. specify the choice of qubit states.
2. provide the required atoms for encoding (*data qubits*) and for syndrome measurement (*ancillary qubits*).
3. realize the prescribed interaction between atoms and ancillas corresponding to the multi-gate operations in the error correction algorithm.
4. supply feasible set of control protocols to implement various gate operations while remaining consistent with the physical layout and constraints of the architecture.

[4]In Rubidium, one would need a 297 nm laser for a direct 5s-ns/nd transition.
[5]Using 780 nm and 480 nm lasers for to the 5s-5p-ns/nd transitions in Rubidium.

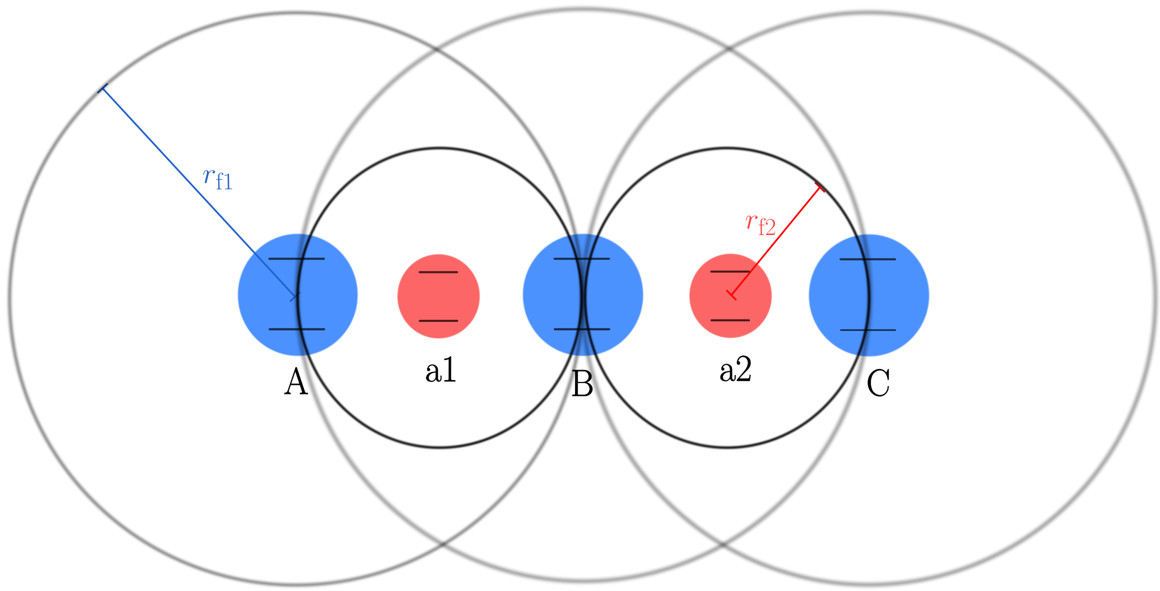


Figure 4.1. Proposed Rydberg System Architecture for BFEC algorithm shown with two-level atoms and their facilitation shells. The atoms are arranged in a linear chain with equal spacing. The blue atoms represent the *data qubits*, while the red atoms represent the *ancillary qubits* with their facilitation radii r_{f1} and r_{f2} respectively.

Following the quantum circuit implementation in the Fig. 3.3, we know that the implementation requires a set of *three* data qubits for encoding the logical qubit, and an additional *two* ancillary qubits for the syndrome measurement. Thus, we design the Rydberg system architecture of *five* atoms, corresponding to each qubit.

In Fig. 4.1, we propose an architecture with two different types of Rydberg atoms (blue and red) for the BFEC algorithm. The *data qubits* (blue), labelled {A, B, C}, are for encoding the logical qubit, while the ancillary qubits (red), labelled {a1, a2} are for syndrome measurement. These two types of atoms are prepared in different Rydberg states, and therefore have different facilitation radii. We place the data qubits at mutual distance r_{f1}, which is the facilitation radius of the data qubits, while the ancillary qubits sits at the midpoint between a data-qubit pair and is chosen so that its facilitation radius r_{f2} is half that of the data qubits, i.e., $r_{f2} = r_{f1}/2$.

As an approximation, we work in a nearest-neighbor interaction regime, where each atom only interacts with its immediate neighbors, as the interaction strength decreases with distance (see Eq. 1.10).

The only multi-qubit gate operations that are required for the BFEC algorithm are CNOT gates, and we will study in detail a efficient implementation of these gates, consistent with our architecture in the next section.

5 Control Protocols for BFEC Algorithm

In this chapter, we will get a better understanding of our system, the various parameters involved, and explore the conditions under which it operates, by studying the dynamics of a toy 2-atom Rydberg system. Further to this, we will study a particularly useful mechanism that we use for implementing the gate operations required for the BFEC algorithm. This mechanism is called Rapid Adiabatic Passage (RAP) and we will be looking into how to design this protocol onto our system of Rydberg atoms, what conditions this protocol poses and how to make this efficient.

5.1 Dynamics of Toy Rydberg System

Let us consider a simple model of two Rydberg atoms $\{\text{Atom 1, Atom 2}\}$ at a fixed facilitation distance r_{f} from each other. The initial state of the system be such that, the Atom 1 is in Rydberg state $|r\rangle$ and Atom 2 is in ground state $|g\rangle$. Now let us define the hamiltonian of the system as

$$\hat{H} = \frac{\Omega}{2}\hat{\sigma}_2^x + \Delta_0 \hat{n}_2 + V_{\text{nn}}\hat{n}_1\hat{n}_2 \tag{5.1}$$

where an arbitrary $\hat{O}_{\text{i}}$ is defined as an operator $\hat{O}$ acting on atom i. Here, $\hat{\sigma}^x = |g\rangle\langle r| + |r\rangle\langle g|$ is the Pauli-X operator and $\hat{n} = |r\rangle\langle r|$ is the number operator. We are driving atom 2 with Rabi frequency Ω and with initial detuning Δ_0, while the Rydberg interaction between the two atoms is modeled by a constant nearest-neighbor interaction V_{nn}. Hence, we have three main parameters $\{\Omega, \Delta_0, V_{\text{nn}}\}$ which define the dynamics of the system and the goal is to understand the role of these parameters and how to choose them optimally for our BFEC protocol.

For the entirety of the work, we deal with the parameters in the order of Ω, which is equivalent to numerically setting $\Omega = 1$. For the rest of the parameters in the Hamiltonian, we formulate certain conditions under which the system operates.

Condition 1: *(Facilitation Condition) An atom at facilitation distance (r_f) from the neighboring atom in Rydberg state, can be driven at resonance iff*

$$\mathbf{\Delta_0} + \mathbf{V}_{\text{nn}} = 0. \tag{5.2}$$

Therefore, the system would exhibit natural resonant Rabi oscillations either when $\Delta_0 = 0$ in the absence of facilitation interaction or when it satisfies **Condition 1** in presence of facilitation interaction.

In addition to this, one should also make sure that there are no residual excitations in Atom 2 in the absence of facilitation interaction from the Atom 1. This puts a condition on Δ_0 to be big enough such that the probability of finding the population at the excited state [48] is almost zero, i.e.,

$$P_e = \frac{\Omega^2}{\Omega^2 + \Delta_0^2}\sin^2\left(\frac{\sqrt{\Omega^2+\Delta_0^2}}{2}t\right) \to 0 \tag{5.3}$$

This is the essence of the following condition.

Condition 2: *For an atom to not have any residual excitations in the absence of facilitation interaction, the initial detuning of the atom should be such that*

$$\boldsymbol{\Delta_0^2 \gg \Omega^2}. \tag{5.4}$$

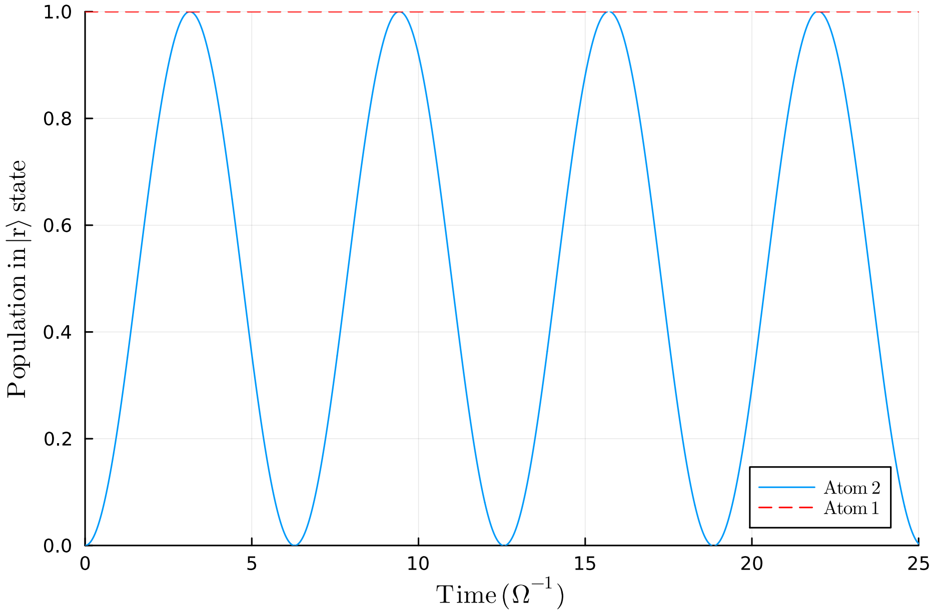


Figure 5.1. Resonant Rabi oscillations of Atom 2 with respect to the condition $\Delta_0 + V_{\text{nn}} = 0$. The numerical values are set to $\Delta_0 = 1000\,\Omega$ and $V_{\text{nn}} = -1000\,\Omega$.

The two conditions defined here are crucial for the system to operate as desired. We proceed further to discuss the CNOT gate implementation, while having these conditions in our consideration.

5.2 Controlled-X (CNOT) gate via Rapid Adiabatic Passage (RAP)

The choice of a suitable protocol for gate operation is very crucial for the implementation of any quantum algorithm. The gate protocol should be efficient, fast and robust to the errors that might occur due to the imperfections in the system. In our case, we need a gate protocol which is a suitable fit for our system of Rydberg atoms, and can be used to implement the BFEC algorithm. The protocol we need is for the Controlled-X (CNOT) gate implementation, and we will be using Rapid Adiabatic Passage (RAP).

In this section, we develop the RAP protocol for our Rydberg atom platform, specify the operating conditions and explore its parameter space for optimal performance. For some representative cases, we also simulate the protocol and visualize the results to aid our understanding.

5.2.1 Rapid Adiabatic Passage (RAP)

Rapid Adiabatic Passage is a quantum control technique that allows for efficient transfer of population between quantum states [49]. This mechanism is very well known for its high fidelity and robustness against variations in system parameters like Rabi frequency or pulse area. This mechanism works by sweeping the detuning of the driving laser field through resonance, while maintaining a constant Rabi frequency. This sweep rate is a crucial parameter in the protocol and is denoted by δ. The detuning $\Delta(t)$ of the system is varied linearly with time as follows

$$\Delta(t) = \Delta_0 - \delta t \tag{5.5}$$

where Δ_0 is the initial detuning at time $t = 0$ and δ is the sweep rate of the detuning.

Dressed-state picture and avoided crossing in RAP: Consider a system of two Rydberg atoms such that one atom (Atom 1) is in the Rydberg state $|R\rangle$ and the other atom (Atom 2) is in the ground state $|G\rangle$. The goal is to transfer the population from Atom 2 to the Rydberg state $|R\rangle$ efficiently and rapidly via this protocol. In the two-state subspace spanned by $\{|RR\rangle, |RG\rangle\}$ and in the rotating frame, the driven dynamics are described by

$$H(t) = \frac{1}{2}\begin{pmatrix} \Delta(t) & \Omega \\ \Omega & -\Delta(t) \end{pmatrix} \tag{5.6}$$

with $\Delta(t)$ from Eq. 5.5. The instantaneous (adiabatic) eigenenergies are

$$E_{\pm}(t) = \pm\tfrac{1}{2}\sqrt{\Delta(t)^2 + \Omega(t)^2}, \tag{5.7}$$

which form an avoided crossing of size $\hbar\Omega$ at $\Delta = 0$. The corresponding dressed states (eigenstates) are given by

$$|+(t)\rangle = \cos\left(\frac{\theta}{2}\right)|RR\rangle + \sin\left(\frac{\theta}{2}\right)|RG\rangle; \qquad |-(t)\rangle = \sin\left(\frac{\theta}{2}\right)|RR\rangle - \cos\left(\frac{\theta}{2}\right)|RG\rangle, \tag{5.8}$$

where $\theta(t) = \arctan(\Omega/\Delta(t))$. As the detuning $\Delta(t)$ is swept from large positive to large negative values, the dressed states continuously morph between the bare states $\{|RR\rangle, |RG\rangle\}$, thus following one branch implements the population transfer.

The mechanism relies on two main criteria:

1. **Adiabaticity Criteria:** The system should evolve slowly enough such that it remains in its instantaneous lowest energy eigenstate throughout the process. This is mathematically expressed as[50]:

$$\left|\frac{\langle m(t)|\frac{dH}{dt}|n(t)\rangle}{(E_n(t) - E_m(t))^2}\right| \ll 1 \tag{5.9}$$

where $|n(t)\rangle$ and $|m(t)\rangle$ are the instantaneous eigenstates of the Hamiltonian $H(t)$ with corresponding eigenvalues $E_n(t)$ and $E_m(t)$. We can also express this criteria from the perspective of Landau-Zener transition [51] which gives the non-adiabatic transition probability between two energy levels during a level crossing, and it can be expressed as

$$\mathrm{P}_{\mathrm{LZ}} = \exp\left(\frac{-2\pi\Omega^2}{\hbar\delta}\right) \approx 0. \tag{5.10}$$

The two expressions Eq. 5.9 and Eq. 5.10 are equivalent and lead to the following condition on the sweep rate δ

$$\delta \ll \Omega^2. \tag{5.11}$$

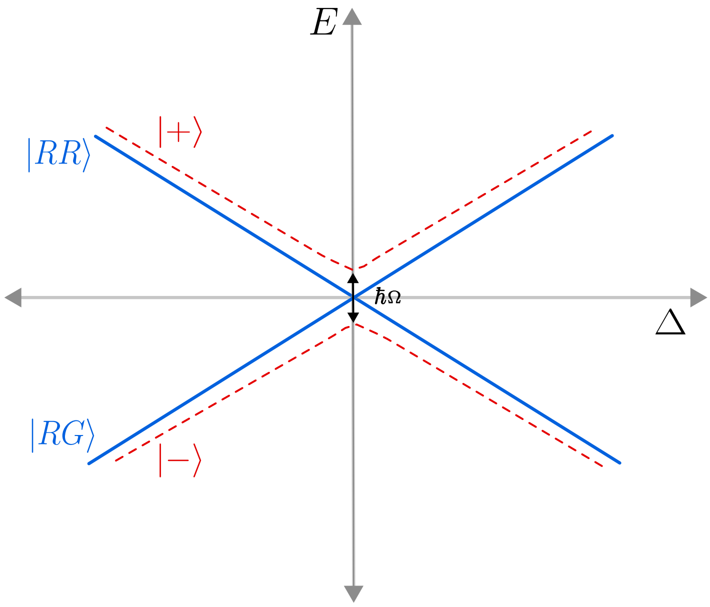


Figure 5.2. Adiabatic crossing picture of the RAP protocol. The detuning $\Delta(t)$ is swept linearly with time passing through the avoided crossing at $\Delta = 0$. The diabatic (bare state) energies are shown as solid lines (blue), while the adiabatic (dressed state) energies are shown as dashed lines (red). The size of the avoided crossing is given by $\hbar\Omega$.

2. **Rapidity Criteria:** The system should evolve fast enough such that the population transfer occurs before any significant relaxation or decay processes can take place. This is mathematically expressed as

$$|\delta| \gg \gamma^2 \tag{5.12}$$

where γ is the decay rate of the excited state.

RAP in presence of decay

Now, we take a look at if and how the presence of decay would affect the RAP protocol. More precisely, we are interested in how decay affects the transition probability of the system at the end of the protocol.

Following the paper by V. M. Akulin et. al. [52], we take a look at the exact analytical solution for the problem of level crossing obtained for decaying levels. According to this, the mathematical equation we are interested in is the number of photons emitted per atom after the passage through the resonance, which is given by

$$I = \frac{2\pi\Omega^2}{\delta} \exp\left(\frac{-\pi\Omega^2}{\delta}\right) \left| W_{\frac{i\Omega^2}{\delta}, \frac{-1}{2}} \left(\frac{-i\gamma^2}{\delta}\right) \right|^2 \tag{5.13}$$

where W corresponds to the special function called the Whittaker function, and is given as

$$W_{\frac{i\Omega^2}{\delta}, \frac{-1}{2}} \left(\frac{-i\gamma^2}{\Omega}\right) = \frac{\exp\frac{i\gamma^2}{2\delta}}{\Gamma\left(\frac{-i\Omega^2}{\delta}\right)} \int_0^\infty \frac{\exp\{\frac{i\gamma^2}{\delta}t - i\frac{\Omega^2}{\delta}\ln\frac{t}{1+t}\}}{t(t+1)}\, dt. \tag{5.14}$$

Consider the limit of slow decay, $\frac{\gamma^2}{\delta} \ll 1$, which is the case we are interested in, as we obey with the *rapidity criteria* (see Eq. 5.12). In this limit, with the asymptotic of Whittaker function given by

$$\lim_{z\to 0} \quad W_{ia,\frac{-1}{2}}(z) = \frac{1}{\Gamma(1+ia)} = \frac{1}{ia\Gamma(ia)} \tag{5.15}$$

which after substituting in Eq. 5.13, we obtain

$$I = 1 - \exp\left(\frac{-2\pi\Omega^2}{\delta}\right). \tag{5.16}$$

We see that the Eq. 5.16 is equivalent to the Eq. 5.10, which is the Landau-Zener transition probability in the absence of decay. This implies that in the limit of slow decay, the irreversible decay of the excited state involved in the transition does not affect the probability of transition at all. This is a very crucial result for our system, as the bit-flip errors we are trying to correct are essentially decay processes from the excited state to the ground state.

5.2.2 Designing RAP Protocol Onto Rydberg System

Here, we will be adapting the RAP protocol for our system of Rydberg atoms. We will take a look at the simplest case of 2-atom system and then extend it to 3-atom system, which will be directly mapped to the BFEC algorithm.

2-atom system

Let us consider a system of two Rydberg atoms {Atom A, Atom 1} placed at facilitation distance r_f from each other. The initial state of the system be $|r\rangle \otimes |g\rangle$. The goal is to drive Atom 1 to the excited state efficiently and rapidly via this protocol. As the Atom A is in the Rydberg state, we start the protocol with positive detuning higher Δ_high than the Rydberg interaction energy V_nn. One should also take into consideration that Δ_high should follow **Condition 2**, discussed previously. With this we can now do a *symmetric sweep* as follows

$$\Delta(t) = \Delta_0 - \delta t = \begin{cases} \Delta_\mathrm{high} + V_\mathrm{nn} & \text{at} \quad t = 0 \\ -\Delta_\mathrm{high} + V_\mathrm{nn} & \text{at} \quad t = t_\mathrm{f}. \end{cases} \tag{5.17}$$

Following the range of detuning from Eq. 5.17, we look into the parameter δ which is crucial for how fast the error correction protocol can be implemented. To reach resonance condition in a symmetric sweep, we have

$$\Delta(t_\mathrm{r}) = |V_\mathrm{nn}| = \Delta_0 - \frac{\delta T_\mathrm{total}}{2} \tag{5.18}$$

The total time taken for the implementation of the RAP protocol and thus the CNOT gate operation can be calculated as follows

$$\begin{aligned} T_\mathrm{CNOT} = T_\mathrm{total} &= \frac{2(\Delta_0 - |V_\mathrm{nn}|)}{\delta} \\ T_\mathrm{total} &= \frac{2\Delta_\mathrm{high}}{\delta}; \quad \delta > 0, \quad \Delta_\mathrm{high} > 0. \end{aligned} \tag{5.19}$$

This gives us a relation between the total time taken for the protocol, the initial detuning and the sweep rate. For the protocol to be efficient, we need to choose the parameters such that it satisfies both the *adiabaticity* and *rapidity* criteria discussed previously. We evaluate these conditions by numerical simulations of the system and choose the parameter ranges optimally.

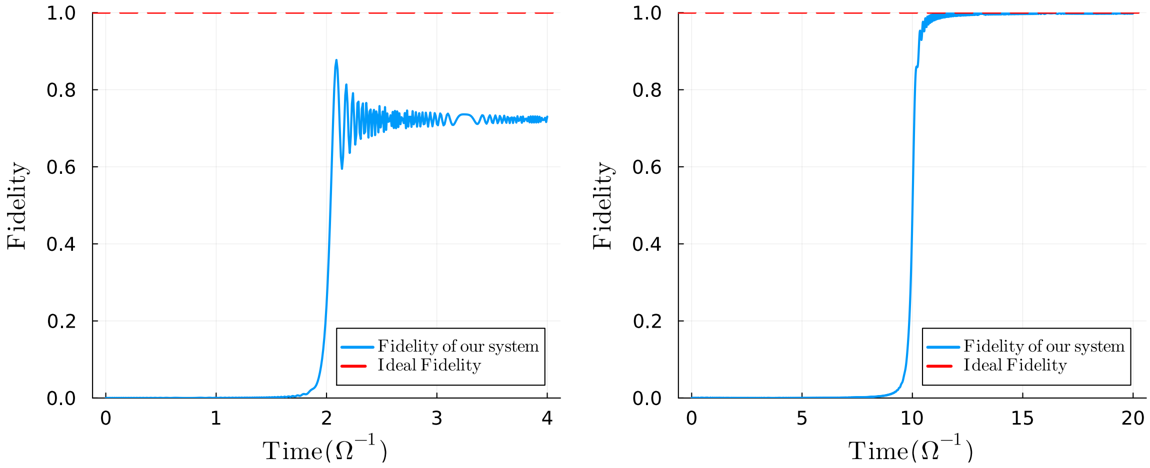


Figure 5.3. Symmetric RAP protocol for 2-atom system. The fidelity is calculated as the overlap with respect to the state $\alpha\,|rr\rangle + \beta\,|gg\rangle$, which is the target state of the protocol for the initial state $(\alpha\,|r\rangle + \beta\,|g\rangle) \otimes |g\rangle$. The numerical values used here are $\alpha = 1$, $\Delta_0 = 3000\,\Omega$, $V_{\text{nn}} = -2000\,\Omega$. (Left) The sweep rate is set to $\delta = 500\,\Omega^2$. The jump in the fidelity occurs at time $T = 2\,\Omega^{-1}$, which is exactly when the Atom 1 meets the resonance condition. (Right) The sweep rate is decreased to $\delta = 100\,\Omega^2$, while keeping all the other parameters same. Here, the jump in the fidelity occurs at time $T = 10\,\Omega^{-1}$.

To note the importance of optimal parameter choice, we simulate the RAP protocol for two different values of the parameter δ and visualize the results in Fig. 5.3. As we can see in the right plot, the fidelity of the system at the end of RAP is high, which is what we desire but at the expense of longer time taken for the protocol. Whereas in the left plot, we see that the fidelity is not as high as the previous case, but the time taken is much less. Thus, there is a tradeoff between the time taken and the fidelity of the protocol. Hence, one needs to find the optimum of the parameter δ for the system to be both fast and efficient. We calculate this numerically further in the section.

3-atom system

Let us now increase the system size to accommodate one more Rydberg atom {Atom 2}, and we would immediately see that the system architecture becomes very crucial. Following the BFEC algorithm, more precisely, the *encoding* step, we need to transfer the population from Atom A to both Atom 1 and Atom 2. Thus, the goal is to perform CNOT gate operations between the three atoms, such that the final state of the system is $\alpha\,|rrr\rangle + \beta\,|ggg\rangle$, where the initial state of the Atom A is $(\alpha\,|r\rangle + \beta\,|g\rangle)$ and $|g\rangle$ is the initial state of both Atom 1 and Atom 2. We explore two different cases of the system architecture depending on the location of Atom 2.

Case 1: Site-Dependent Detuning

Consider the system of 3-atoms {Atom A, Atom 1, Atom 2} placed in a linear chain at facilitation distance from each other in the given order. The initial state of the system be $|r\rangle \otimes |g\rangle \otimes |g\rangle \equiv |rgg\rangle$.

Let the initial detunings of {Atom 1, Atom 2} be $\{\Delta_0^1, \Delta_0^2\}$ and be driven with sweep rates $\{\delta_1, \delta_2\}$ respectively. Now we determine the total time taken to implement the gate operations on both atoms. Under the simpler case of only nearest neighbor interactions, one can do the CNOT gate operation between {Atom A, Atom 1} and {Atom 1, Atom 2} as the resonant drive of Atom 2 occurs only after Atom 1 has been put in the $|r\rangle$ state.

Thus, the total time taken to resonantly drive Atom 1 is given by Eq. 5.19. Following this, sequentially we do the protocol on Atom 2 which takes the same time. Thus the total time taken in such an architecture would be,

$$T_{\text{total}} = \frac{2(\Delta_0^1 - |V_{\text{nn}}|)}{\delta_1} + \frac{2(\Delta_0^2 - |V_{\text{nn}}|)}{\delta_2} \tag{5.20}$$

If we consider the symmetric conditions for Atom 1 and Atom 2, i.e., $\Delta_0^1 = \Delta_0^2 = \Delta_0$ and $\delta_1 = \delta_2 = \delta$, then we effectively have twice the total time taken in Eq. 5.19 i.e.,

$$T_{\text{total}} = \frac{4(\Delta_0 - |V_{\text{nn}}|)}{\delta}. \tag{5.21}$$

Such configuration for sequential RAP implementation, where the detuning sweep from Eq. 5.17 is dependent for each atom site is referred to as *site-dependent detuning*.

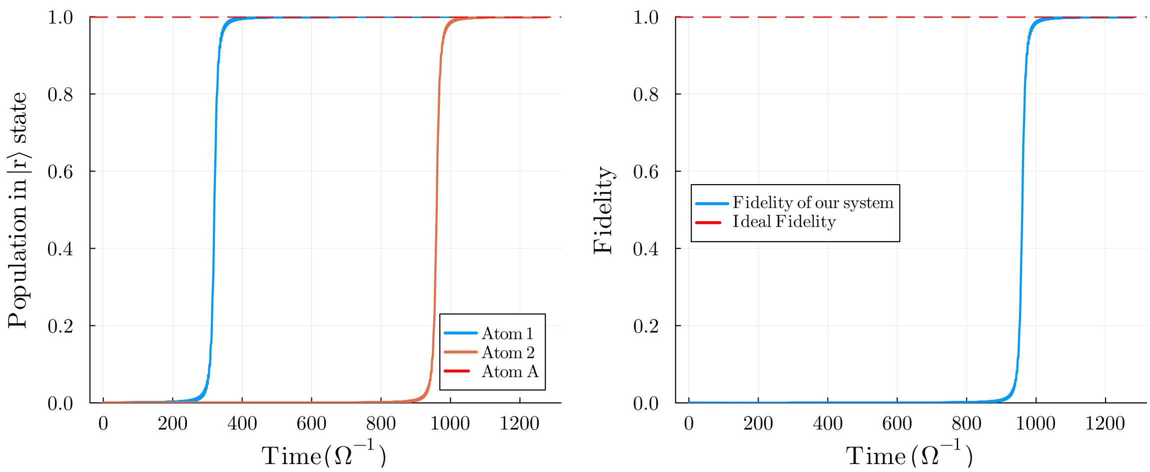


Figure 5.4. Symmetric RAP protocol for 3-atom system for the case of *site-dependent detuning*. Here, the initial state of the system is $(\alpha\,|r\rangle + \beta\,|g\rangle) \otimes |g\rangle \otimes |g\rangle$. The numerical values used are $\alpha = 1$, $\Delta_0^1 = \Delta_0^2 = 1032\,\Omega$, $V_{\text{nn}} = -1000\,\Omega$ and $\delta_1 = \delta_2 = 0.1\,\Omega^2$. (Left) Populations in $|r\rangle$ state for each atom. (Right) Fidelity of the system. The fidelity is calculated as the overlap with the target state $\alpha\,|rrr\rangle + \beta\,|ggg\rangle$.

Case 2: Site-Independent Detuning

One of the major drawbacks of the above architecture is the total time taken to reach the target state. The longer the protocol takes, the more errors can accumulate in the system, defeating the purpose of error correction. Thus, it is crucial to implement them at greater speeds, for which every gate operation must be engineered for time efficiency. Retaining the idea of efficient population transfer via symmetric RAP, we can reduce the total time in Eq. 5.21, by modifying the system layout.

Consider the atoms arranged in the order {Atom 1, Atom A, Atom 2}. With this, both Atom 1 and Atom 2 can be driven parallelly at the same time, thus we would effectively slash the total time taken for implementing CNOT gates by a factor of 0.5 compared to the previous layout, and thus it would take the same time as in the case of the 2-atom system in Eq. 5.19. This configuration for parallel RAP implementation, is referred to as *site-independent detuning*.

This is a very crucial improvement in the system architecture, as it would lead to a significant reduction in the time taken for multiple CNOT gate implementations, especially when scaled to larger system sizes.

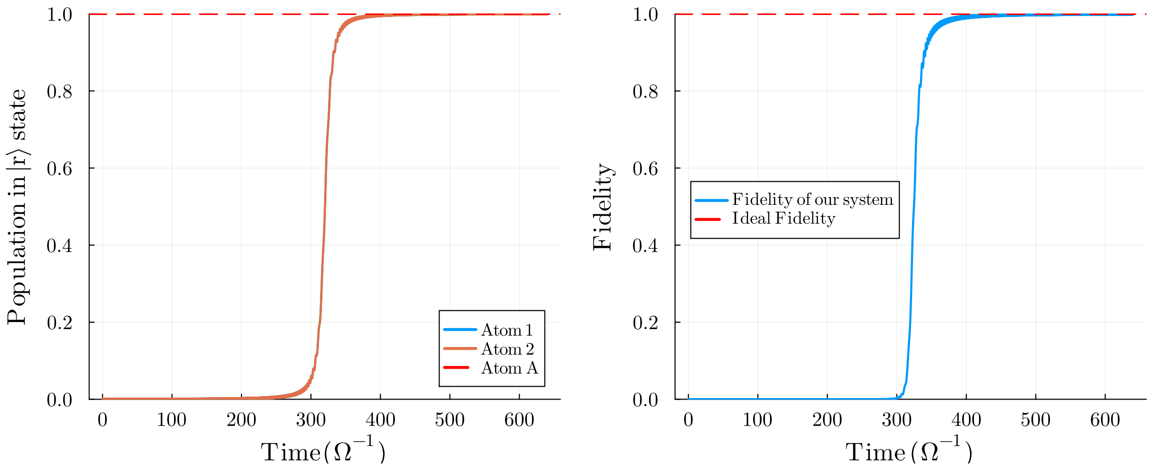


Figure 5.5. Symmetric RAP protocol for 3-atom system for the case of *site-independent detuning*. Here, the initial state of the system is $|g\rangle \otimes (\alpha |r\rangle + \beta |g\rangle) \otimes |g\rangle$. The numerical values used are $\alpha = 1$, $\Delta_0^1 = \Delta_0^2 = 1032\,\Omega$, $V_{\rm nn} = -1000\,\Omega$ and $\delta_1 = \delta_2 = 0.1\,\Omega^2$. (Left) Populations in $|r\rangle$ state for each atom. Atom 1 and Atom 2 follow the same dynamics, thus the lines overlap. (Right) Fidelity of the system. The fidelity is calculated as the overlap with the target state $\alpha |rrr\rangle + \beta |ggg\rangle$.

5.2.3 Optimal Parameter Search via Numerical Simulations

Despite the conditions defined and system layout optimizations we made in this section, there are still loose ends in the parameter space which can lead to a number of issues in our error correction algorithm including, having residual excitations in the system, non-zero infidelity with the target state at the end of the RAP protocol or simply taking too long for implementation. Here, we aim to find the optimal parameter space for these free parameters, via numerical simulations of 3-atom system, aiming to directly map it to the BFEC algorithm.

1. Detuning Parameter Δ_0

Consider the detuning parameter Δ_0 subject to **Condition 2**, i.e., $\Delta_0^2 \gg \Omega^2$. Referring to Eq. 5.3, the excited state population under a pulse of duration t is

$$P_e(t;\Omega,\Delta_0) \;=\; \frac{\Omega^2}{\Omega^2+\Delta_0^2}\sin^2\left(\frac{\sqrt{\Omega^2+\Delta_0^2}}{2}\,t\right) \;=\; \frac{\Omega^2}{\Omega_R^2}\sin^2\left(\frac{\Omega_R t}{2}\right) \tag{5.22}$$

where $\Omega_R \equiv \sqrt{\Omega^2+\Delta_0^2}$. We fix the fidelity tolerance of the RAP protocol to three decimal places by imposing the error bound

$$P_e(t;\Omega,\Delta_0) \;\leq\; 10^{-3}. \tag{5.23}$$

There are two common design choices:
(i) Time-synchronized pulse: Choose the pulse length to coincide with a revival of the detuned Rabi oscillation,

$$t_n = \frac{2\pi n}{\Omega_R}, \qquad n \in \mathbb{N}, \tag{5.24}$$

which gives $\sin^2(\Omega_R t_n/2) = 0$ and hence $P_e(t_n) = 0$ exactly. This realizes Eq. 5.23 with margin, at the cost of tighter timing control.
(ii) Time-agnostic pulse: If t cannot be synchronized, impose $\max_t \sin^2(\cdot) = 1$ in Eq. 5.22. Then Eq. 5.23 is guaranteed, provided

$$\frac{\Omega^2}{\Omega^2 + \Delta_0^2} \le 10^{-3} \iff \Delta_0 \ge \Omega\sqrt{10^3 - 1} \approx 31.6\,\Omega. \tag{5.25}$$

This conservative choice bounds P_e by 10^{-3} for any pulse length t.

Unless otherwise specified, we adopt the time-synchronized design from Eq. 5.24 as we aim to optimize the time efficiency of the protocol. Following this, we model the detuning numerically as follows for subsequent simulations

$$\text{For} \quad V_{\text{nn}} = -1000\,\Omega; \quad \text{we get} \quad \Delta_0^1 = \Delta_0^2 = 1032\,\Omega, \tag{5.26}$$

where V_{nn} is the nearest neighbor interaction energy between the atoms, Ω is the resonant Rabi frequency of the driving laser field and $\Delta_0^{1,2}$ are the initial detunings of {Atom 1, Atom 2} respectively.

2. Sweep Rate δ and Decay Rate γ_{decay}

Following the Eq. 5.19, the total time taken for CNOT gate operation is dependent on δ, with other parameters being choosen as Eq. 5.26. As discussed previously from the *rapidity criteria*, the parameter γ_{decay} of the Rydberg state is a crucial parameter affecting the optimal δ.

Let us try to find the optima for $\{\delta_1, \delta_2\}$ for {Atom 1, Atom 2} respectively, for varying decay rates γ_{decay}. We work in the following range of these parameters

$$\delta_1 = \delta_2 = \delta \in [0.01\,\Omega^2, 0.50\,\Omega^2] \quad \text{and} \quad \gamma_{\text{decay}} \in [10^{-5}\,\Omega, 10^{-1}\,\Omega]. \tag{5.27}$$

Following the *site-independent architecture*, consider the initial state of the system to be

$$|\psi(t=0)\rangle = |g\rangle \otimes (\alpha\,|r\rangle + \beta\,|g\rangle) \otimes |r\rangle \tag{5.28}$$

with $\alpha = \beta = \sqrt{0.5}$. Let us observe the fidelity attained by the symmetric RAP protocol at the end of the sweep, for the parameters range from Eq. 5.27. The fidelity is calculated as the overlap with the target state as follows

$$F = |\,\langle\phi_{\text{ideal}}|\,\rho(t=t_{\text{f}})\,|\phi_{\text{ideal}}\rangle\,| \quad \text{where} \quad |\phi_{\text{ideal}}\rangle = \alpha\,|rrr\rangle + \beta\,|ggg\rangle \tag{5.29}$$

and $\rho(t=t_{\text{f}})$ is the density matrix of the system obtained at the end of the protocol.

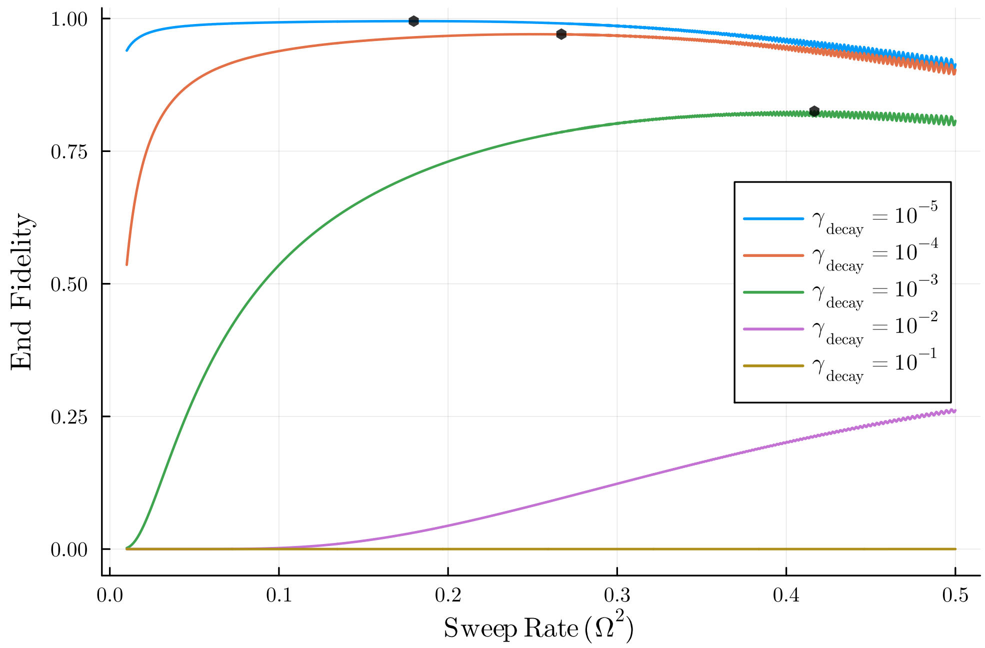


Figure 5.6. For 3-atom system with site-independent detuning undergoing symmetric RAP protocol under varying ranges of sweep rate (δ) and decay rate (γ_{decay}). End fidelity of the system at $t = t_{\mathrm{f}}$. The black dots correspond to the highest fidelity attained by the system for a particular decay rate.

In the Fig. 5.6, we notice that at very slow sweep rates, the end fidelity is low signifying that the system is dominated by the decay and thus the system is not able to reach the target state. This decrease is more pronounced for higher decay rates. As we increase the sweep rate, the end fidelity increases and reaches a maximum at some optimal value of δ for a particular γ_{decay}. As we increase the sweep rate further, the protocol loses the *adiabaticity* criteria and thus the end fidelity starts to decrease again.

We can also model an analytical formula for the end fidelity in the presence of decay, which should approximately map the result we see above. If P_0 is the probability of the system being in the excited state at $t = 0$, then we know that for a system with decay, the probability of the system being in excited state after the time t with decay rate γ_{decay} is given by

$$\mathrm{P}(t) = \mathrm{P}_0 e^{-\gamma_{\mathrm{decay}} t}. \tag{5.30}$$

As we have seen previously, the presence of decay does not affect the Landau-Zener transition of the system in the slow decay limit. The P_0 of the system is given by Eq. 5.16 and for our convenience we consider the effect of decay at $0.5\, t_{\mathrm{f}}$, where t_{f} is the total time taken for the RAP mechanism to complete given by Eq. 5.19. With this, we can write the analytical formula for the probability of the system being in the excited state at the end of the protocol as follows

$$\mathrm{P}(t_{\mathrm{f}}) = \left(1 - e^{\frac{-2\pi\Omega^2}{\hbar\delta}}\right) e^{\frac{-\gamma_{\mathrm{decay}} t_{\mathrm{f}}}{2}}. \tag{5.31}$$

As we are focussed on symmetric RAP protocol, we cannot control our system to maintain the maximum fidelity it attains till the end of the protocol; however, there are certain works which point in the direction of asymmetric adiabatic sweep [49, 53, 54], which can be used

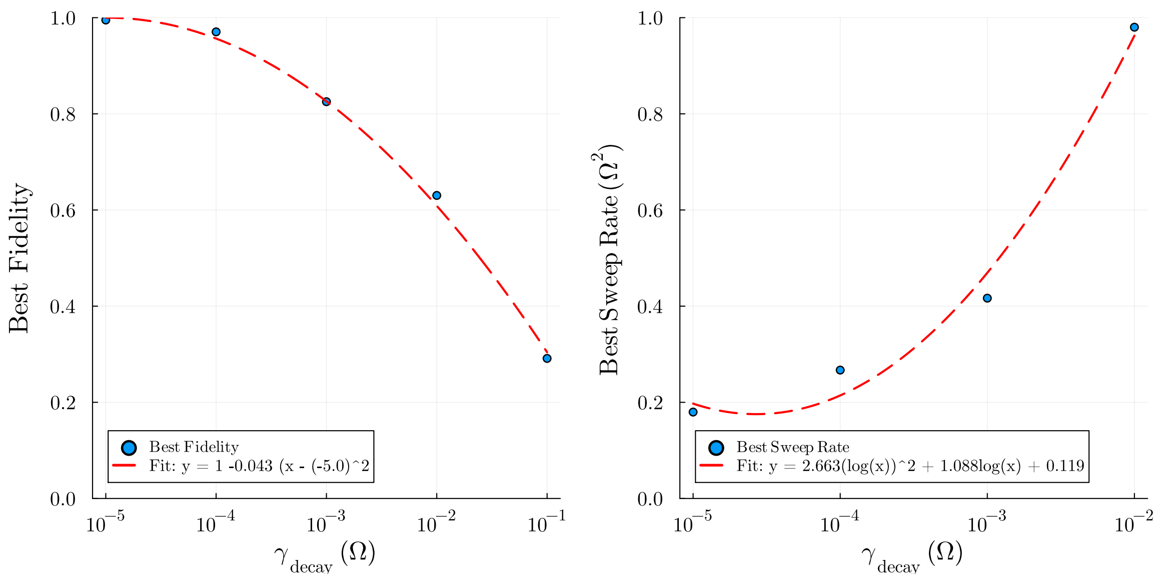


Figure 5.7. (Left) Maximum end fidelity attained by the system for varying range of decay rate (γ_{decay}) fit with a polynomial curve $y = 1 - c\,(x - x_0)^2$, with $x = \log_{10}(\gamma_{\text{decay}})$. (Right) Best sweep rate (δ) used to achieve the maximum end fidelity the system can attain for varying range of decay rate (γ_{decay}) fit with function $y = c_2(\log(x))^2 + c_1 \log(x) + c_0$.

to stop the RAP protocol soon as it reaches the maximum fidelity. The struggle with this would be to make the protocol robust and efficient, which it is with the symmetric sweep.

In the Fig. 5.7 (Left), we see the highest fidelity the system can attain at the end time t_f for various decay rates (γ_{decay}). We observe that as the decay rate increases, the best fidelity attained by the system decreases as expected. In the Fig. 5.7 (Right), we see the corresponding sweep rate (δ) which is responsible to achieve the best fidelity for the decay rate γ_{decay}. The best sweep rate follows a polynomial trend with respect to logarithm of the decay rate with a certain slope.

5.2.4 Conclusion

In this section we motivated the choice of RAP to implement the CNOT gate for the BFEC algorithm, we detailed how to embed the protocol in our Rydberg-atom platform, and specified the conditions required for efficient operation in the presence of decay. We compared two-atom and three-atom layouts and found that the *site-independent detuning* architecture achieves shorter gate times than the *site-dependent detuning* alternative. We then numerically optimized the *site-independent* layout to identify a fast, high-fidelity parameter set for RAP. In the remainder of the BFEC study, we will use these results, particularly Eq. 5.27 and Fig. 5.7, as the basis for selecting RAP parameters.

6 Implementation

In this chapter, we present the implementation of the BFEC algorithm by designing the Hamiltonians corresponding to various steps in the algorithm. Prior to this, we will lay out a well-thought framework for the algorithmic implementation. This framework will later allow us to simulate it via the MCWF method discussed in the Sec. 2.3. We will also discuss how to model bit-flip errors in a realistic manner, and optimize the parameters of the error model to ensure the success of the protocol.

As a recap, the computational qubit states are chosen to be the ground state $|g\rangle$ as $|0\rangle$ and the rydberg state $|r\rangle$ as $|1\rangle$, and we will use this notation through out the rest of the work. We will follow the system layout and notation as established in the Ch. 4, and the reader is advised to refer to it for clarity.

6.1 Algorithm Workflow

The implementation of the BFEC algorithm can be structured as a sequence of steps that correspond to quantum circuit operations. The workflow here is designed based on that, and is intended to be modular, allowing for easy adjustments, optimizations and usage of different physical processes as needed for gate operations. Below is the outline of the workflow:

1. **Encoding:** T_1 *timesteps*
   - Prepare the atom B in the state $\alpha|1\rangle + \beta|0\rangle$, and atoms A, C in the ground state $|0\rangle$.
   - Drive atoms A and C through Rydberg facilitation to implement the CNOT gate operation with atom B as the control qubit. This encodes the state of atom B into the logical state $\alpha|111\rangle + \beta|000\rangle$.
2. **Syndrome Measurement:** T_2 *timesteps*
   - Prepare ancillas a1 and a2 in the ground state $|0\rangle$.
   - Drive ancillas a1 and a2 through Rydberg facilitation to implement the CNOT gate operations such that {A, a1}, {B, a1}, {B, a2} and {C, a2} are the pairs of control and target qubits respectively.
   - For the projective measurement of ancillas a1 and a2 on to the qubit states, we will do the following:
     - Check the population of ancillas a1 and a2 in the excited state $|1\rangle$. Say the population is p_1 and p_2 respectively.
     - Now, we pick a random number r in the range $[0, 1]$, such that:
If $r \geq p_1$, then ancilla a1 is projected to $|0\rangle$, else a1 is projected to $|1\rangle$.
Similarly, if $r \geq p_2$, then ancilla a2 is projected to $|0\rangle$, else a2 is projected to $|1\rangle$.
3. **Correction:** T_3 *timesteps*

- Based on the measurement outcomes of ancillas a1 and a2, we will apply correction operations such as:
  - If only ancilla a1 is projected to $|1\rangle$, then atom A has undergone a bit-flip error and thus we drive atom A through Rydberg facilitation from atom B and ancilla a1 to correct it.
  - If only ancilla a2 is projected to $|1\rangle$, then atom C has undergone a bit-flip error and thus we drive atom C through Rydberg facilitation from atom B and ancilla a2 to correct it.
  - If both ancillas a1 and a2 are projected to $|1\rangle$, then atom B has undergone a bit-flip error and thus we drive atom B through Rydberg facilitation from ancillas a1, a2 and atoms A, C to correct it.

Because the described workflow is modular, we can adjust the durations T_1, T_2 and T_3 of the three steps, to match the control processes in implementing these steps. This flexibility allows for optimization and adaptation to various experimental constraints. In what follows, we adhere to the symmetric RAP technique for all steps.

Note that in the correction step, we explicitly couple ancillas and data qubits, to implement the correction operations rather than relying solely on single-qubit operations (see Sec. 3.3.1). This choice respects realistic constraints imposed by closely spaced atoms and the facilitation/blockade geometry. It also brings two practical advantages:

- Multi-qubit interactions can replace a longer sequence of single-qubit gates, reducing the total cycle time and exposure to decoherence.
- We can keep a fixed optical set-up and let the RAP protocol provide the required effective detunings, reducing laser re-tuning overhead.

We plan to simulate this via the MCWF method, and there are few additional considerations we need to take into account to ensure the algorithm is simulated correctly. That is, all the above steps are to be implemented for a single trajectory, and since the simulation technique is stochastic, we will repeat the above steps for a large number of trajectories to obtain statistical average results. We rely on this average to understand the error correction capabilities of the BFEC algorithm and the constraints imposed by a successful correction.

6.2 Modelling Bit Flip Errors

In this section, we understand and model bit-flip errors for our system of Rydberg atoms, under realistic conditions. We will look into the error channel description, the jump operator and find the optimal range of the decay rate parameter γ_{decay} for the success of the protocol.

6.2.1 Amplitude Damping Channel

In quantum computing, bit-flip errors are a common type of error that can occur during the manipulation and storage of quantum information. Generally, a bit-flip error occurs when a qubit experiences a flip around the X-axis of the Bloch sphere, i.e., the state $|0\rangle$ flips to $|1\rangle$ and vice versa. As the scope of this work is to model a realistic scenario for our system of Rydberg atoms, we only consider the case of relaxation from the excited state $|r\rangle$ to the ground state $|g\rangle$. In other words, our model of bit-flip errors occurs via an *Amplitude Damping* error channel [3], where the state $|r\rangle$ can flip to $|g\rangle$ with a probability p_{f}, while the state $|g\rangle$ remains unchanged i.e.,

$$\begin{aligned} |r\rangle \rightarrow |g\rangle \quad & \text{with probability } p_{\mathrm{f}} \\ |g\rangle \rightarrow |g\rangle \quad & \text{with probability } 1-p_{\mathrm{f}}. \end{aligned} \tag{6.1}$$

The asymmetric nature of the channel arises from the fact that in our physical system, the excited state $|r\rangle$ is prone to spontaneous emission and other decay processes, while the ground state $|g\rangle$ is relatively stable. This can be represented using the Kraus operators as

$$\mathcal{E}_{\mathrm{AD}}(\rho) = E_0 \rho E_0^{\dagger} + E_1 \rho E_1^{\dagger} \tag{6.2}$$

where E_k are Kraus operators defined as

$$E_0 = \begin{pmatrix} 1 & 0 \\ 0 & \sqrt{1-p_{\mathrm{f}}} \end{pmatrix}, \quad E_1 = \begin{pmatrix} 0 & \sqrt{p_{\mathrm{f}}} \\ 0 & 0 \end{pmatrix}. \tag{6.3}$$

It is important to observe this asymmetry, as it plays a crucial role in finding the optimal range of the decay rate parameter γ_{decay}.

Equivalently, we can define a jump operator $\hat{C}_{\mathrm{bf}}$ that describes our bit-flip error, as

$$\hat{C}_{\mathrm{bf}} = \sqrt{\gamma_{\mathrm{decay}}}\,|g\rangle\langle r| = \sqrt{\gamma_{\mathrm{decay}}}\,\sigma^{-} \tag{6.4}$$

where γ_{decay} is the decay rate parameter that controls the probability of the bit-flip error occurring.

In a realistic setting, the jump operator Eq. 6.4 operates on *all* the atoms, both the data qubits and the ancillas. This introduces errors in ancillas, which might lead to *false positives* or *false negatives* in the syndrome measurement. Consequently, the admissible range of the decay rate parameter γ_{decay} is affected and is constrained by the need to provide reliable syndrome measurements for the overall success of the algorithm. We will analyze this in detail in the next section, where we will optimize the decay rate parameter γ_{decay} for the algorithm. However, adopting this system-wide error model is crucial for simulating a faithful experimental scenario, and we will retain this condition through out rest of the work.

6.2.2 Optimizing Decay Rate Parameter

The decay rate parameter γ_{decay} plays a crucial role in the performance of the BFEC algorithm. It determines the success rate of the protocol. We should therefore spend some time and effort to understand the expected range of this parameter, before we proceed to actual implementation.

As briefly discussed before, the decay rate parameter γ_{decay} controls the probability of the bit-flip error occuring for an atom. The higher the value of γ_{decay}, the higher the probability of the bit-flip error occuring, and vice versa. On an average, this leads to exponential decay of the population in the excited state $|r\rangle$, given as

$$p_r(t) = p_r(0) e^{-\gamma_{\mathrm{decay}} t} \tag{6.5}$$

where $p_r(t)$ is the population in the excited state $|r\rangle$ at time t, and $p_r(0)$ is the initial population at time $t = 0$. The above equation is a simple exponential decay equation, which is a common model for the decay of populations in quantum systems.

Following this understanding, we move on to the condition for our error-correction procedure to work perfectly. We know that the three-qubit repetition codes can correct only single bit-flip errors, and this straightforward condition leads to us working in the regime where

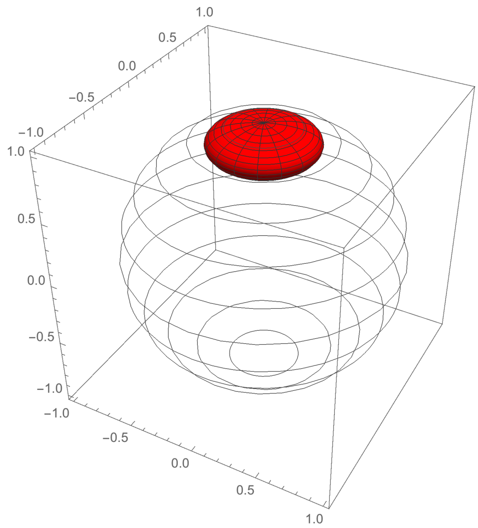


Figure 6.1. The effect of the amplitude damping error channel on the Bloch sphere. The outermost sphere represents the initial states of the Bloch sphere, and the inner sphere represents the states after the amplitude damping error channel has acted on them. Notice that the entire Bloch sphere shrinks towards the state $|0\rangle$ at the north pole. Figure reproduced from [55]. No changes made.

the probability of more than a single atom undergoing a bit-flip error is smaller than the probability of a single atom undergoing a bit-flip error. As, we deal with 5 atoms undergoing errors in the protocol, we can write this mathematically as

$$P_{\geq 2 \text{ errors}} < p \tag{6.6}$$

where p is the probability of an error, given by

$$p = 1 - e^{-\gamma_{\text{decay}} T} \tag{6.7}$$

for time T and $P_{\geq 2 \text{ errors}}$ is the probability of two or more atoms undergoing a bit-flip error, given by

$$\begin{aligned} P_{\geq 2 \text{ errors}} &= \binom{5}{2} P^2 (1-P)^3 + \binom{5}{3} P^3 (1-P)^2 + \binom{5}{4} P^4 (1-P) + \binom{5}{5} P^5 \\ &= 1 - \binom{5}{0} P^0 (1-P)^5 - \binom{5}{1} P^1 (1-P)^4 \\ &= 1 - (1-P)^5 - 5P(1-P)^4 \end{aligned} \tag{6.8}$$

Solving for the Eq. 6.6, we get the optimized range of the decay rate parameter γ_{decay} as

$$0 < \gamma_{\text{decay}} T < 0.1404. \tag{6.9}$$

The Eq. 6.9 is a crucial aspect of any error correction protocol, as long as we deal with 5 atoms subject to errors in the system. This limit further extends to decay rate parameter γ_{decay} being inversely proportional to the total time taken T by an error correction protocol. Thus, the speed of an error correction protocol is very crucial for the range of errors we can successfully correct. More precisely, the faster gate operations we have, the higher decay rate values we can work with. We obey the above condition for both the BFEC algorithm and also the PFEC algorithm, which will be discussed in the next part of the work.

6.3 Hamiltonian Design for Gate Operations

In this section, we design the Hamiltonians corresponding to various steps in the algorithm workflow outlined in the previous section. The BFEC algorithm mainly consists of CNOT gate operations, which can be implemented using Rydberg facilitation and the RAP technique. As we deal with time evolution of multiple atoms, we have to carefully work in different Rydberg facilitation regimes, and RAP conditions at different steps of the algorithm.

The notation used for the operators in the following Hamiltonians is X_i^j, where X_i is the single atom operator of shape (2×2) and j is the atom {A, B, C} or ancilla {a1, a2} on which the operator acts. The overall Hamiltonian lives in the Hilbert space of the entire system dimension $(2^5 \times 2^5)$.

Following the workflow, the Hamiltonian design is structured as follows:

1. **Encoding:** CNOT gate operation between atoms {B, A} and {B, C} with first atom as control and second atom as target.

$$H_{\text{enc}} = \frac{\Omega}{2}\left(\sigma_x^{\text{A}} + \sigma_x^{\text{C}}\right) + \Delta(t)\left(n_r^{\text{A}} + n_r^{\text{C}}\right) + V_{\text{nn}}\left(n_r^{\text{A}} n_r^{\text{B}} + n_r^{\text{B}} n_r^{\text{C}}\right) \tag{6.10}$$

   where the range of detuning $\Delta(t)$, starting from Δ_0 at $t = 0$, is symmetric over V_{nn}. Thus T_1 can be calculated similarly to Eq. 5.19

$$T_1 = \frac{2(\Delta_0 - |V_{\text{nn}}|)}{\delta}.$$

2. **Syndrome Measurement:** CNOT gate operation between {A, a1}, {B, a1} , {B, a2} and {C, a2} with first qubit as control and second qubit as target.

$$\begin{aligned} H_{\text{syn}} = &\frac{\Omega}{2}\left(\sigma_x^{\text{a1}} + \sigma_x^{\text{a2}}\right) + \Delta(t)\left(n_r^{\text{a1}} + n_r^{\text{a2}}\right) \\ &+ \frac{V_{\text{nn}}}{2^6}\left(n_r^{\text{A}} n_r^{\text{a1}} + n_r^{\text{B}} n_r^{\text{a1}} + n_r^{\text{B}} n_r^{\text{a2}} + n_r^{\text{C}} n_r^{\text{a2}}\right) \end{aligned} \tag{6.11}$$

   where the range of detuning $\Delta(t)$, starting from Δ_0' at $t = 0$, is symmetric over $V_{\text{nn}}/2^6$. Thus T_2 can be calculated as

$$T_2 = \frac{2\left(\Delta_0' - |\frac{V_{\text{nn}}}{2^6}|\right)}{\delta}$$

3. **Correction:**
   - **For atom A:** If ancilla a1 is projected to $|r\rangle$,

$$H_{\text{correct\_A}} = \frac{\Omega}{2}\sigma_x^{\text{A}} + \Delta(t) n_r^{\text{A}} + V_{nn}\left(n_r^{\text{A}} n_r^{\text{B}} + \frac{1}{2^6} n_r^{\text{A}} n_r^{\text{a1}}\right) \tag{6.12}$$

     where the range of detuning $\Delta(t)$, starting from Δ_0^{A} at $t = 0$, is symmetric over $(V_{\text{nn}} + V_{\text{nn}}/2^6)$. Thus the time taken for the correction of atom A can be calculated as

$$T_{\text{correct\_A}} = \frac{2\left(\Delta_0^{\text{A}} - \left|V_{\text{nn}} + \frac{V_{\text{nn}}}{2^6}\right|\right)}{\delta}.$$

- **For atom B:** Both ancillas a1 and a2 are projected to $|r\rangle$,

$$H_{\text{correct\_B}} = \frac{\Omega}{2}\sigma_x^{\text{B}} + \Delta(t)n_r^{\text{B}} + V_{nn}\left(n_r^{\text{B}}n_r^{\text{A}} + n_r^{\text{B}}n_r^{\text{C}} + \frac{1}{2^6}\left(n_r^{\text{B}}n_r^{\text{a1}} + n_r^{\text{B}}n_r^{\text{a2}}\right)\right) \quad (6.13)$$

  where the range of detuning $\Delta(t)$, starting from Δ_0^{B} at $t = 0$, is symmetric over $2\left(V_{nn} + V_{\text{nn}}/2^6\right)$. Thus the time taken for the correction of atom B can be calculated as

$$T_{\text{correct\_B}} = \frac{2\left(\Delta_0^{\text{B}} - 2\left|V_{\text{nn}} + \frac{V_{\text{nn}}}{2^6}\right|\right)}{\delta}.$$

- **For atom C:** If ancilla a2 is projected to $|r\rangle$.

$$H_{\text{correct\_C}} = \frac{\Omega}{2}\sigma_x^{\text{C}} + \Delta(t)n_r^{\text{C}} + V_{nn}\left(n_r^{\text{C}}n_r^{\text{B}} + \frac{1}{2^6}n_r^{\text{C}}n_r^{\text{a2}}\right) \quad (6.14)$$

  where the range of detuning $\Delta(t)$, starting from Δ_0^{C} at $t = 0$, is symmetric over $\left(V_{nn} + V_{nn}/2^6\right)$. Thus the time taken for the correction of atom C can be calculated as

$$T_{\text{correct\_C}} = \frac{2\left(\Delta_0^{\text{C}} - |V_{nn} + \frac{V_{nn}}{2^6}|\right)}{\delta}.$$

According to the above equations, the time taken to correct atoms A, B and C are different, but since atoms A and C are symmetric, we can have $\Delta_0^{\text{A}} = \Delta_0^{\text{C}}$, and thus the it takes the same time to correct atoms A and C. The total time taken by our BFEC algorithm is sum of the time taken by each step in the workflow and is given as

$$T_{\text{bf}} = T_1 + T_2 + T_{\text{correct\_A/B/C}} \quad (6.15)$$

The total time taken T_{bf} is dependent on the initial detunings, Rydberg interaction strength and the sweep rate of the RAP technique. In order to minimize this so we can correct for higher γ_{decay} values, one can optimize the parameters: $\{\Delta_0,\ \Delta_0',\ \Delta_0^{\text{A}},\ \Delta_0^{\text{B}},\ \Delta_0^{\text{C}},\ V_{\text{nn}},\ \delta\}$. For this, we follow the numerical optimization procedure discussed in Sec. 5.2.3 and pick the optimal set of parameters from here.

With this, we are all set to simulate the BFEC algorithm using the MCWF method, which will be discussed in the next section.

6.4 MCWF Simulations

In this chapter, we present the results of our MCWF simulations for the BFEC protocol. We analyze the performance of the protocol under various conditions. The results are presented in terms of fidelity and error rates. Fidelity measures how closely the final state of the system matches the desired state after error correction and is defined as

$$F\left(|\psi\rangle_{\text{id}}, \rho_{\text{final}}\right) = \langle\psi_{\text{id}}|\rho_{\text{final}}|\psi_{\text{id}}\rangle, \quad (6.16)$$

where $|\psi_{\text{id}}\rangle$ is the ideal state and ρ_{final} is the density matrix of the final state after error correction.

We mainly look at the results of the case where the initial state of the atom B is $|\psi\rangle_{\text{B}} = |1\rangle$ i.e., $\alpha = 1$, simply because the results are more illustrative. Whereas one can easily extend the work to different values of α and the results will be qualitatively similar.

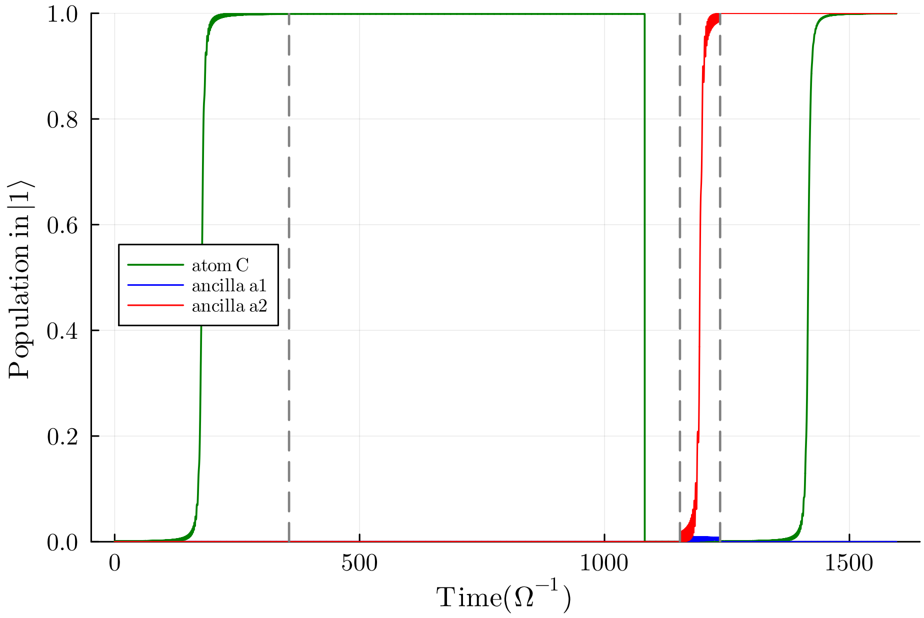


Figure 6.2. Population dynamics trajectory for entire runtime of BFEC algorithm for the decay rate $\gamma_{\text{decay}} = 10^{-5}$ and storage time $t_{\text{storage}} = 1\,T_{\text{bf}}$. The vertical dashed lines indicate the transitions between different steps in the algorithm: encoding, storage, syndrome measurement, and correction. This particular trajectory corresponds to trajectory number 378, where a bit-flip error occurs on atom C during the storage step, which is successfully detected and corrected in the syndrome measurement and correction steps, respectively.

The measure of success of the error correction protocol, lies in the necessity of *storage of encoded data qubits*, before performing syndrome measurement and correction steps. Any error correction protocol will correct errors that occur during the storage of the encoded qubits, but will not correct errors that occur during the syndrome measurement and correction steps themselves. Therefore, we introduce a variable called the *storage time*, t_{storage}, which is the time duration for which the encoded data qubits are stored before performing syndrome measurement and correction steps.

We first analyze the individual trajectories generated by the MCWF simulations. Each trajectory represents a possible evolution of the quantum state under the influence of bit-flip errors and the error correction protocol. We observe that the trajectories exhibit stochastic behavior, with some trajectories successfully correcting errors while others fail to do so. The Fig. 6.2 illustrates an example for the evolution of atom C in the BFEC algorithm. It shows the dynamics of the population in $|1\rangle$ over time for a particular trajectory, highlighting the moments when error has occurred, detected, and when the error correction protocol has been applied. This trajectory corresponds to one of the case when a bit-flip error occurs on atom C, which is successfully detected by jump in the population of ancilla a2 during the syndrome measurement step, and thus later corrected during the correction step.

We then analyze the distribution of fidelities at different steps during the protocol, over multiple trajectories for a particular case of bit-flip error rate and storage time. The Fig. 6.3 shows the histogram of fidelities over 1000 trajectories for the decay rate $\gamma_{\text{decay}} = 10^{-5}\,\Omega$ and storage time $t_{\text{storage}} = 50\,T_{\text{bf}}$. We observe that the distribution of fidelities is quite concentrated around 1.0 after the *encoding* step, indicating that the encoding process is highly effective in preserving the quantum information for this case. Following this, we see a

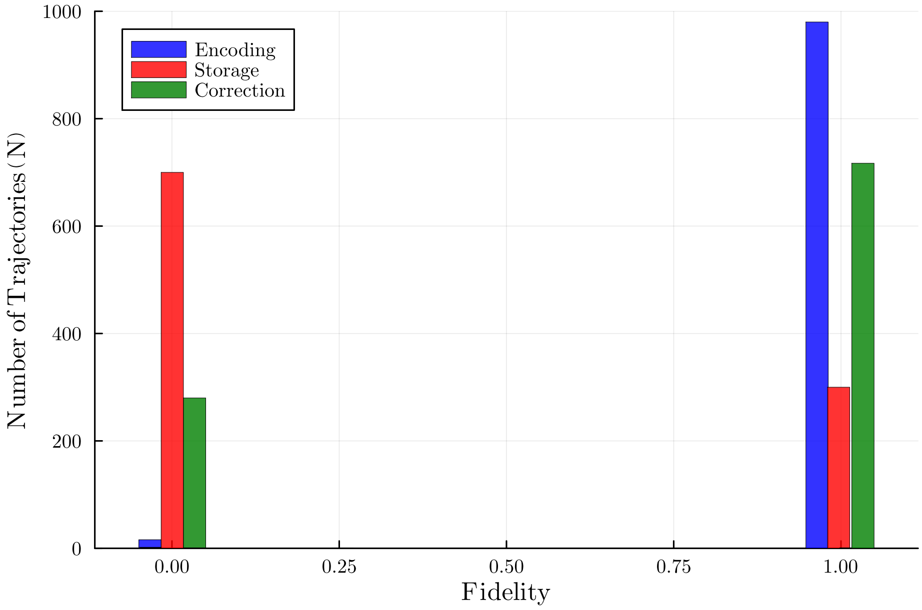


Figure 6.3. Analysis of the distribution of fidelities over 1000 trajectories for the decay rate $\gamma_{\text{decay}} = 1\text{e}^{-5}$ and storage time $t_{\text{storage}} = 50\,T_{\text{bf}}$. The initial state of the Atom B is chosen to be $|\psi\rangle_{\text{B}} = |1\rangle$.

spread in the distribution of the fidelities after the *storage* step, where many trajectories have zero fidelity. Among them, we see that a significant number of them are corrected during the *correction* step, as indicated by the peak around 1.0 fidelity after the correction step. However, there are still considerable number of trajectories that remain at zero fidelity even after the correction step, which prominently correspond to cases where two or more bit-flip errors have occurred during the storage step, which the protocol is unable to correct. It is important to note other causes for these zero fidelity cases, which can arise due to errors occuring during any gate operations, errors on the ancilla qubits leading to incorrect syndrome measurements, and finally also due to the inherent probabilistic nature of measurement itself. These are the prominent factors that can lead to the failure of a QEC protocol, due to the *realistic* nature of the implementation.

Following this, we analyze the average fidelity over multiple trajectories to obtain a statistical measure of the protocol's performance. In this analysis, we stick to simulating 10 K trajectories for various cases, as the statistical error for MCWF simulations scales as $\left(1/\sqrt{N_{\text{traj}}}\right)$, where N_{traj} is the number of trajectories simulated. Thus, simulating 10 K [6] trajectories ensures that the statistical error for the average fidelity become sufficiently small at this number of trajectories.

Extrapolating from the single trajectory analysis, we expect the average fidelity over multiple trajectories to show a general trend of decreasing fidelity over time due to the accumulation of bit-flip errors, with a noticeable bump in the fidelity during the error correction step of the protocol. This can be seen in the Fig. 6.4. It indicates the effectiveness of the error correction strategy employed, whereas the success of the protocol is to be determined in more detail.

In general, one can define the success of an error correction protocol as follows:

[6] 1K = 1000

Success Criterion of QEC Protocol: *Any error correction protocol is deemed to be successful if it provides a net positive difference between the average fidelities of the final state of the system with and without the error correction protocol.*

The success of the protocol is then analyzed as a function of t_{storage}, and the bit-flip error rate, γ_{decay}. That is, we determine the range of storage times for which the protocol satisfies the success criterion for a given γ_{decay}.

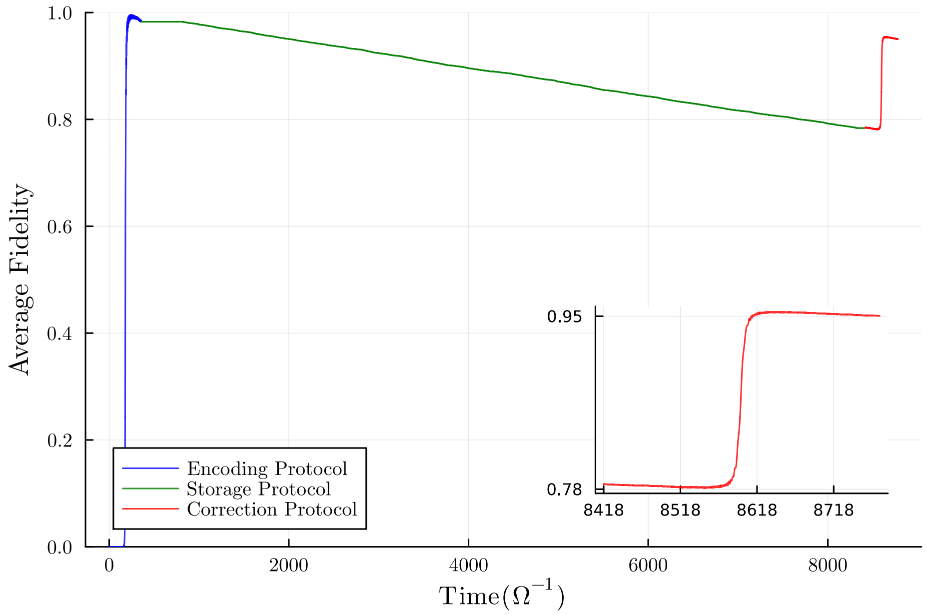


Figure 6.4. Average Fidelity over 10K trajectories for $\gamma_{\mathrm{decay}} = 10^{-5}\,\Omega$ and for storage time of $t_{\mathrm{storage}} = 10\,T_{\mathrm{bf}}$. The main plot shows the fidelity evolution over the entire algorithm duration, while the inset zooms in on the gain in fidelity during the correction step, highlighting the effectiveness of the error correction process.

To calculate for the case of no error correction protocol, we simply let the atom B evolve under the influence of bit-flip errors for a time duration equal to the total time taken i.e., $t_{\mathrm{bf}} + t_{\mathrm{storage}}$. Consider the initial state of the Atom B to be $|\psi\rangle_{\mathrm{B}} = \alpha\,|1\rangle + \beta\,|0\rangle$, then the density matrix of the Atom B after a time duration T, under the influence of bit-flip errors is given by

$$\rho_T = \begin{pmatrix} |\beta|^2 & \alpha^*\beta \\ \alpha\beta^* & |\alpha|^2 e^{-\gamma_{\mathrm{decay}} T} \end{pmatrix}. \tag{6.17}$$

Calculating the fidelity of ρ_{T} with the initial state $|\psi\rangle_{\mathrm{B}}$ of the Atom B, we obtain

$$F^{(1)} = |\beta|^2 + 2|\alpha|^2|\beta|^2 + |\alpha|^4 e^{-\gamma_{\mathrm{decay}} T}. \tag{6.18}$$

where $F^{(1)}$ is defined the single particle evolution fidelity under the influence of bit-flip errors for a time duration T in the *Amplitude Damping* Channel. The success criterion of our BFEC protocol can therefore be mathematically defined as

$$F^{\mathrm{corrected}} > F^{(1)} \tag{6.19}$$

where $F^{\mathrm{corrected}}$ is the average fidelity of the final state of the system after the error correction protocol.

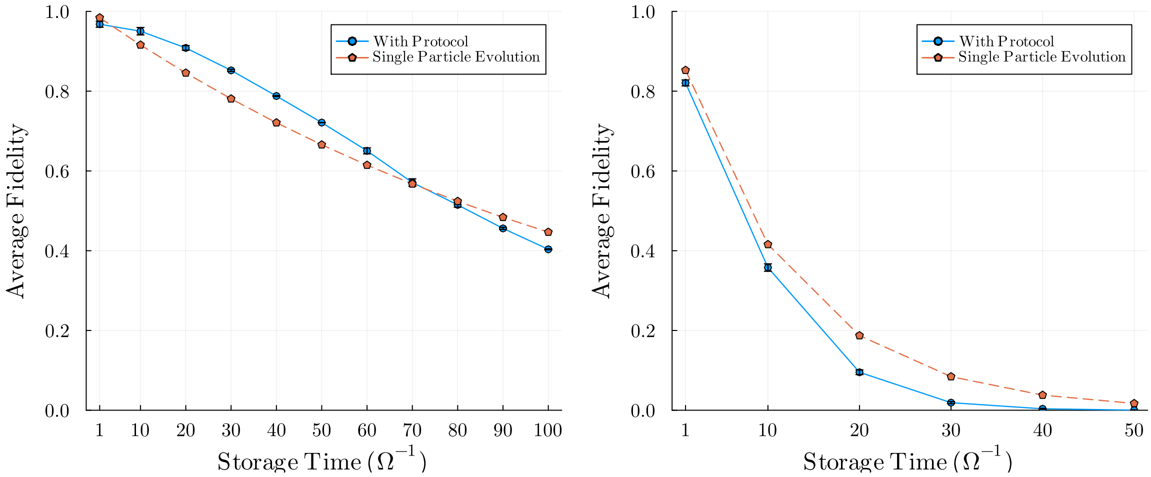


Figure 6.5. Comparision between average fidelity calculated after the BFEC protocol over 10K trajectories (Blue) and the single qubit evolution fidelity (Red). The error bars represent the standard deviation of the fidelity due to the finite number of MCWF trajectories simulated. (Left) For $\gamma_{\text{decay}} = 10^{-5}\,\Omega$ and for storage times ranging from $1\,T_{\text{bf}}$ to $100\,T_{\text{bf}}$. (Right) For $\gamma_{\text{decay}} = 10^{-4}\,\Omega$ and for storage times ranging from $1\,T_{\text{bf}}$ to $50\,T_{\text{bf}}$.

In the Fig. 6.5, we see the average fidelity for $\gamma_{\text{decay}} = 10^{-5}\,\Omega$, for different storage times ranging from $1\,T_{\text{bf}}$ to $100\,T_{\text{bf}}$, in comparision to the fidelity of a single qubit evolving under the influence of bit-flip errors for a time duration equal to $T_{\text{bf}} + t_{\text{storage}}$. It shows that for storage times between $\approx 5\,T_{\text{bf}}$ and $\approx 70\,T_{\text{bf}}$, our BFEC algorithm satisfies the success criterion, thus proving its effectiveness.

Whereas for $\gamma_{\text{decay}} = 10^{-4}\,\Omega$, the protocol is never successful for any storage time. Thus, we can conclude that the BFEC algorithm is effective for low bit-flip error rates, specifically $\gamma_{\text{decay}} \leq 10^{-5}\,\Omega$, and for storage times ranging from 10 to 70 times the duration of the algorithm itself. During this range, the BFEC algorithm can be employed iteratively to further maintain high fidelity of the quantum state over extended periods.

Part IV

Phase-Flip Error Correction Protocol

7 System Architecture

In this part of the thesis work, we design the Phase-Flip Error Correction (PFEC) algorithm, with our system of Rydberg atoms. We follow the same approach as we did for the BFEC case, and thus this work goes hand in hand with the previous part.

We first verify if the qubit-state choice used for the BFEC algorithm is also suitable here. We then design the corresponding system architecture, specify the control processes for gate operations, implement the PFEC algorithm, and finally assess the protocol's error-correction performance together with the constraints required for successful correction.

7.1 Dynamical Phase: The Problem

We adopt the same computational basis as in the BFEC case, using the ground state $|g\rangle$ and a Rydberg state $|r\rangle$. To illustrate the encoding idea, consider three atoms $\{1, 2, 3\}$ with initial state

$$|\psi_0\rangle = |g\rangle \otimes (\alpha |r\rangle + \beta |g\rangle) \otimes |g\rangle ,$$

and target mapping $|\psi_0\rangle \mapsto \alpha |rrr\rangle + \beta |ggg\rangle$, preserving the initial relative phase between the α and β components.

We use a symmetric RAP to perform the said mapping. As the sequence proceeds, atoms 1 and 2 are driven into $|r\rangle$ and further continued adiabatic evolution accumulates a *dynamical phase* between the $|g\rangle$ and $|r\rangle$ branches [48]. In our case it becomes most apparent in the second half of the RAP sweep, once atoms 1 and 2 occupy $|r\rangle$ and the evolution continues (see Fig. 7.1). In practice, this phase is deterministic i.e., it is set by the pulse shapes and detunings.

By definition, the dynamical phase accumulated by the system after atoms 1 and 2 occupy the $|r\rangle$, can be mathematically modelled as

$$\begin{aligned} |\psi(t)\rangle &= e^{-i\frac{\mathrm{H}t}{\hbar}} |\psi(0)\rangle \\ &= e^{-\frac{i}{\hbar}\int_0^t \mathrm{E_r}(\tau)d\tau} \alpha |rrr\rangle + e^{-\frac{i}{\hbar}\int_0^t \mathrm{E_g}(\tau)d\tau} \beta |ggg\rangle \\ &= \alpha e^{i\phi_d} |rrr\rangle + \beta |ggg\rangle \end{aligned} \tag{7.1}$$

where ϕ_d is the dynamical phase accumulated between the two states $|ggg\rangle$ and $|rrr\rangle$, and is given by

$$\phi_d = -\frac{1}{\hbar}\int_0^t (\mathrm{E_r}(\tau) - \mathrm{E_g}(\tau))\, d\tau. \tag{7.2}$$

Here, $\mathrm{E_r}(\tau)$ and $\mathrm{E_g}(\tau)$ are the instantaneous energies of the Rydberg state and the ground state respectively, at time τ. Technically, one can also apply for the correction of the dynamical phase, for which one has to keep track of instantaneous energies of the states during the time evolution of the system until the end of the protocol and then apply a relative phase correction of $-\phi_d$ to the state $|rrr\rangle$. This ensures that the final state accurately accounts for the accumulated dynamical phase. This correction is feasible if the instantaneous energies of the states are known at all times. One can also neglect the dynamical phase if the instantaneous energies of the states are quite small i.e., comparable with the linewidth of the laser used to

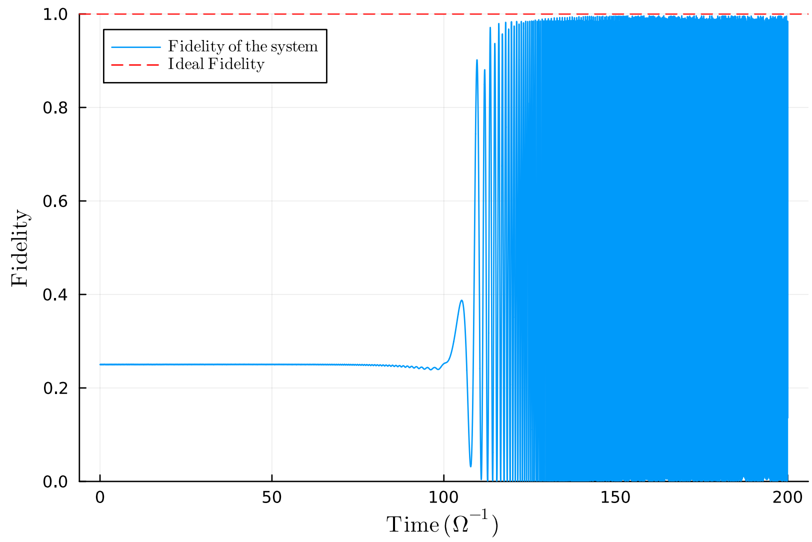


Figure 7.1. Dynamical Phase accumulated during the symmetric RAP protocol for $\alpha = \sqrt{0.5}$. The fidelity of the system is calculated as the overlap between the actual state and the target state.

excite the atom. Since we deal with the Rydberg states, which have large energy difference with the ground state, the dynamical phase is quite large and thus we cannot neglect it. Thus it calls for a different choice of qubit states, which can help us overcome this relative phase intrinsically accumulated by the system.

As a quick note, the dynamical phase doesn't affect the populations of the states, but only the relative phase between them, thus one can avoid this problem in our implementation of the BFEC algorithm by redefining the *fidelity measure* with *populations* in the qubit states, rather than the full quantum state including phases.

7.2 Redefined Qubit States and System Design

For the PFEC algorithm we work with a *three-level* (Λ) system: two degenerate ground states serve as the computational basis, $|0\rangle$ and $|1\rangle$, and a Rydberg state $|r\rangle$ provides strong, switchable interactions (blockade/facilitation) and a coupling channel for multi-qubit gate operations. Using these qubit states gives excellent coherence and enables high-fidelity single-qubit control, while $|r\rangle$ is only populated during gates.

This choice changes the control toolbox. The standard two-level RAP (adiabatic rapid passage) used for $|g\rangle \leftrightarrow |r\rangle$ does not apply to $|0\rangle \leftrightarrow |1\rangle$, since these states are not directly coupled. Entangling operations still use the Rydberg pathway, mapped to CNOT gate operation. We will discuss the control protocols in detail in the next chapter.

In practice, the two degenerate ground states can be hyperfine states of the atomic ground manifold (for example, in ^{87}Rb; see Fig. 7.2) [23]. These hyperfine states are split by GHz, so they are not strictly "degenerate". In these cases, the choice of qubit states still mitigates the large, uncontrollable dynamical phases encountered in the direct two-level $|g\rangle \leftrightarrow |r\rangle$ drives, whereas the hyperfine splitting can be handled with standard techniques (e.g., working in a rotating frame, spin-echo, composite pulses, etc.) [9, 56, 57]. Also it is important to note that handling these hyperfine states is much easier as their energy splitting is fairly robust against external perturbation except magnetic fields.

For simplicity, we model the three-level system as an idealized Λ-scheme with perfectly degenerate $|0\rangle$ and $|1\rangle$. Thus the three-level Λ-scheme preserves phase coherence required for PFEC while enabling robust Rydberg-mediated gates.

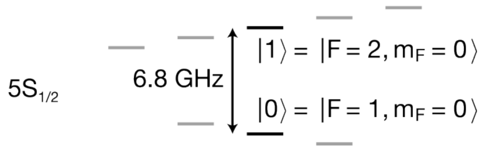


Figure 7.2. Ground state energy levels of ^{87}Rb atom showing the hyperfine splitting. The two levels marked in black are used as qubit states. Image is adapted from [23].

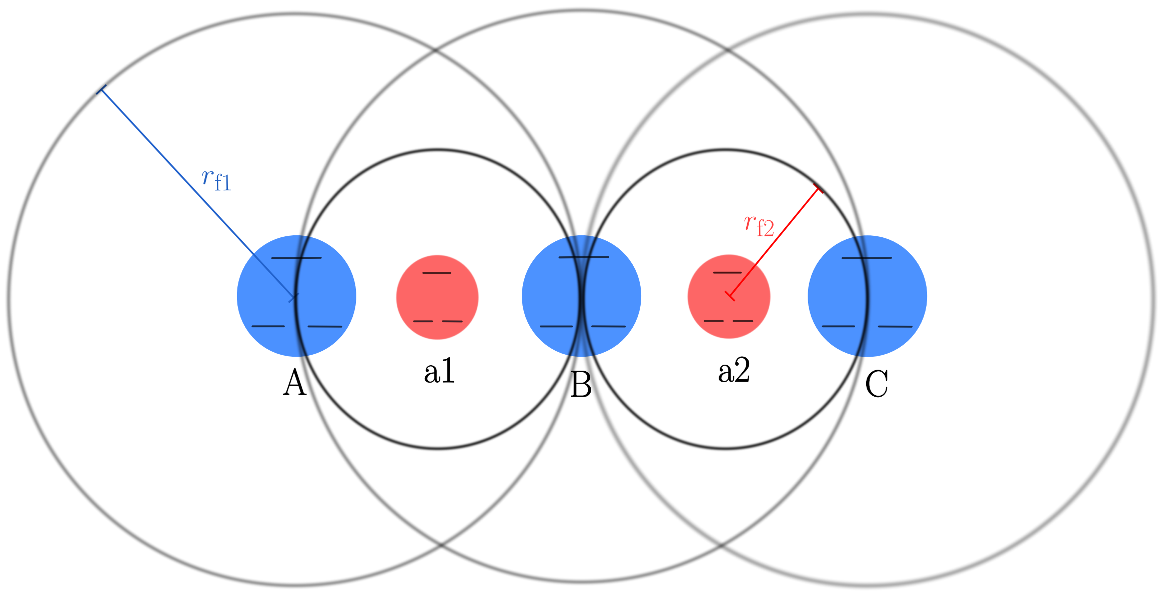


Figure 7.3. Proposed Rydberg System Architecture for PFEC algorithm shown with three-level atoms and their facilitation shells. The atoms are arranged in a linear chain with equal spacing. The blue atoms represent the *data qubits*, while the red atoms represent the *ancillary qubits* with their facilitation radii r_{f1} and r_{f2} respectively.

Following this, we propose the system architecture for the PFEC algorithm, shown in Fig. 7.3. The architecture is similar to the one used for the BFEC algorithm, with the only difference being that we now deal with three-level atoms instead of two-level atoms. To reiterate, the *data qubits* (blue), labelled $\{A, B, C\}$, are for encoding the logical qubit, while the ancillary qubits (red), labelled $\{a1, a2\}$ are for syndrome measurement. The data qubits are placed at mutual distance r_{f1}, which is the facilitation radius of the data qubits, while the ancillary qubits sit at the midpoint between a data-qubit pair and are chosen so that its facilitation radius r_{f2} is half that of the data qubits, i.e., $r_{\mathrm{f2}} = r_{\mathrm{f1}}/2$. We again work in a nearest-neighbor interaction regime, where each atom only interacts with its immediate neighbors.

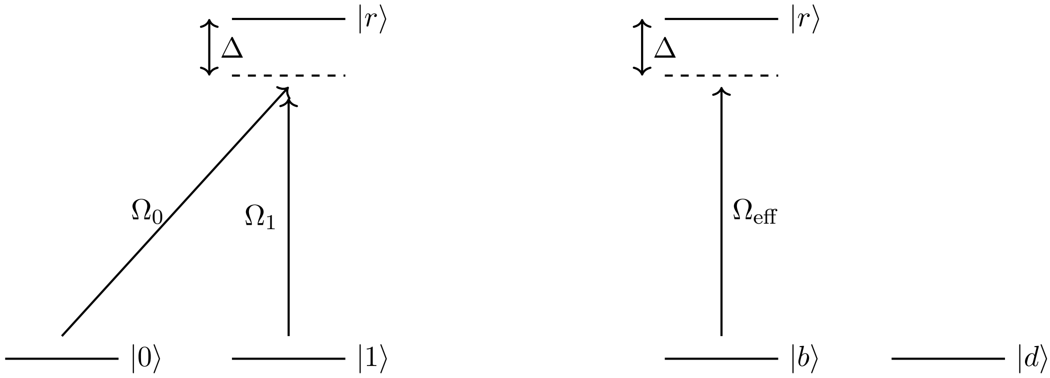
8 Control Protocols for PFEC Algorithm

In this chapter, we will deepen our understanding of 3-level systems, and see how we can use a basis transformation technique to describe its dynamics by an equivalent 2-level system representation. Following this, we will explore certain control processes that can be used to implement the equivalents of required gate operations for the PFEC algorithm.

8.1 Λ-scheme and Morris-Shore Transformation

In 1982, James R. Morris and Bruce W. Shore introduced a transformation of basis states that allows the problem of examining the dynamics of a degenerate two-level system coupled to an excited state to be reduced to a set of independent nondegenerate two-level systems and residue one-level systems [58]. This transformation came to be known as the Morris-Shore (MS) transformation, and is particularly useful in the context of 3-level systems in a Λ-configuration.

A Λ-system consists of two lower energy states, say $|0\rangle$ and $|1\rangle$, and an excited Rydberg state $|r\rangle$, such that two lower states are coupled to the excited state, while there is no direct coupling between themselves. The energy level diagram of a Λ-system and its equivalent 2-level system after the MS transformation is shown in Fig. 8.1.



(a) Λ-system **(b)** Equivalent 2-level system

Figure 8.1. Energy level diagrams of a Λ-system and its equivalent 2-level system after the MS transformation.

The Hamiltonian of our Λ-system in the $\{|0\rangle, |1\rangle, |r\rangle\}$ basis and in the rotating wave approximation (RWA) is given by

$$\hat{H} = \frac{\hbar}{2}\begin{pmatrix} 0 & 0 & \Omega_0 \\ 0 & 0 & \Omega_1 \\ \Omega_0 & \Omega_1 & \Delta \end{pmatrix} \tag{8.1}$$

where Ω_0 and Ω_1 are the Rabi frequencies of the transitions $|0\rangle \leftrightarrow |r\rangle$ and $|1\rangle \leftrightarrow |r\rangle$ respectively, and Δ is the detuning of the laser from the excited state.

We can show the equivalence of such a system to a 2-level system by solving the set of Bloch equations for its density matrix elements. Consider a general state of the system given by

$$|\psi\rangle = c_0 |0\rangle + c_1 |1\rangle + c_r |r\rangle \tag{8.2}$$

and following the Schrödinger equation, the time evolution of the above state is given by

$$\frac{d}{dt} |\psi\rangle = -\frac{i}{\hbar}\hat{H} |\psi\rangle . \tag{8.3}$$

From this, we can derive the equations of motion for the coefficients c_0, c_1 and c_r as

$$\begin{aligned}
\dot{c_0} &= -i\Omega_0 c_r \\
\dot{c_1} &= -i\Omega_1 c_r \\
\dot{c_r} &= -i\Omega_0 c_0 - i\Omega_1 c_1 - i\Delta c_r .
\end{aligned} \tag{8.4}$$

In principle, we can solve this set of coupled differential equations by approximating one of the states to be weakly populated, i.e., assuming $\dot{c_r} \approx 0$.

Whereas here, we take a different approach by transforming the basis states to simplify the equations of motion. We introduce new basis states known as the "bright" and "dark" states. The bright state $|b\rangle$ is a superposition of the two lower states that couples to the excited state, while the dark state $|d\rangle$ is orthogonal to the bright state and does not couple to the excited state. They are defined as

$$\begin{aligned}
|b\rangle &= \frac{\Omega_0 |0\rangle + \Omega_1 |1\rangle}{\sqrt{\Omega_0^2 + \Omega_1^2}} \\
|d\rangle &= \frac{\Omega_1 |0\rangle - \Omega_0 |1\rangle}{\sqrt{\Omega_0^2 + \Omega_1^2}} .
\end{aligned} \tag{8.5}$$

In this new basis, the state of the system can be expressed as

$$|\psi\rangle = c_b |b\rangle + c_d |d\rangle + c_r |r\rangle \tag{8.6}$$

where c_b and c_d are the coefficients of the bright and dark states respectively. The equations of motion for the coefficients in this new basis are given by

$$\begin{aligned}
\dot{c_b} &= -i\sqrt{\Omega_0^2 + \Omega_1^2} c_r \\
\dot{c_d} &= 0 \\
\dot{c_r} &= -i\sqrt{\Omega_0^2 + \Omega_1^2} c_b - i\Delta c_r .
\end{aligned} \tag{8.7}$$

From these equations, we can see that the dark state $|d\rangle$ is decoupled from the dynamics of the system, and the system effectively behaves like a 2-level system consisting of the bright state $|b\rangle$ and the excited state $|r\rangle$ with an effective Rabi frequency $\Omega_{\text{eff}} = \sqrt{\Omega_0^2 + \Omega_1^2}$.

The Hamiltonian of this effective 2-level system in the bright-dark basis is given by

$$\hat{H}_{\text{eff}} = \frac{\hbar}{2} \begin{pmatrix} 0 & \Omega_{\text{eff}} \\ \Omega_{\text{eff}} & \Delta \end{pmatrix} \tag{8.8}$$

and one can write this Hamiltonian in original computational basis as,

$$\hat{H}_{\text{eff}} = \frac{\hbar}{2} \begin{pmatrix} \frac{|\Omega_0|^2}{\Delta} & \frac{\Omega_0\Omega_1^*}{\Delta} \\ \frac{\Omega_1\Omega_0^*}{\Delta} & \frac{|\Omega_1|^2}{\Delta} \end{pmatrix} . \tag{8.9}$$

This transformation greatly simplifies the analysis of the system and allows us to interpret the dynamics of the Λ-system in terms of the well-known dynamics of a 2-level system.

8.2 Hadamard Gate using Raman Transition

In this section, we will explore certain mechanisms that can be used for implementing the required gate operations for the PFEC algorithm. As discussed in the Ch. 2, the algorithm requires the implementation of single qubit Hadamard (H) and two-qubit controlled-X (CNOT) gates.

Let us first consider the implementation of the single qubit Hadamard gate on a Λ-system. In 2017, David C. McKay et al. demonstrated the idea of zero-duration arbitrary "virtual" Z-gates that can be used for quantum state manipulation [59]. We'll extend this idea to implement the Hadamard gate in our Rydberg atom system. But first, let us briefly review the concept of virtual Z-gates.

Implementation of a Z-gate, i.e., a rotation around the Z-axis of the Bloch sphere, can be achieved by changing the relative phase between the two basis states $|0\rangle$ and $|1\rangle$ of a qubit. The result is that the qubit state rotates with respect to X- and Y-axes. However, one can also achieve the same effect by equivalently rotating the X- and Y-axes of the Bloch sphere around the Z-axis by the same angle in the opposite direction. This is the idea behind *virtual Z-gates* (VZ), where the Z-rotation is implemented without any physical operation on the qubit itself. In this implementation, no population is moved, the "gate" is *classical frame update* in the control electronics. This means that VZ gates are noiseless and effectively instantaneous, making them a very powerful tool for quantum state manipulation.

Consider an arbitrary SU(2) gate operation, which is given by

$$U(\theta,\phi,\lambda) = \begin{pmatrix} \cos\left(\frac{\theta}{2}\right) & -ie^{i\lambda}\sin\left(\frac{\theta}{2}\right) \\ -ie^{i\phi}\sin\left(\frac{\theta}{2}\right) & e^{i(\lambda+\phi)}\cos\left(\frac{\theta}{2}\right) \end{pmatrix} \tag{8.10}$$

where θ, ϕ and λ are the Euler angles. This gate can be decomposed into a sequence of rotation around the X-axis and VZ gate, upto a global phase as

$$U(\theta,\phi,\lambda) = VZ(\phi)\cdot R_x(\theta)\cdot VZ(\lambda). \tag{8.11}$$

where,

$$VZ(\phi) = \mathrm{e}^{-i\frac{\phi}{2}\hat{Z}}; \qquad R_X(\theta) = \mathrm{e}^{-i\frac{\theta}{2}\hat{X}} \tag{8.12}$$

with $\hat{Z}$ and $\hat{X}$ being the Pauli-Z and Pauli-X operators respectively. Naively, the Hadamard gate can be implemented as $H = XY^{1/2}$ i.e., it requires combination of rotations. Whereas following the VZ-gate implementation, the Hadamard gate can be decomposed as

$$H = VZ\left(\frac{\pi}{2}\right)\cdot R_x\left(\frac{\pi}{2}\right)\cdot VZ\left(\frac{\pi}{2}\right). \tag{8.13}$$

Thus, we have a simpler implementation of Hadamard gate which only requires physical rotation about the X-axis, with the total wall-clock time equals to the time taken for the $R_x(\pi/2)$ pulse.

Now, let us look at the implemenation of the $R_x(\pi/2)$ pulse on our Λ-system. As discussed in the previous section, we'll use the equivalent 2-level system after the MS transformation to analyze the dynamics of our Λ-system. One can rewrite the Hamiltonian of the effective 2-level system, in terms of the Pauli matrices as

$$\hat{H}_{\text{eff}} = \frac{\hbar}{2}\left(\frac{\Omega_0\Omega_1}{\Delta}\right)\sigma_x + \frac{\hbar}{2}\left(\frac{|\Omega_1|^2-|\Omega_0|^2}{2\Delta}\right)\sigma_z \tag{8.14}$$

where we have assumed Ω_0 and Ω_1 to be real for simplicity. For the implementation of the $R_x(\pi/2)$ pulse, which is given as $R_x(\pi/2) = e^{-i\frac{\pi}{4}\sigma_x}$, we need to ensure that the Hamiltonian

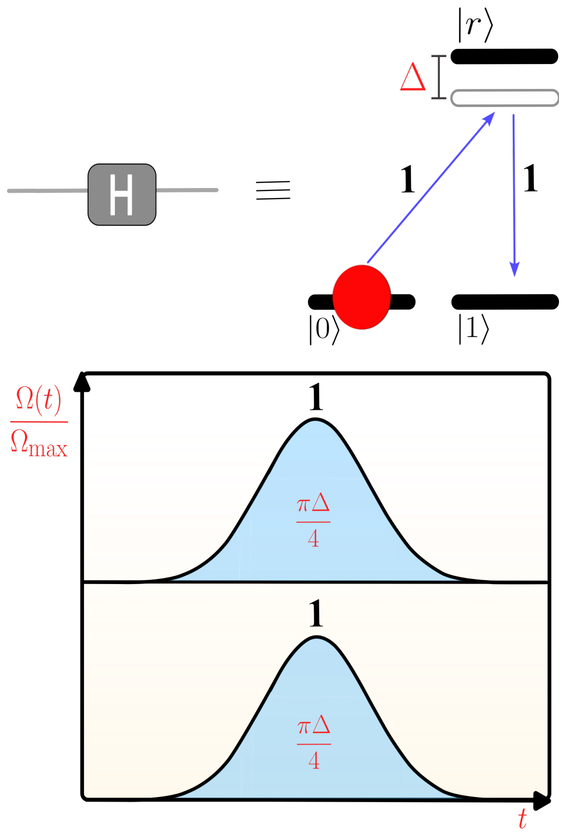


Figure 8.2. (Top) Schematic representation of the proposed Hadamard gate operation (*far-detuned Raman Transition*). (Bottom) Required pulse schedule for the Hadamard gate operation.

has no σ_z component. This can be achieved by setting $|\Omega_0| = |\Omega_1| = \Omega$. With this condition, the time evolution operator of the system is given by

$$U(t) = e^{-i\hat{H}_{\text{eff}}T} = e^{-i\frac{\Omega^2}{\Delta}T\sigma_x} \tag{8.15}$$

Equating this to the desired $R_x(\pi/2)$ operation, we find that the pulse duration T should be set to

$$T_{\text{H}} = T = \frac{\pi\Delta}{4\Omega^2} \tag{8.16}$$

Such a transition is called the *far-detuned Raman Transition*, where the relative detuning δ between the two ground states is zero and the excited state is detuned with respect to ground states (see Fig. 8.2) [49]. This is a widely used technique for implementing single-qubit gates in neutral atom systems. Therefore, on a Λ-system, the Hadamard gate can be implemented by applying two simultaneous laser pulses with equal Rabi frequencies $|\Omega_0| = |\Omega_1|$ between the states $|0\rangle \leftrightarrow |r\rangle$ and $|1\rangle \leftrightarrow |r\rangle$ respectively with detuning Δ, for a duration given in Eq. 8.16, along with the appropriate virtual Z-gates before and after the pulse.

Modified Hadamard Operation

It is important to note that the Hadamard operation implemented above is not the standard Hadamard operation, but rather a modified version of it. To see this, let us look at the action of the defined Hadamard gate on the basis states $|0\rangle$ and $|1\rangle$ of our Λ-system, which is given

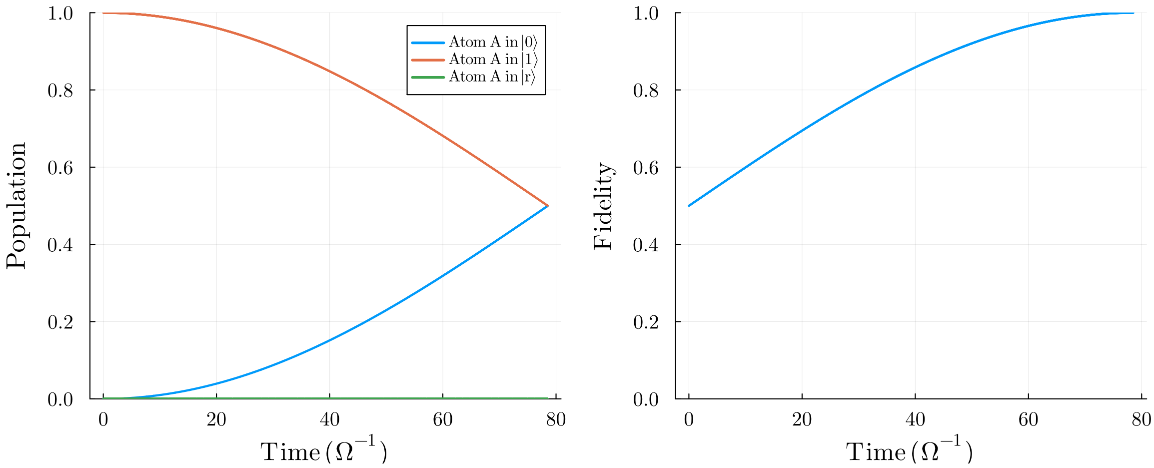


Figure 8.3. Simulation of the Hadamard operation on a 3-level Λ-system using the proposed method, corresponding to the Fig. 8.2. The initial state of the system is $|1\rangle$. (Left) Population dynamics of atom A. (Right) Fidelity dynamics of atom A with respect to the state $\sqrt{0.5}(|0\rangle + |1\rangle)$.

by

$$\begin{aligned} |0\rangle &\xrightarrow{H} -\frac{1}{\sqrt{2}}(|0\rangle - |1\rangle) = -\,|-\rangle \\ |1\rangle &\xrightarrow{H} \frac{1}{\sqrt{2}}(|0\rangle + |1\rangle) = +\,|+\rangle \,. \end{aligned} \tag{8.17}$$

Therefore for any arbitrary state $|\psi\rangle = \alpha\,|0\rangle + \beta\,|1\rangle$, we have

$$|\psi\rangle \xrightarrow{H} -\alpha\,|-\rangle + \beta\,|+\rangle \,. \tag{8.18}$$

The additional phase factor of "-1" is important to note, as it can affect the overall fidelity of the system during the error correction protocol. Since we apply the Hadamard operation over 3 data qubits in the protocol (see Fig. 3.4), the overall phase factor will be $(-1)^3 = -1$, and thus the relative phase still persists.

8.3 Controlled-X (CNOT) Gate using Rydberg Facilitation

In this section, we will explore the implementation of the two-qubit CNOT gate using the combination of *Rydberg facilitation* mechanism and the *Fully-Resonant Raman transition* (FRR) mechanism, in the context of our Λ-system. To understand this, let us first briefly review the FRR transition mechanism.

The FRR transition is a technique used to transfer population between two ground states of a Λ-system via an intermediate excited state, while avoiding population in the excited state at the end of the pulse [49]. It is similar to the *far-detuned Raman transition* discussed earlier, but in this case, the detuning Δ of the excited state is set to zero, i.e., the laser fields are resonant with the transitions between the ground states and the excited state.

Consider an atom of our Λ-system, with two ground states $|0\rangle$ and $|1\rangle$, and an excited state $|r\rangle$. The two ground states are coupled to the excited state by two laser fields with Rabi frequencies Ω_0 and Ω_1, with no direct coupling between the two ground states. Under the condition that $|\Omega_0| = |\Omega_1| = \Omega$, we can effectively transfer the population between the states $|0\rangle$ and $|1\rangle$ via the excited state $|r\rangle$, while having some transient population in the excited state during the pulse. Using the MS transformation, this can be seen as analogous to a 2π Rabi pulse in a 2-level system, with an effective Rabi frequency $\Omega_{\text{eff}} = \sqrt{2}\Omega$. Thus the time taken for such an operation is given by

$$T_{\text{frr}} = \frac{2\pi}{\Omega_{\text{eff}}} = \frac{\sqrt{2}\pi}{\Omega}. \tag{8.19}$$

With this understanding, the CNOT gate can be implemented using the Rydberg facilitation mechanism to mediate the interaction between the control and target atoms. Consider two atoms, the control atom A and the target atom B, both of which are Λ-systems and are placed at *facilitation distance* from each other. Let us consider the following sequence of operations to implement the CNOT gate:

1. Apply a π-pulse on the control atom A to excite it from the state $|1\rangle$ to the Rydberg state $|r\rangle$.
2. Use FRR mechanism on the target atom B to transfer the population between the states $|0\rangle$ and $|1\rangle$, conditioned on the state of the control atom A.
3. Apply a π-pulse on the control atom A to de-excite it from the Rydberg state $|r\rangle$ back to the state $|1\rangle$.

Effectively, the Raman pulse on the target atom B is applied with a detuning Δ_0 and Rabi frequency Ω, such that if the control atom A is in the state $|1\rangle$, thus after the first π-pulse, it is in the Rydberg state $|r\rangle$, the target atom B experiences a shift in its Rydberg state due to facilitation from control atom A. This satisfies the *resonant* condition for the Raman pulse on the target atom B, thus allowing the population transfer between the states $|0\rangle$ and $|1\rangle$. Else, if the control atom A is in the state $|0\rangle$, the target atom B does not experience any facilitation and thus the Raman pulse is off-resonant, and no effective population transfer occurs in the driven time T_{frr}. This entire sequence of operations for CNOT implementation is shown in the Fig. 8.4.

The total time taken for the CNOT operation is given by the sum of the time taken for each operation, which is given by

$$T_{\text{CNOT}} = 2 \cdot \frac{\pi}{\Omega_{\text{control}}} + \frac{\sqrt{2}\pi}{\Omega_{\text{target}}} = (2 + \sqrt{2})\frac{\pi}{\Omega} \tag{8.20}$$

where we have assumed that the Rabi frequencies of the control and target atoms are equal, i.e., $\Omega_{\text{control}} = \Omega_{\text{target}} = \Omega$ for simplicity.

Modified CNOT Operation

It is important to note that the CNOT operation implemented as above comes with a caveat. From the corresponding 2-level representation after the MS transformation, the FRR pulse performed on the target atom B, in the bright-dark basis is equivalent to a 2π effective Rabi pulse between the bright state $|b\rangle$ and the Rydberg state $|r\rangle$, while the dark state $|d\rangle$ remains unaffected. We know that the action of a 2π Rabi pulse ends up adding a phase factor of "-1" to the state, i.e., in the target atom B, we have

$$\begin{aligned} |b\rangle &\xrightarrow{2\pi \text{ pulse}} -|b\rangle \\ |d\rangle &\xrightarrow{2\pi \text{ pulse}} +|d\rangle \end{aligned} \tag{8.21}$$

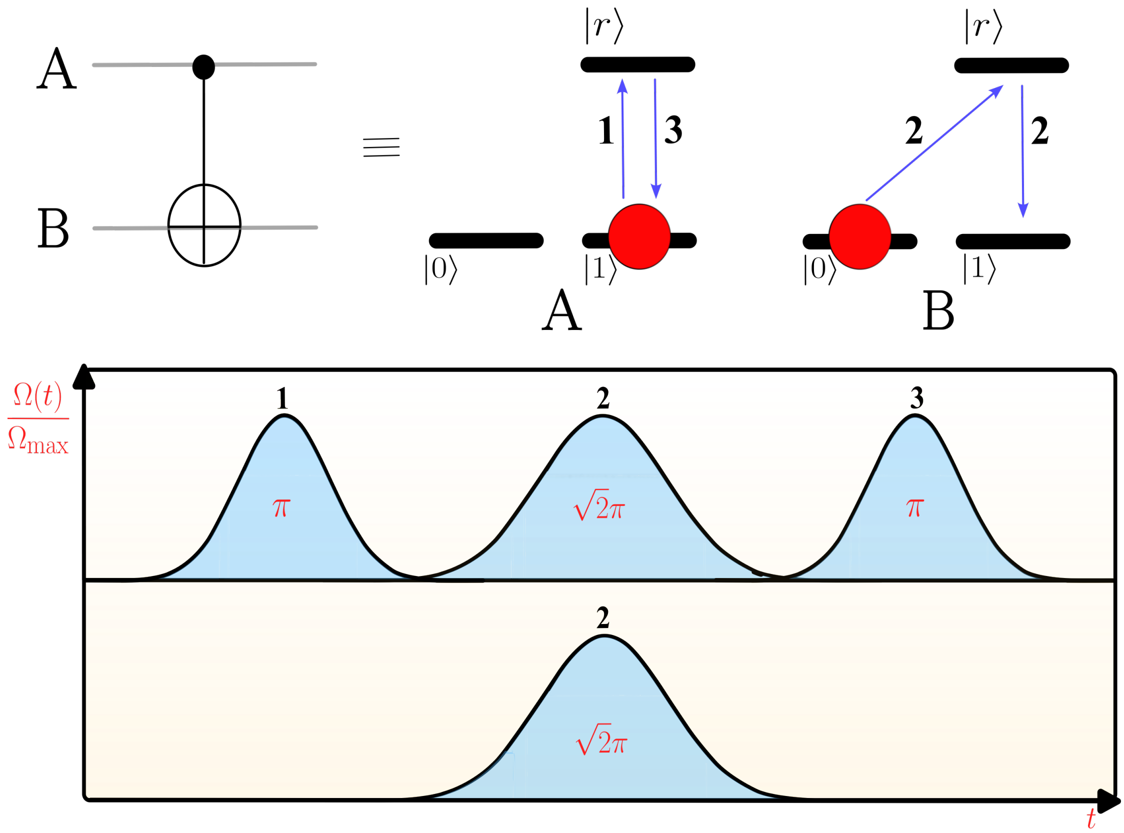


Figure 8.4. (Top) Schematic representation of the proposed CNOT gate operation. (Bottom) Required pulse schedule for the CNOT operation corresponding to the sequence of operations.

Now, consider our Rydberg system architecture for the PFEC algorithm (see Fig. 7.3). Consider the atom B as the control and we have two target atoms A and C on either side of it, both at facilitation distance from the control atom B. Let the initial state of the 3-atom system to be $|\psi\rangle = |0\rangle_A \otimes |1\rangle_B \otimes |0\rangle_C$, then after the first π-pulse on the control atom B, the CNOT operation performed can be seen as

$$\begin{aligned} |\psi\rangle &= |0\rangle_A \otimes |r\rangle_B \otimes |0\rangle_C \\ &= \frac{1}{\sqrt{2}} \left(|b\rangle + |d\rangle\right)_A |r\rangle_B \frac{1}{\sqrt{2}} \left(|b\rangle + |d\rangle\right)_C \\ &= \frac{1}{2} \left(|brb\rangle + |brd\rangle + |drb\rangle + |drd\rangle\right). \end{aligned} \tag{8.22}$$

After applying the $\sqrt{2}\pi$ pulse on atoms A and C, we get

$$\begin{aligned} |\psi\rangle &\propto (-1)^2 |brb\rangle - |brd\rangle - |drb\rangle + |drd\rangle \\ &\propto (|b\rangle - |d\rangle) |r\rangle (|b\rangle - |d\rangle). \end{aligned} \tag{8.23}$$

Applying π pulse on atom B, we get

$$|\psi\rangle = -|111\rangle. \tag{8.24}$$

If we extend this to the general case, where the initial state of the control atom B is $\alpha|0\rangle + \beta|1\rangle$, then the final state of the 3-atom system after the CNOT operation is given by

$$|\psi\rangle = \alpha|000\rangle - \beta|111\rangle \tag{8.25}$$

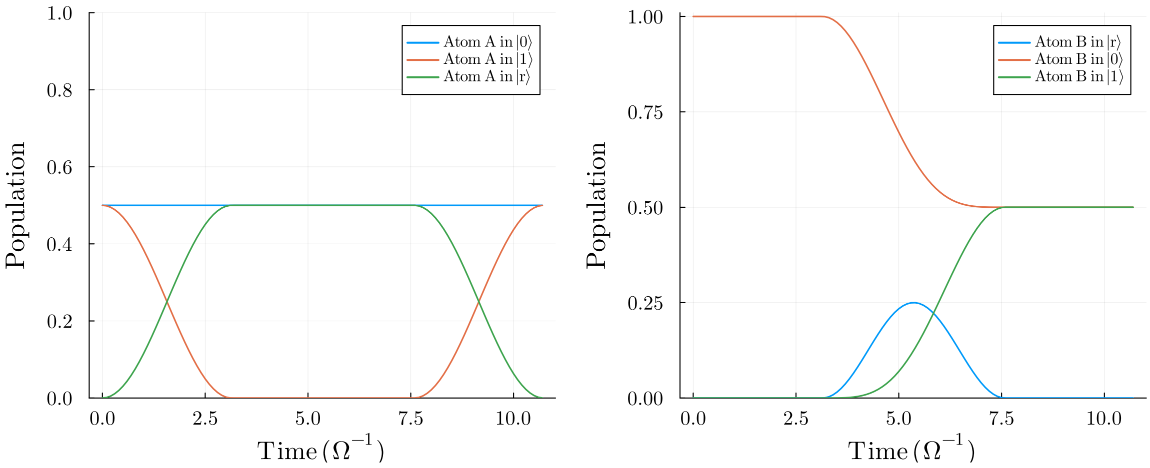


Figure 8.5. Simulation of the CNOT operation on a 3-level Λ-system using the proposed method, corresponding to the Fig. 8.4. The initial state of the system is $\sqrt{0.5}(|0\rangle + |1\rangle)_{\text{A}} \otimes |1\rangle_{\text{B}}$. (Left) Population dynamics of control atom A. (Right) Population dynamics of target atom B.

Thus, we see that the CNOT operation implemented above comes with an additional relative phase factor of "-1". This is an important point to note for the further fidelity analysis of the PFEC algorithm.

9 Implementation

In this chapter, we discuss the implementation of the PFEC protocol using the MCWF method. We begin by laying out a well-thought algorithmic framework for the implementation, then further discuss the modelling of phase-flip errors and finally design the Hamiltonians corresponding to the various gate operations required for the algorithm.

We follow the system layout and the nomenclature established in Ch. 7 and the reader is encouraged to refer to it for clarity.

9.1 Algorithm Workflow

Similar to the BFEC algorithm, the implementation here can also be structured as a sequence of steps that correspond to various quantum circuit operations. The workflow is again designed to be modular, allowing for easy adjustments and optimizations. This allows one to experiment with different control protocols and error models, without having to redesign the entire simulation framework. The workflow is as follows:

1. **Encoding:** T_1 *timesteps*
    - Prepare the atom B in the state $|\psi\rangle = \alpha|1\rangle + \beta|0\rangle$, where α and β are complex coefficients satisfying $|\alpha|^2 + |\beta|^2 = 1$, and atoms A, C in the ground state $|0\rangle$.
    - Drive atoms A and C through Rydberg facilitation mimicking CNOT gates with atom B as the control qubit. The resulting state of the three atoms is $\alpha|111\rangle - \beta|000\rangle$.
    - Apply the Hadamard gate operations on all three atoms which leads to the state $\alpha|+++\rangle + \beta|---\rangle$.
2. **Syndrome Measurement:** T_2 *timesteps*
    - Apply the Hadamard gate operations on all three atoms to convert errors (if any) from Pauli X-basis to Pauli Z-basis.
    - Prepare ancillas a1 and a2 in the ground state $|0\rangle$.
    - Drive ancillas a1 and a2 through Rydberg facilitation with respect to the atoms {A, B, C} such that {A, a1}, {B, a1}, {B, a2} and {C, a2} are the pairs of control and target qubits respectively.
    - For the projective measurement of the ancillas a1 and a2, we use the same technique as discussed in BFEC algorithm, i.e.,
        - Check the population of ancillas a1 and a2 in the state $|1\rangle$. Say the population is p_1 and p_2 respectively.
        - Now, we pick a random number r in the range $[0, 1]$, such that:
          If $r \geq p_1$, then ancilla a1 is projected to $|g\rangle$, else ancilla a1 is projected to $|1\rangle$. Similarly, if $r \geq p_2$, then ancilla a2 is projected to $|g\rangle$, else ancilla a2 is projected to $|1\rangle$.

3. **Correction:** T_3 *timesteps*
   - Based on the measurement outcomes of ancillas a1 and a2, we will apply correction operations as:
     - If only ancilla a1 is projected to $|1\rangle$, then atom A has undergone a phase flip error and thus we drive atom A through Rydberg Facilitation from atom B and ancilla a1 to correct it.
     - If only ancilla a2 is projected to $|1\rangle$, then atom C has undergone a phase flip error and thus we drive atom C through Rydberg Facilitation from atom B and ancilla a2 to correct it.
     - If both ancilla a1 and ancilla a2 are projected to $|1\rangle$, then atom B has undergone a phase flip error and thus we drive atom B through Rydberg Facilitation from ancillas a1, a2 and atoms A, C to correct it.
4. **Decoding:** T_4 *timesteps*
   - Apply the Hadamard gate operations on all three atoms A, B, and C to get back in the original basis.

As highlighted in the Sec. 3.3.2, we exchange the recovery operation and the decoding operation, as compared to the standard quantum error correction protocol. This enables us to use exactly the same correction operations as in the BFEC protocol, rather than designing new ones for the phase-flip errors. This can be seen in the workflow above, where the correction operations are identical to those in the BFEC protocol in Sec. 6.1.

9.2 Modelling Phase-Flip Errors

In this section, we examine the nature of phase-flip errors and develop a model tailored to our system of Rydberg atoms, which will be instrumental in simulating the system dynamics under realistic conditions.

9.2.1 Symmetric Phase-Flip Channel

In general, under the influence of phase-flip errors, the qubit state flips the sign of the $|1\rangle$ component of the qubit state or in other words, a phase-flip is just a rotation about the Z-axis of the Bloch sphere by π radians. Its action can be summarized as

$$\begin{aligned} |1\rangle &\longrightarrow -|1\rangle \quad \text{and} \quad |0\rangle \longrightarrow |0\rangle \\ |+\rangle &\longrightarrow |-\rangle \quad \text{and} \quad |-\rangle \longrightarrow |+\rangle\,. \end{aligned} \tag{9.1}$$

For a more general state in the computational basis, the phase-flip error can be represented as

$$\alpha\,|1\rangle + \beta\,|0\rangle \rightleftarrows -\alpha\,|1\rangle + \beta\,|0\rangle\,. \tag{9.2}$$

This can be represented using the Kraus operators as

$$\mathcal{E}(\rho) = (1-p)\rho + pZ\rho Z^\dagger \tag{9.3}$$

where we have two Kraus operators defined as

$$E_0 = \sqrt{p}\begin{pmatrix}1 & 0\\ 0 & 1\end{pmatrix}, \quad E_1 \quad = \sqrt{1-p}\begin{pmatrix}1 & 0\\ 0 & -1\end{pmatrix} \tag{9.4}$$

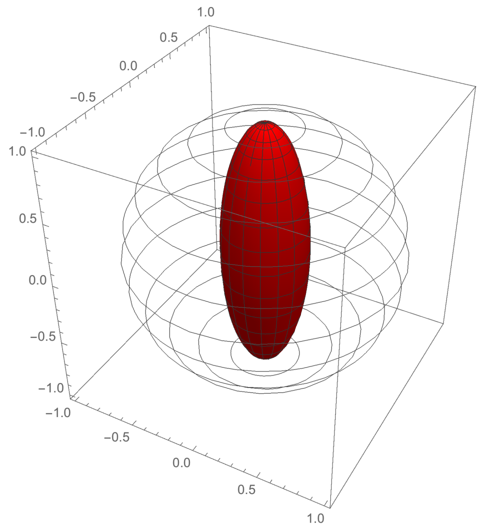


Figure 9.1. The effect of the phase-flip error channel on the Bloch sphere. The outermost sphere represents the initial states of the Bloch sphere, and the inner sphere represents the states after the phase flip error channel has acted on them. Notice that the states on the z-axis remain unaffected. Figure reproduced from [55]. No changes made.

with p as the probability of a phase flip error occurring, and Z is the Pauli-Z operator. This leads to a *symmetric* quantum channel, unlike the *Amplitude Damping* channel, we discussed in Sec. 6.2.1.

In the jump operator formalism, the phase-flip error can be modelled as

$$\hat{C}_{\text{pf}} = \sqrt{\gamma_\perp}\left(|0\rangle\langle 0| - |1\rangle\langle 1|\right) = \sqrt{\gamma_\perp}\hat{Z} \tag{9.5}$$

where $\gamma_\perp$ is the dephasing rate. Similar to the bit-flip error channel, the jump operator in Eq. 9.5 acts on all the data qubits and ancillas in the system for the entire duration of the algorithm.

9.2.2 Optimizing the Dephasing Rate Parameter

The dephasing rate parameter $\gamma_\perp$ is a crucial parameter that quantifies the strength of phase-flip errors in the system. Similar to the case of bit-flip errors, we need to operate in the optimal regime of $\gamma_\perp$ for the success of the protocol.

The realization of the symmetric phase-flip channel is very crucial because operationally, this means that when phase-flips occur as rare, independent kicks, multiple kicks within one cycle of error correction need not compound linearly i.e., some pairs cancel. This can modestly relax the effective error budget compared to the BFEC algorithm. Thus, we can afford a slightly higher error rate or a slightly longer algorithm runtime, while still remaining within the error-correction capability of the code.

To model the impact of these phase-flip errors mathematically, we consider a simple microscopic model where phase-flip errors occur as independent Poissonian errors with rate λ per qubit. According to this model, the probability of k errors occurring in a fixed time window is given by the Poisson distribution

$$\Pr(N = k) = e^{-\lambda}\frac{\lambda^k}{k!}, \quad k = 0, 1, 2, \ldots \tag{9.6}$$

where N is the variable representing the number of errors and in our case, $\lambda = \gamma_\perp T$, with T being the runtime of the error correction algorithm. With this, we are only focused on the *odd* parity of the number of errors on each qubit. This can be calculated as

$$P_{\text{odd}} = q = \sum_{k \in \text{odd}} \Pr(N = k) = \sum_{m=0}^{\infty} e^{-\lambda} \frac{\lambda^{2m+1}}{(2m+1)!} = e^{-\lambda} \sinh \lambda = \frac{e^{\lambda} - e^{-\lambda}}{2} e^{-\lambda} = \frac{1 - e^{-2\lambda}}{2}. \tag{9.7}$$

We know that our phase-flip code can correct up to *one net error*, thus leading to the condition

$$P_{\geq 2 \text{ errors}} < q. \tag{9.8}$$

Since we deal with 5-atoms in our architecture, we can expand $P_{\geq 2 \text{ errors}}$ as

$$P_{\geq 2 \text{ errors}} = 1 - P_0 - P_1 = 1 - \binom{5}{0} q^0 (1-q)^5 - \binom{5}{1} q^1 (1-q)^4 = 1 - (1-q)^5 - 5q(1-q)^4. \tag{9.9}$$

Solving for Eq. 9.8, we get the condition on $\gamma_\perp T$ as

$$0 < \gamma_\perp T < 0.1519. \tag{9.10}$$

9.3 Hamiltonian Design for Gate Operations

Following the workflow outlined previously, we now design the Hamiltonians corresponding to the various steps in the algorithm, motivated by the control processes discussed in the Ch. 8.

Here the operators σ_{x_0r} and σ_{x_1r} represent the transitions between the ground states $\{|0\rangle, |1\rangle\}$ and $\{|0\rangle, |r\rangle\}$ respectively. The operator n_r represents the number operator for the Rydberg state, which is used to account for the detuning and interaction terms in the Hamiltonian. In the notation O^i, the superscript i denotes the atoms or ancillas {A, B, C, a1, a2} on which the operator O acts. The total Hamiltonian is of the shape $(3^5 \times 3^5)$.

The Hamiltonians for the various steps are as follows:

1. **Encoding:**
   a. CNOT operation between atoms {B, A} and {B, C} with first atom as the control qubit and second atom as the target qubit.

$$\begin{aligned} H_{\text{E1}} =& \frac{\Omega_1}{2} \sigma_{x\_1r}^{\text{B}} + \frac{\Omega_2}{2} \Bigg[\left(\sigma_{x\_0r}^{\text{A}} + \sigma_{x\_1r}^{\text{A}} \right) + \left(\sigma_{x\_0r}^{\text{B}} + \sigma_{x\_1r}^{\text{B}} \right) \\ & + \Delta_0 \left(n_r^{\text{A}} + n_r^{\text{B}} \right) + V_{\text{nn}} \left(n_r^{\text{A}} n_r^{\text{B}} + n_r^{\text{C}} n_r^{\text{B}} \right) \Bigg] \end{aligned} \tag{9.11}$$

   b. Hadamard operation on all three atoms A, B and C.
      * Apply virtual Z gate, $VZ(\pi/2)$ on all three atoms A, B and C.
      * Apply $R_x(\pi/2)$ on all three atoms A, B and C, given by the Hamiltonian:

$$\begin{aligned} H_{\text{E2}} =& \frac{\Omega}{2} \left(\sigma_{x\_0r}^{\text{A}} + \sigma_{x\_1r}^{\text{A}} + \sigma_{x\_0r}^{\text{B}} + \sigma_{x\_1r}^{\text{B}} + \sigma_{x\_0r}^{\text{C}} + \sigma_{x\_1r}^{\text{C}} \right) \\ & + \Delta_0 \left(n_r^{\text{A}} + n_r^{\text{B}} + n_r^{\text{C}} \right) \end{aligned} \tag{9.12}$$

* Apply $VZ(\pi/2)$ on all three atoms A, B and C.

The time duration for the encoding can be shown as,

$$T_1 = T_{\mathrm{E1}} + T_{\mathrm{E2}} = \frac{(2+\sqrt{2})\pi}{\Omega} + \frac{\Delta_0 \pi}{4\Omega_1 \Omega_2}$$

2. **Syndrome Measurement:**
   a. Hadamard operation on all three atoms A, B and C.
      * Apply virtual Z gate, $VZ(\pi/2)$ on all three atoms A, B and C.
      * Apply $R_x(\pi/2)$ on all three atoms A, B and C, given by the Hamiltonian:

$$\begin{aligned} H_{\mathrm{S1}} =& \frac{\Omega}{2}\left(\sigma^{\mathrm{A}}_{x\_0r} + \sigma^{\mathrm{A}}_{x\_1r} + \sigma^{\mathrm{B}}_{x\_0r} + \sigma^{\mathrm{B}}_{x\_1r} + \sigma^{\mathrm{C}}_{x\_0r} + \sigma^{\mathrm{C}}_{x\_1r}\right) \\ &+ \Delta_0\left(n^{\mathrm{A}}_r + n^{\mathrm{B}}_r + n^{\mathrm{C}}_r\right) \end{aligned} \tag{9.13}$$

      * Apply $VZ(\pi/2)$ on all three atoms A, B and C.

   b. CNOT operation between {A, a1}, {B, a1} and {B, a2}, {C, a2} with first atom as the control qubit and second ancilla atom as the target qubit.

$$\begin{aligned} H_{\mathrm{S2}} =& \sum_{\mathrm{i}\in\{\mathrm{A,B,C}\}} \frac{\Omega_1}{2}\sigma^{\mathrm{i}}_{x\_1r} + \sum_{\mathrm{j}\in\{\mathrm{a1,a2}\}} \frac{\Omega_2}{2}\left[\left(\sigma^{\mathrm{j}}_{x\_0r} + \sigma^{\mathrm{j}}_{x\_1r}\right) + \Delta_0 n^{\mathrm{j}}_r\right] \\ &+ V_{\mathrm{nn}}\left(n^{\mathrm{A}}_r n^{\mathrm{a1}}_r + n^{\mathrm{B}}_r n^{\mathrm{a1}}_r + n^{\mathrm{B}}_r n^{\mathrm{a2}}_r + n^{\mathrm{C}}_r n^{\mathrm{a2}}_r\right) \end{aligned} \tag{9.14}$$

The time duration for the syndrome measurement can be shown as,

$$T_2 = T_{\mathrm{S1}} + T_{\mathrm{S2}} = \frac{\Delta_0 \pi}{4\Omega_1 \Omega_2} + \frac{(2+\sqrt{2})\pi}{\Omega}$$

3. **Correction:**
   a. **For atom A:** If ancilla a1 is projected to $|1\rangle$, then apply CNOT operation between {B, A} and {a1, A} with first atom as the control qubit and second atom as the target qubit.

$$H_{\mathrm{correct\_A}} = \frac{\Omega_1}{2}\sigma^{\mathrm{a1}}_{x\_1r} + \frac{\Omega_2}{2}\left(\sigma^{\mathrm{A}}_{x\_0r} + \sigma^{\mathrm{A}}_{x\_1r}\right) + \Delta_0 n^{\mathrm{A}}_r + V_{\mathrm{nn}} n^{\mathrm{A}}_r n^{\mathrm{a1}}_r \tag{9.15}$$

   b. **For atom B:** If both ancillas a1 and a2 are projected to $|1\rangle$, then apply CNOT operation between {B, a1}, {B, a2} with first atom as the control qubit and second atom as the target qubit.

$$\begin{aligned} H_{\mathrm{correct\_B}} =& \frac{\Omega_1}{2}\left(\sigma^{\mathrm{a1}}_{x\_1r} + \sigma^{\mathrm{a2}}_{x\_1r}\right) + \frac{\Omega_2}{2}\left(\sigma^{\mathrm{B}}_{x\_0r} + \sigma^{\mathrm{B}}_{x\_1r}\right) + \Delta_0 n^{\mathrm{B}}_r \\ &+ V_{\mathrm{nn}}\left(n^{\mathrm{B}}_r n^{\mathrm{a1}}_r + n^{\mathrm{B}}_r n^{\mathrm{a2}}_r\right) \end{aligned} \tag{9.16}$$

c. **For atom C:** If ancilla a2 is projected to $|1\rangle$, then apply CNOT operation between $\{$B, C$\}$ and $\{$a2, C$\}$ with first atom as the control qubit and second atom as the target qubit.

$$H_{\text{correct\_C}} = \frac{\Omega_1}{2}\sigma^{\text{a2}}_{x\_1r} + \frac{\Omega_2}{2}\left(\sigma^{\text{C}}_{x\_0r} + \sigma^{\text{C}}_{x\_1r}\right) + \Delta_0 n^{\text{C}}_r + V_{\text{nn}} n^{\text{C}}_r n^{\text{a2}}_r \tag{9.17}$$

The time duration for the correction can be shown as,

$$T_3 = T_{\text{correct\_A/B/C}} = \frac{(2+\sqrt{2})\pi}{\Omega}$$

4. **Decoding:**
   - Hadamard operation on all three atoms A, B and C.
     * Apply virtual Z gate, $VZ(\pi/2)$ on all three atoms A, B and C.
     * Apply X-rotation $R_x(\pi/2)$ on all three atoms A, B and C, given by the Hamiltonian:

$$H_{\text{D}} = \frac{\Omega}{2}\left(\sigma^{\text{A}}_{x\_0r} + \sigma^{\text{A}}_{x\_1r} + \sigma^{\text{B}}_{x\_0r} + \sigma^{\text{B}}_{x\_1r} + \sigma^{\text{C}}_{x\_0r} + \sigma^{\text{C}}_{x\_1r}\right) + \Delta_0\left(n^{\text{A}}_r + n^{\text{B}}_r + n^{\text{C}}_r\right) \tag{9.18}$$

     * Apply $VZ(\pi/2)$ on all three atoms A, B and C.

   The time duration for the decoding can be shown as,

$$T_4 = \frac{\Delta_0 \pi}{4\Omega_1\Omega_2}$$

The total time duration for the implementation of the PFEC algorithm can be summarized as,

$$T_{\text{pf}} = T_1 + T_2 + T_3 + T_4. \tag{9.19}$$

9.4 MCWF Simulations

In this section, we present the results of our MCWF simulations for the PFEC algorithm, where we analyze its performance under various conditions of dephasing error rates, storage times, and find the optimal range for the algorithm to be successful. The initial state of the atom B is represented as $|\psi\rangle_B = \alpha\,|1\rangle + \beta e^{i\phi}\,|0\rangle$, where the simulation results are also presented for different values of α and ϕ, analyzing the performance of the algorithm for different initial states.

The results are presented in terms of fidelity, and we use the same definition of the fidelity measure defined in Eq. 6.16 i.e.,

$$F\left(|\psi\rangle_{\text{id}}, \rho_{\text{final}}\right) = \langle\psi_{\text{id}}|\rho_{\text{final}}|\psi_{\text{id}}\rangle,$$

where $|\psi_{\text{id}}\rangle$ is the ideal state and ρ_{final} is the density matrix of the final state after error correction.

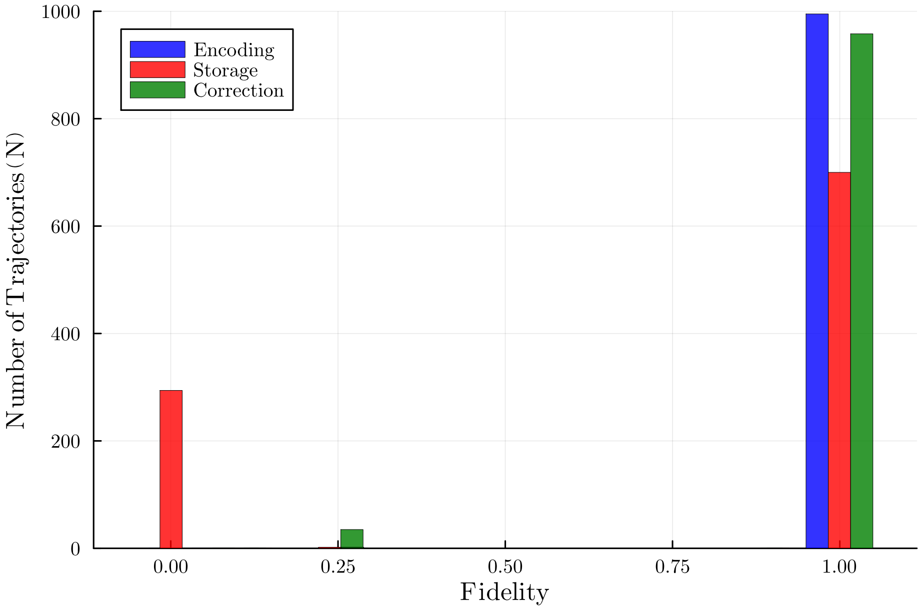


Figure 9.2. Analysis of the distribution of fidelities over 1000 trajectories for the dephasing rate $\gamma_\perp = 10^{-5}\,\Omega$ and storage time $t_{\text{storage}} = 50\,T_{\text{pf}}$. The initial state of the Atom B is chosen to be $\alpha = \sqrt{1/5}$ and $\phi = 0.915$.

Analogous to the BFEC study, we examine the fidelity distributions at successive stages of the PFEC algorithm by running multiple trajectories for a fixed dephasing rate and storage time. Fig. 9.2 shows histograms over 1000 trajectories for $\gamma_\perp = 10^{-5}\,\Omega$ and $t_{\text{storage}} = 50\,T_{\text{pf}}\,\Omega^{-1}$.

After *encoding* step, the distribution is mostly concentrated near 1.0, again indicating highly effective state preparation. Following the *storage* interval, the distribution broadens and a significant fraction of trajectories reach zero fidelity. After the *correction* and *decoding* steps, many of these trajectories are recovered, producing a prominent peak again near 1.0. However, a secondary peak persists at ≈ 0.27, which matches with the analytically expected fidelity for cases with exactly two phase-flip errors that the PFEC code cannot correct. Overall, we observe a dominant peak near 1.0 (successful correction), a smaller peak near 0.27 (uncorrectable two-error events), and a very small residual mass at zero fidelity corresponding to trajectories with three or more phase flips.

As discussed in Sec. 6.4, additional nonzero infidelity can arise from imperfections during gate application, syndrome extraction, and ancilla readout, which can reduce the final fidelity below 1.0 even in otherwise correctable instances.

Before, we move on to success criterion analysis of the algorithm, we would like to discuss the effect of the initial state parameters α and ϕ on the performance of the protocol. One can define the *Trust Threshold* of the protocol, defined as the minimum fidelity below which the protocol is not trusted to perform at all and is to be discarded. In other words, it is the fidelity limit of the single qubit evolution under the influence of the dephasing errors for a time duration $T \to \infty$. It is easy to see that the trust threshold is given by only the diagonal elements of the density matrix ρ_T as $T \to \infty$, as the system completely loses coherence. Thus, we have

$$F^{(1)}_{\text{min}} = |\beta|^4 + |\alpha|^4. \tag{9.20}$$

It is important to note that the Trust Threshold is dependent on the initial state of the atom B, specifically the value of α. Thus, this gives us the first insight into the effect of the initial state parameters on the performance of the algorithm. The Fig. 9.3 highlights this point by showing the average infidelity $(1 - F)$ of the final state after the error-correction protocol for different values of α, averaged over 10 random values of $\phi \in [0, 2\pi)$, for different dephasing rates. One can see that for the dephasing rate $\gamma_\perp = 10^{-1}\,\Omega$, all the cases fail to satisfy the *Trust Threshold*, whereas for the dephasing rate $\gamma_\perp = 10^{-2}\,\Omega$, only the case of $\alpha = \sqrt{2/5}$ and $\alpha = \sqrt{3/5}$ satisfies the trust threshold. Thus, we can immediately disregard these cases while analyzing the success of the protocol.

Finally, similar to the Sec. 6.4, we are interested in the success of the protocol, defined as the net positive difference between average fidelities of the final state of the system with and without the error correction protocol, for a given dephasing error rate, $\gamma_\perp$. We follow the same definition of the fidelity measure defined in Eq. 6.16. Let us calculate the fidelity of the single qubit state under the influence of dephasing errors for a time duration T. Consider the initial state of the atom B to be $|\psi\rangle_\mathrm{B} = \alpha\,|1\rangle + \beta\,|0\rangle$, then the density matrix of the atom B after a time duration T, under the influence of dephasing errors is given by

$$\rho_T = \begin{pmatrix} |\beta|^2 & \alpha^*\beta e^{-\gamma_\perp T} \\ \alpha\beta^* e^{-\gamma_\perp T} & |\alpha|^2 \end{pmatrix}. \tag{9.21}$$

Calculating the fidelity of ρ_T with the initial state $|\psi\rangle_\mathrm{B}$ of the atom B, we obtain

$$F^{(1)} = |\beta|^2 + 2|\alpha|^2|\beta|^2 e^{-\gamma_\perp T} + |\alpha|^4. \tag{9.22}$$

where $F^{(1)}$ is defined as the single particle evolution fidelity under the influence of dephasing errors for a time duration T in the *Symmetric Phase-flip* channel. The success criterion of the algorithm is therefore defined as before,

$$F^{\text{corrected}} > F^{(1)},$$

where $F^{\text{corrected}}$ is the average fidelity of the final state of the system after the error correction protocol.

In the Fig. 9.4, we present the results of our simulations for the dephasing rate $\gamma_\perp = 10^{-5}\,\Omega$, as a function of storage time t_storage. We see the average fidelity for different storage times ranging from $1\,T_\text{pf}$ to $100\,T_\text{pf}$, in comparision to the fidelity of a single qubit evolving under the influence of dephasing errors for a time duration equal to $T_\text{pf}+t_\text{storage}$. It shows that our PFEC algorithm satisfies the success criterion for the storage times between $\approx 1\,T_\text{pf}$ and $\approx 80\,T_\text{pf}$, thus proving its effectiveness. Whereas, the Fig. 9.5 shows the similar simulation results for the dephasing rates $\gamma_\perp = 10^{-3}\,\Omega$ and $\gamma_\perp = 10^{-4}\,\Omega$. We see that for higher dephasing rates, the algorithm fails to satisfy the success criterion for almost all storage times, except for very low storage times in the case of $\gamma_\perp = 10^{-4}\,\Omega$, as shown in the inset of the right plot. Thus, we can conclude that our PFEC algorithm is effective for low dephasing error rates, specifically $\gamma_\perp \leq 10^{-4}\,\Omega$, and for the storage times as discussed above. In this range, the algorithm can be employed iteratively to further maintain high fidelity of the quantum state over extended periods.

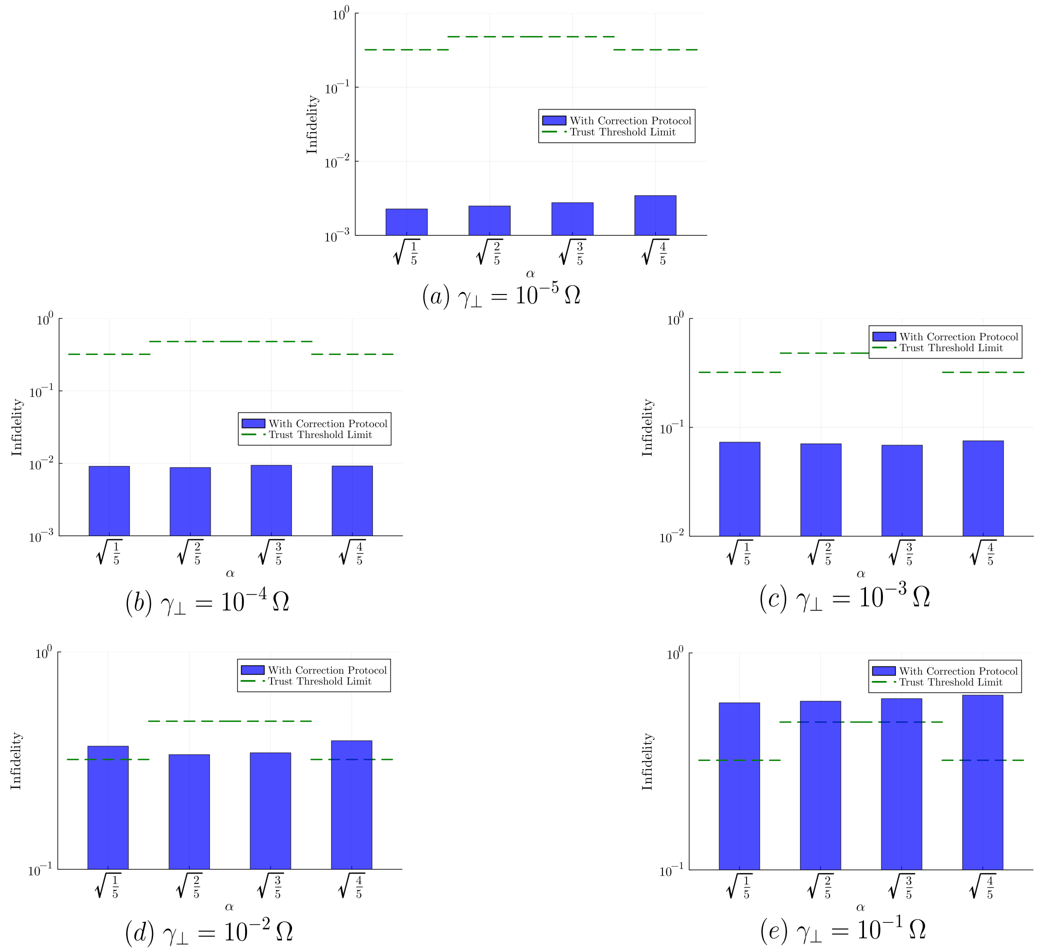


Figure 9.3. Effect of four initial amplitudes for atom B, $\alpha \in \{\sqrt{1/5}, \sqrt{2/5}, \sqrt{3/5}, \sqrt{4/5}\}$, averaged over 10 random values of $\phi \in [0, 2\pi)$, on the performance of the PFEC algorithm. The y-axis shows the average infidelity $1 - F$ of the final state after the error correction protocol (blue bars), compared with the *Trust Threshold* (green lines) for each case. Each panel corresponds to a different dephasing rate.

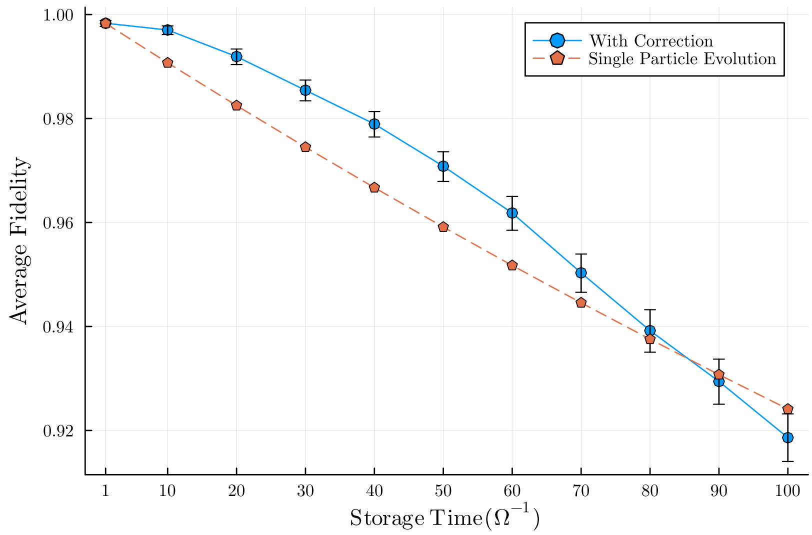


Figure 9.4. Average fidelity over 10 K trajectories, of the final state of the system after the error correction protocol (blue dots) and the single qubit evolution fidelity under the influence of dephasing errors, for the storage times ranging from $1\,T_{\text{pf}}$ to $100\,T_{\text{pf}}$ and dephasing rate $\gamma_{\perp} = 10^{-5}\,\Omega$. The error bars represent the standard deviation of the fidelity due to the finite number of MCWF trajectories simulated. The initial state of the atom B is chosen to be $\alpha = \sqrt{1/5}$ and $\phi = 0.915$.

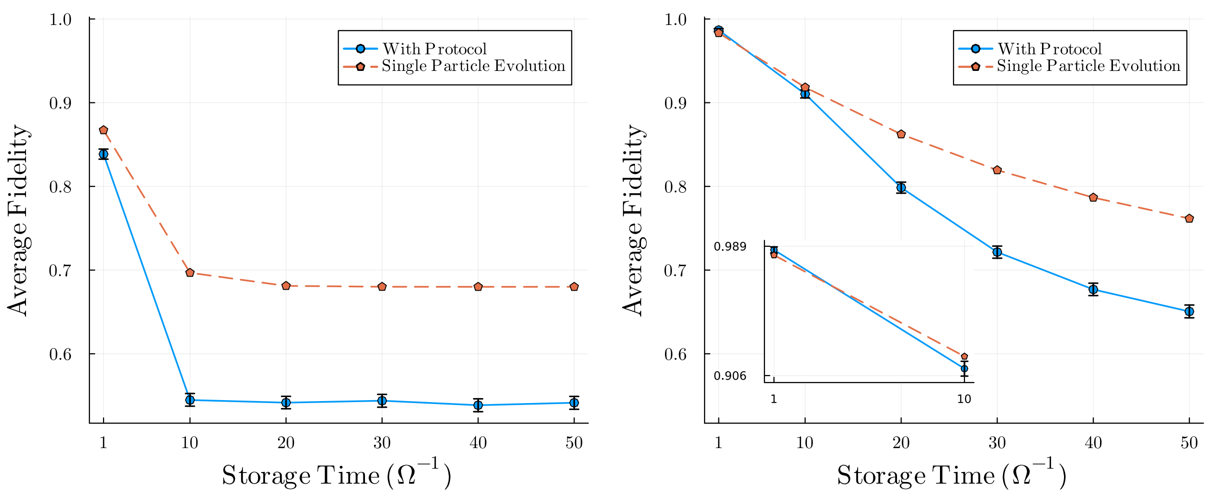


Figure 9.5. Average fidelity over 10 K trajectories, of the final state of the system after the error correction protocol (blue dots) and the single qubit evolution fidelity under the influence of dephasing errors, for the storage times ranging from $1\,T_{\text{pf}}$ to $50\,T_{\text{pf}}$. The error bars represent the standard deviation of the fidelity due to the finite number of MCWF trajectories simulated. The initial state of the atom B is chosen to be $\alpha = \sqrt{1/5}$ and $\phi = 0.915$. (Left) The dephasing rate is chosen to be $\gamma_{\perp} = 10^{-3}\,\Omega$. (Right) The dephasing rate is chosen to be $\gamma_{\perp} = 10^{-4}\,\Omega$. The inset shows a zoomed in version of the plot to highlight the success of the protocol at lower storage times.

Part V

Conclusion and Outlook

In this thesis, we have explored the implementation and performance of quantum error correction protocols, specifically the Bit-Flip Error Correction (BFEC) and Phase-Flip Error Correction (PFEC) codes, for a neutral Rydberg atom-based quantum systems. The error correction protocols designed and simulated in this work leverage the unique properties of Rydberg atoms, mainly *Rydberg blockade* and *Rydberg facilitation*, in addition to their favorable characteristics such as long coherence times and strong interactions, to perform high-fidelity quantum operations necessary for effective error correction.

For both protocols, we started by designing the physical architecture of the system, which includes the arrangement of Rydberg atoms serving as data and ancillary qubits. Following this, we conducted a detailed study of various physical processes that can be utilized to implement the series of quantum gates, including single-qubit rotations and multi-qubit entangling gates, required for the error correction protocols. We then developed a comprehensive implementation strategy, by desiging the Hamiltonians for various stages of the protocols, and Lindbladian operators to model the realistic noise processes affecting the system. Finally, we performed extensive numerical simulations using the MCWF method to evaluate the performance of the protocols under different error rates and operational conditions. In conclusion, we have demonstrated the effectiveness of these protocols in mitigating bit-flip and phase-flip errors in a quantum system, thereby enhancing the fidelity of quantum states over time. Additionally, we have analyzed the impact of various parameters on the performance of these protocols and identified optimal conditions for their operation.

Following this, we also outline the implementation of the Shor code, which combines both bit-flip and phase-flip error correction capabilities. The Shor code is a more complex protocol that requires a larger number of qubits and more intricate gate operations. We have not performed numerical simulations for the Shor code in this work since the minimum system size required for its implementation is larger than what can be simulated using Exact Diagonalization or Quantum Trajectories methods within reasonable computational resources. However, we have provided a detailed theoretical framework for its implementation using Rydberg atoms (see Sec. B), most of which can be borrowed from the BFEC and PFEC protocols discussed earlier.

Looking ahead, there are several avenues for future research and development in this area. We aim to explore the scalability of these protocols and investigate their performance in larger quantum systems. This includes studying the effects of increased qubit numbers and more complex error models on the efficacy of the error correction strategies. Furthermore, one could refine the physical implementation of the protocols by optimizing gate fidelities and minimizing operational times, which are crucial for practical quantum computing applications. Another important direction is to explore the integration of these error correction protocols with quantum algorithms and applications, to assess their impact on overall computational performance. Finally, it would be desirable to extend our numerical simulations to include the Shor code and other advanced error correction schemes such as *Topological Codes* [44], to provide a comprehensive understanding of their capabilities and limitations in real-world quantum systems. One can also combine all these ideas and implement a *fault-tolerant* [60] quantum computing architecture using Rydberg atoms, which would be a significant step towards realizing practical and reliable quantum computers.

Appendix

A Derivation of the Lindblad Master Equation

In this Appendix, we derive the Lindblad Master equation by first deriving the Born-Markov master equation for a general open quantum system. The derivation presented here is based on the work of Brasil et al. [61] and D. Manzano [62].

Consider a quantum system S, whose dynamics is the object of interest, interacting with an environment, which can be modelled as a bath B. Let the system S belong to the Hilbert space $\mathcal{H}_S$ and the bath B belong to the Hilbert space $\mathcal{H}_B$. The combined system-bath is then described by the Hilbert space $\mathcal{H}_{SB} = \mathcal{H}_S \otimes \mathcal{H}_B$. The total Hamiltonian of the system-bath can be written as

$$\hat{H}_{SB} = \hat{H}_S \otimes \hat{I}_B + \hat{I}_S \otimes \hat{H}_B + \alpha \hat{H}_{SB} \tag{A.1}$$

where $\hat{H}_S$ is the Hamiltonian of the system, $\hat{H}_B$ is the Hamiltonian of the bath, and $\hat{H}_{SB}^{int}$ is the interaction Hamiltonian between the system and the bath. The parameter α is a coupling constant that quantifies the strength of the interaction between the system and the bath. $\hat{I}_S$ and $\hat{I}_B$ are the identity operators in the Hilbert spaces of the system and the bath, respectively. The time evolution of the combined system-bath can be described by the von Neumann equation as,

$$\mathrm{i}\hbar \frac{d}{dt}\hat{\rho}_{SB}(t) = \left[\hat{H}_S \otimes \hat{I}_B + \hat{I}_S \otimes \hat{H}_B + \alpha \hat{H}_{SB}, \hat{\rho}_{SB}(t)\right] \tag{A.2}$$

By choosing to work in the interaction picture, we can write the global Hamiltonian as

$$\hat{H}(t) = e^{\frac{\mathrm{i}}{\hbar}(\hat{H}_S+\hat{H}_B)t}\hat{H}_{SB}e^{-\frac{\mathrm{i}}{\hbar}(\hat{H}_S+\hat{H}_B)t} \tag{A.3}$$

The density matrix of the system-bath at any time t can be expressed as

$$\hat{\rho}(t) = e^{\frac{\mathrm{i}}{\hbar}(\hat{H}_S+\hat{H}_B)t}\hat{\rho}_{SB}(t)e^{-\frac{\mathrm{i}}{\hbar}(\hat{H}_S+\hat{H}_B)t} \tag{A.4}$$

The equation A.2 can be rewritten in the interaction picture as

$$\mathrm{i}\hbar \frac{d}{dt}\hat{\rho}(t) = \alpha \left[\hat{H}(t), \hat{\rho}(t)\right] \tag{A.5}$$

The solution for the equation A.5 can be given as the following integral equation

$$\hat{\rho}(t) = \hat{\rho}(0) - \frac{\mathrm{i}}{\hbar}\alpha \int_0^t \left[\hat{H}(t'), \hat{\rho}(t')\right] dt' \tag{A.6}$$

To derive the master equation, we will make the following assumptions:

1. **Assumption 1:** The system is weakly coupled to the bath, i.e., α is small. This would allow us to introduce perturbative theory. This is known as the Born approximation.

2. **Assumption 2:** The bath is in a thermal equilibrium state, which is described by the Gibbs state $\hat{\rho}_B = \frac{e^{-\beta \hat{H}_B}}{Z_B}$, where $\beta = \frac{1}{k_B T}$ is the inverse temperature, k_B is the Boltzmann constant, and $Z_B = \mathrm{tr}_B\left(e^{-\beta \hat{H}_B}\right)$ is the partition function of the bath.

3. **Assumption 3:** The system-bath interaction is Markovian, meaning that the system's future evolution depends only on its present state and not on its past states.

4. **Assumption 4:** The system-bath interaction is weakly dependent on the system's state, meaning that the system's state does not significantly affect the bath's state.

We proceed by substituting the equation A.6 into von Neumann equation A.5,

$$\frac{d}{dt}\hat{\rho}(t) = -\frac{\mathrm{i}}{\hbar}\alpha\left[\hat{H}(t), \hat{\rho}(0)\right] - \frac{1}{\hbar^2}\alpha^2\left[\hat{H}(t), \int_0^t \left[\hat{H}(t'), \hat{\rho}(t')\right] dt'\right] \tag{A.7}$$

For the **Assumption 1**, the above substitution is enough to obtain the first order correction to the density matrix. We can now obtain the density matrix of the system at time t by taking the partial trace over the bath degrees of freedom, leading to

$$\frac{d}{dt}\hat{\rho}_s(t) = -\frac{\mathrm{i}}{\hbar}tr_B\left(\left[\hat{H}(t), \hat{\rho}(0)\right]\right) - \frac{1}{\hbar^2}\alpha^2 tr_B\left(\left[\hat{H}(t), \int_0^t \left[\hat{H}(t'), \hat{\rho}(t')\right] dt'\right]\right) \tag{A.8}$$

From equation A.3, $\hat{H}(t)$ is only dependent on $\hat{H}_{SB}$ and it can defined in a manner such that the first term on the right hand side vanishes, leading to

$$\frac{d}{dt}\hat{\rho}_S(t) = -\frac{1}{\hbar^2}\alpha^2 tr_B\left(\left[\hat{H}(t), \int_0^t \left[\hat{H}(t'), \hat{\rho}(t')\right] dt'\right]\right) \tag{A.9}$$

From the **Assumption 2** and **Assumption 4**, we can say that the state of the bath will not change over time and no correlations between the system and the bath will be created. Thus, the bath state at anytime t can be approximated as,

$$\hat{\rho}_B(t) \approx \hat{\rho}_B(0) = \hat{\rho}_B \tag{A.10}$$

This allows us to write the equation A.9 as

$$\frac{d}{dt}\hat{\rho}_S(t) = -\frac{1}{\hbar^2}\alpha^2 tr_B\left(\left[\hat{H}(t), \int_0^t \left[\hat{H}(t'), \hat{\rho}_S(t')\hat{\rho}_B\right] dt'\right]\right) \tag{A.11}$$

We can now use the **Assumption 3**, which allows us to replace the integral over the past states with the present state, and taking $\alpha = 1$ leads to

$$\frac{d}{dt}\hat{\rho}_S(t) = -\frac{1}{\hbar^2}\int_0^t dt' tr_B\left(\left[\hat{H}(t), \left[\hat{H}(t'), \hat{\rho}_S(t')\hat{\rho}_B\right]\right]\right) \tag{A.12}$$

This is the Born-Markov Master Equation [63], which describes the time evolution of the density matrix of the system in the presence of a bath.

The Born-Markov Master Equation is a general equation that describes the time evolution of the density matrix of a quantum system in the presence of a bath. However, it is not always easy to work with, especially when the system-bath interaction is complex. To address this issue, we can derive a more specific form of the Born-Markov Master Equation, known as the Lindblad Master Equation, which is a more tractable equation that can be used to describe the dynamics of open quantum systems. The Lindblad Master Equation is a special case of the Born-Markov Master Equation, which can be derived by making some additional assumptions about the system-bath interaction and the form of the system-bath Hamiltonian.

Let us assume that the system-bath interaction Hamiltonian can be simplified as

$$\hat{H}_{SB} = \hbar\sum_i \hat{S}_i \otimes \hat{B}_i \tag{A.13}$$

where $\hat{S}_i$ is any general system operator and $\hat{B}_i$ is any general bath operator. To proceed, we make one more assumption, which is at $t = 0$ the system and the bath have a seperable state of the form $\rho_t(0) = \rho_S(0) \otimes \rho_B(0)$. This means that there are no correlations between the system and the bath at the initial time. This assumption is valid if the system and the bath have not interacted previously or if the correlations between them are short-lived and decay quickly. Following this, from Eq. A.12, by changing the integral variable $t' \rightarrow t - t'$, we obtain the famous Redfield Equation [64],

$$\frac{d}{dt}\hat{\rho}_S(t) = -\frac{1}{\hbar^2}\int_0^\infty dt' tr_B\left(\left[\hat{H}(t), \left[\hat{H}(t-t'), \hat{\rho}_S(t')\hat{\rho}_B\right]\right]\right) \tag{A.14}$$

The above equation does not guarantee the complete positivity of the density matrix $\hat{\rho}_S(t)$, which is a crucial property for any physical state. To ensure complete positivity, we can make use of the *rotating wave approximation (RWA)*. We use the spectrum of the superoperator $\tilde{H}A \equiv [\hat{H}_S, A]$ to decompose the system operators $\hat{S}_i$ into complete basis of space $\mathcal{H}_{sys}$ as

$$\hat{S}_i = \sum_\omega \hat{S}_i(\omega) \tag{A.15}$$

such that ω are the eigenvalues of the superoperator $\tilde{H}$ and $\hat{S}_i(\omega)$ are the corresponding eigenoperators. The eigenoperators satisfy the following commutation relations,

$$[\hat{H}_S, \hat{S}_i(\omega)] = -\hbar\omega\hat{S}_i(\omega) \quad \text{and} \quad [\hat{H}_S, \hat{S}_i^\dagger(\omega)] = \hbar\omega\hat{S}_i^\dagger(\omega) \tag{A.16}$$

Switching back to the Schrödinger picture, we get

$$\tilde{H}_{SB}(t) = \sum_{k,\omega} e^{-\mathrm{i}\omega t}\hat{S}_k(\omega)\otimes\tilde{B}_k(t) = \sum_{k,\omega} e^{+\mathrm{i}\omega t}\hat{S}_k^\dagger(\omega)\otimes\tilde{B}_k^\dagger(t) \tag{A.17}$$

Expanding the commutators in the Redfield equation A.14,

$$\begin{aligned}\frac{d}{dt}\hat{\rho}_S(t) = -\frac{1}{\hbar^2}tr_B\Bigg[&\int_0^\infty dt'\hat{H}_{SB}(t)\hat{H}_{SB}(t-t')\hat{\rho}_S(t)\otimes\hat{\rho}_B - \int_0^\infty dt'\hat{H}_{SB}(t)\hat{\rho}_S(t)\otimes\hat{\rho}_B\hat{H}_{SB}(t-t') \\ &- \int_0^\infty dt'\hat{H}_{SB}(t-t')\hat{\rho}_S(t)\otimes\hat{\rho}_B\hat{H}_{SB}(t) + \int_0^\infty dt'\hat{\rho}_S(t)\otimes\hat{\rho}_B\hat{H}_{SB}(t-t')\hat{H}_{SB}(t)\Bigg]\end{aligned} \tag{A.18}$$

Substituting for $\hat{H}_{SB}(t-t')$ in terms of $\hat{S}_i(\omega)$ and for $\hat{H}_{SB}(t)$ in terms of $\hat{S}_i^\dagger(\omega)$, we obtain

$$\frac{d}{dt}\hat{\rho}_S(t) = \sum_{\omega,\omega'}\sum_{k,l}\left[e^{\mathrm{i}(\omega'-\omega)t}\Gamma_{kl}(\omega)\left(\hat{S}_l(\omega)\hat{\rho}_S(t)\hat{S}_k^\dagger(\omega') - \hat{S}_k^\dagger(\omega')\hat{S}_l(\omega)\hat{\rho}_S(t)\right) + h.c.\right] \tag{A.19}$$

where the effect of the bath is captured in the coefficients $\Gamma_{kl}(\omega)$, which are defined as

$$\Gamma_{kl}(\omega) = \int_0^\infty dt' e^{\mathrm{i}\omega t'} tr_B\left(\tilde{B}_k^\dagger(t)\tilde{B}_l(t-t')\hat{\rho}_B\right) \tag{A.20}$$

We can now make the rotating wave approximation, by considering that the terms with $|\omega - \omega'| \gg 1$ oscillate rapidly than the typical timescale of the system dynamics, and thus their contribution to the dynamics averages out to zero over time. Thus, we only keep the terms with $\omega = \omega'$, in the low coupling regime. This reduces the Eq. A.19 to

$$\frac{d}{dt}\hat{\rho}_S(t) = \sum_\omega\sum_{k,l}\left(\Gamma_{kl}(\omega)\left[\hat{S}_l(\omega)\hat{\rho}_S(t)\hat{S}_k^\dagger(\omega) - \hat{S}_k^\dagger(\omega)\hat{S}_l(\omega)\hat{\rho}_S(t)\right] + h.c.\right) \tag{A.21}$$

Dividing the coefficients $\Gamma_{kl}(\omega)$ into their Hermitian and non-Hermitian parts, we can write

$$\Gamma_{kl}(\omega) = \frac{1}{2}\gamma_{kl}(\omega) + \mathrm{i}\pi_{kl}(\omega) \tag{A.22}$$

Substituing this into the Eq. A.21, transforming back to the Schrödinger picture and simplifying into diagonal form, we obtain the Lindblad Master Equation,

$$\frac{d}{dt}\hat{\rho}_S(t) = -\frac{\mathrm{i}}{\hbar}[\hat{H}_S+\hat{H}_{LS}, \hat{\rho}_S(t)]+\sum_i \gamma_i \left(\hat{A}_i(\omega)\hat{\rho}_S(t)\hat{A}_i^\dagger(\omega) - \frac{1}{2}\left\{\hat{A}_i^\dagger(\omega)\hat{A}_i(\omega), \hat{\rho}_S(t)\right\}\right) \quad \text{(A.23)}$$

where $\hat{H}_{LS} = \sum_\omega \sum_{k,l} \pi_{kl}(\omega)\hat{S}_k^\dagger(\omega)\hat{S}_l(\omega)$ is the Lamb shift Hamiltonian, which describes the renormalization of the system's energy levels due to its interaction with the bath. In the simplest case, for one relevant frequency ω_0, the Lindblad Master Equation can be written as

$$\frac{d}{dt}\hat{\rho}_S(t) = -\frac{\mathrm{i}}{\hbar}[\hat{H}_S, \hat{\rho}_S(t)] + \sum_i \gamma_i \left(\hat{A}_i\hat{\rho}_S(t)\hat{A}_i^\dagger - \frac{1}{2}\left\{\hat{A}_i^\dagger\hat{A}_i, \hat{\rho}_S(t)\right\}\right) \equiv \hat{\mathcal{L}}\hat{\rho}_S(t) \quad \text{(A.24)}$$

where $\hat{A}_i$ are the jump operators, which describe the effect of the bath on the system, and γ_i are the corresponding rates.

B Extension to Shor Code Algorithm

In this Appendix, we extend the QEC protocols presented in the main text to the Shor code, which can correct arbitrary single qubit errors. We borrow heavily from the two error correction protocols we have designed, and combine the best mentioned ideas and techniques explored so far.

As usual, we start by describing the system architecture for the protocol. For the qubit states, we stick to the same 3-level Λ-system as defined in the PFEC algorithm and use the same notation as before. This choice is motivated by the same reason as seen before, i.e., to overcome the dynamical phase problem. Following this, we describe the system architecture in detail.

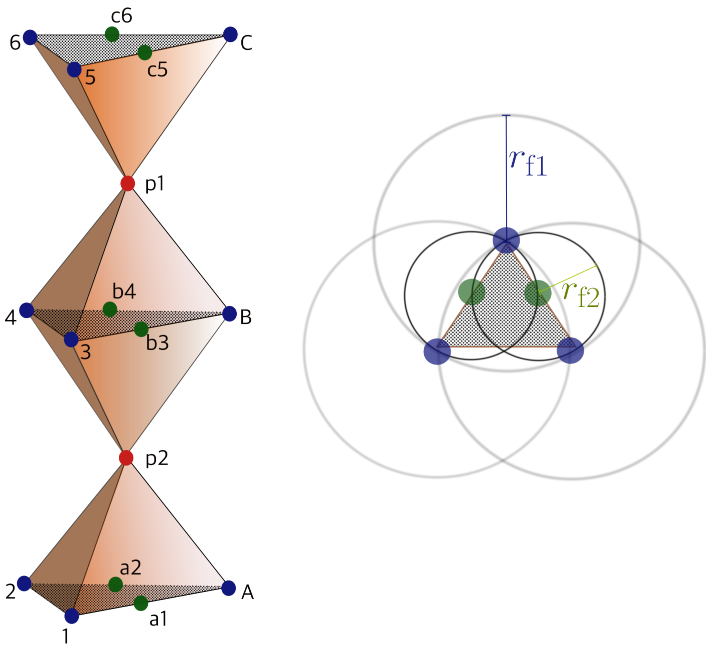


Figure B.1. (Left) Proposed 3D architecture for the Shor code protocol. The 9 data qubits are shown in blue, the 6 ancillary atoms for bit-flip error syndrome measurement are shown in green, and 2 ancillary atoms for phase-flip error syndrome measurement are shown in red. (Right) The distances between the atoms and their facilitation radii corresponding to a particular plane from the 3D architecture.

As we have seen previously in Sec. 3.3.3, it is a 9-qubit code, which utilizes 8 more ancillary qubits for syndrome measurement and error correction. The proposed 3-dimensional architecture of the system, is shown in the Fig. B.1. We arrange the atoms such that 9 qubits are placed in three parallel planes, with 3-atoms in each plane forming an equilateral triangle. This is shown by the blue atoms {A, B, C, 1, 2, 3, 4, 5, 6} in the figure. The six ancillary qubits for bit-flip error syndrome measurement are placed two in each plane as shown by the green atoms {a1, a2, b3, b4, c5, c6}. The two ancillary qubits for phase-flip error syndrome measurement are placed in between the planes as shown by the red atoms {p1, p2}. Thus, we propose a model of three different types of atoms, each with different Rydberg states, such that their facilitation radii are r_{f1}, r_{f2} and r_{f3}.

The data qubits in the same plane are at each other's facilitation radii r_{f1}, while the bit-flip ancillary qubits have the facilitation radius $r_{f2} = r_{f1}/2$ and are placed midway between the data qubits. The phase-flip ancillary qubits are placed such that they are equidistant from the plane of data qubits above and below them, and are at its facilitation radii r_{f3} from all the data qubits in both the planes. This arrangement allows us to perform the necessary gate operations, syndrome measurements efficiently utilizing the facilitation/blockade effects.

For the implementation of the gate operations, we have already explored the efficient techniques for various gate implementation in the main parts of the thesis. Of both the sets of techniques, we choose the same ones as in the phase-flip protocol as we are dealing with the same qubit layout. More precisely, we use the *Hadamard gate using Raman Transition* technique and the *CNOT using the Rydberg Facilitation* technique for single-qubit and two-qubit gates respectively.

Further to this, we describe the modular Shor code algorithm workflow as follows:

1. **Encoding:** T_1 *timesteps*
   - Prepare the atom B in the state $|\psi\rangle = \alpha|1\rangle + \beta|0\rangle$, where α and β are complex coefficients satisfying $|\alpha|^2 + |\beta|^2 = 1$ and rest of the atoms in the state $|0\rangle$.
   - Drive atoms A and C through Rydberg facilitation mimicking CNOT gates with atom B as the control qubit. The resulting state of the three atoms is $\alpha|111\rangle - \beta|000\rangle$.
   - Apply the Hadamard gate operations on all three atoms A,B and C to prepare the system into equal superposition of all basis states i.e., $\alpha|+++\rangle - \beta|+++\rangle$.
   - Drive atoms 1 - 6 through Rydberg facilitation mimicking CNOT gates with atoms A, B and C such that {(A,1,2), (B,3,4), (C,5,6)} are the {(control, target1, target2)} qubit triplets. The resulting state of the nine atoms is $\alpha|1\rangle_L + \beta|0\rangle_L$.

2. **Syndrome Measurement:** T_2 *timesteps*

Case 1: **Bit-Flip error syndrome measurement:** T_b *timesteps*

i. Drive ancillary atoms a1 - c6 through Rydberg facilitation mimicking CNOT gates such that {(a1, A, 1), (a2, A, 2), (b3, B, 3), (b4, B, 4), (c5, C, 5), (c6, C, 6)} are the {(target, control1, control2)} qubit triplets.

Case 2: **Phase-Flip error syndrome measurement:** T_p *timesteps*

i. Apply the Hadamard gate operations on all nine data atoms {A, B, C, 1, 2, 3, 4, 5, 6} to convert the state to the bit-flip basis.

ii. Drive ancillary atoms p1 - p2 through Rydberg facilitation mimicking CNOT gates such that {(p1, A, 1, 2, B, 3, 4), (p2, B, 3, 4, C, 5, 6)} are the {(target, control1, control2, control3, control4, control5)} qubit sextuplets.

For the readout of the ancillary atoms, we use the same projective measurement technique as described in Sec. 9.1. These syndrome measurements are done *sequentially* such that, the BFEC syndrome measurement is done first, followed by the PFEC syndrome measurement. Thus the total time taken for the syndrome measurement step is $T_2 = T_\mathrm{b} + T_\mathrm{p}$.

3. **Error Correction:** T_3 *timesteps*

Case 1: **BFEC:** T_b *timesteps*

– If any of the ancillary atoms a1 - c6 is found in the state $|1\rangle$, drive the corresponding data qubit through Rydberg facilitation mimicking a CNOT gate, with the ancillary atom as the control qubit and the data qubit as the target qubit.

Case 2: **PFEC:** T_p *timesteps*

- If any of the ancillary atoms p1 - p2 is found in the state $|1\rangle$, drive the corresponding data qubits through Rydberg facilitation mimicking CNOT gates, with the ancillary atom as the control qubit and the data qubits as the target qubits.

4. **Decoding:** T_4 *timesteps*
   - Apply the Hadamard gate operations on all nine data qubits to get the state back to the logical basis.

For this algorithm, the quantum error channel is the combination of the previously discussed *Amplitude Damping* and *Symmetric Phase-Flip Channel*. Thus, we use the same Lindblad operators as described in Eq. 6.4 and Eq. 9.5. The total time taken for the protocol is $T = T_1 + T_2 + T_3 + T_4$. The design of Hamiltonians for each step of the protocol can be designed in a similar manner as described in Sec. 9.3, with the only difference being the number of atoms involved.

C Computational Feasibility of Simulating the Shor Code Algorithm

In this Appendix, we discuss the difficulties in simulating the Shor code algorithm using the MCWF method, as described in Sec. 2.3.

We model $N = 17$ atoms, each as a 3-level system. The many-body Hilbert space has dimension

$$d = 3^{17} = 1.29 \times 10^8. \tag{C.1}$$

Even a single state vector $|\psi\rangle \in \mathbb{C}^d$ in **Complex64** costs $\sim d \times 8$ bytes ≈ 1 GB of memory. A *dense* Hamiltonian matrix $\hat{H} \in \mathbb{C}^{d\times d}$ would cost $\sim d^2 \times 8$ bytes $\approx 1.3 \times 10^{16}$ bytes $\approx 13,000$ TB of memory, which is infeasible to store in memory. Thus, we use a *sparse* matrix representation of the Hamiltonian.

We simulate the algorithm using the MCWF method, which evolves the system under the non-Hermitian effective Hamiltonian

$$\hat{H}_{\text{eff}} = \hat{H} - \frac{i\hbar}{2}\sum_i \hat{A}_i^\dagger \hat{A}_i \tag{C.2}$$

interrupted by stochastic quantum jumps $\hat{A}_i$. Each step requires a **matvec** (matrix-vector multiplication) operation $\hat{H}_{\text{eff}}\,|\psi\rangle$, which is the most computationally expensive operation in the simulation. Because our controls are time-dependent, the ODE solver must take small steps set by the fastest pulses, and every quantum jump resets the integrator, preventing reuse of step history or factorizations.

Now, moving from phase-flip code to the Shor code, the number of qubits increases from $N = 5$ to $N = 17$, thus the growth is $\sim 3^{12} \approx 531441$ times in the Hilbert space dimension. This leads to a similar growth of $\sim 5.3\times10^5$ times in the time taken for a single **matvec** operation. This inflation is before considering longer protocol time, more complex Hamiltonians, and more jumps.

For emperical reference:

- BFEC protocol: $N = 5$ (qubits), single trajectory runtime ~ 40 s for $T_{\text{wait}} = 0$.
- PFEC protocol: $N = 5$ (qutrits), single trajectory runtime ~ 60 s for $T_{\text{wait}} = 0$.
- Memory requirement for simulating 10^3 trajectories (with sparse **Complex64** matrix representation) was: ~ 1 GB and ~ 1.5 GB for the protocols respectively.

For the Shor code protocol ($N = 17$ qutrits), the runtime for simulating the *encoding* step alone was ~ 10 hours. A naive dimensional scaling from the 5-qutrit baseline (60 s$\times 3^{12}$) would already predict $\mathcal{O}(1year)$ per trajectory, consistent with our observation that the full protocol is impractical on a typical workstation.

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Declaration of Authorship

I hereby declare that I have written this Master's thesis independently and have used only the sources and aids indicated. All material that is quoted directly or paraphrased from published or unpublished sources have been clearly cited. I further acknowledge that AI-based language tools such as ChatGPT, were solely used to help improve grammar and phrasing.

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